

The Thinking Hand: Existential and Embodied Wisdom in Architecture

Juhani Pallasmaa

In our current global networked culture that puts so much emphasis on the virtual and the visual, the mind and the body have become detached and ultimately disconnected. Though physical appearance is idolised for its sexual appeal and its social identity, the role of the body in developing a full understanding of the physical world and the human condition has become neglected. The potential of the human body as a knowing entity – with all our senses as well as our entire bodily functions being structured to produce and maintain silent knowledge together – fails to be recognised.

It is only through the unity of mind and body that craftsmanship and artistic work can be fully realised. Even those endeavours that are generally regarded as solely intellectual, such as writing and thinking, depend on this union of mental and manual skills.

In *The Thinking Hand*, Juhani Pallasmaa reveals the miraculous potential of the human hand. He shows how the pencil in the hand of the artist or architect becomes the bridge between the imagining mind and the emerging image. The book surveys the multiple essences of the hand, its biological evolution and its role in the shaping of culture, highlighting how the hand–tool union and eye–hand–mind fusion are essential for dexterity and how ultimately the body and the senses play a crucial role in memory and creative work. Pallasmaa here continues the exploration begun in his classic work *The Eyes of the Skin* by further investigating the interplay of emotion and imagination, intelligence and making, theory and life, once again redefining the task of art and architecture through well-grounded human truths.

Juhani Pallasmaa is one of Finland's most distinguished architects and architectural thinkers. His previous positions include: Rector of the Institute of Industrial Arts, Helsinki; Director of the Museum of Finnish Architecture, Helsinki; and Professor and Dean of the Faculty of Architecture, Helsinki University of Technology. He has also held visiting professorships in several universities internationally. Pallasmaa is the author/editor of 24 books, including *The Eyes of the Skin: Architecture and the Senses* (Academy, 1995 and John Wiley & Sons, 2005), *The Architecture of Image: Existential Space in Cinema* (Helsinki, 2001) and *Encounters* (Helsinki, 2005).

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The Thinking Hand

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Existential and Embodied Wisdom in Architecture

JUHANI PALLASMAA



Hands are part of a person's individual persona and character, but they also perform their independent actions, and are crucial in human communication with a language of their own. Matteo Zambelli, *You Are More Than One*, 2006. Collage made by Photoshop 7.0. The process of making the Collage:

1. I dropped a rectangular glass wrapped in a cardboard

- envelope from the third level of my house. After this, I put each piece of glass on a glass table, where I recomposed the original rectangular glass, now broken into pieces.
2. I asked my parents and relatives to put their left or right hand under the pieces of glass, and took several shots.
3. I downloaded the photos on my computer, and assembled them in Photoshop 7.0. When

assembling the photos, I didn't always use the entire hands, some of the images had just one or two details of the whole hand, others were doubled and blurred. I created different layers for each piece of the hand, superimposed all the layers with different gradients of transparency, and merged all these layers for the final photos.

Introduction

Embodied Existence and Sensory Thought

'Briefly what I am proposing is that the psychology of the mature human being is an unfolding, emergent, oscillating, spiralling process marked by progressive subordination of older, lower-order behaviour systems to newer, higher-order systems as man's existential problems change.'

Clare W Graves¹

Western consumer culture continues to project a dualistic attitude towards the human body. On the one hand we have an obsessively aestheticised and eroticised cult of the body, but on the other, intelligence and creative capacity are equally celebrated as totally separate, or even exclusive individual qualities. In either case, the body and the mind are understood as unrelated entities that do not constitute an integrated unity. This separation is reflected in the strict division of human activities and work into physical and intellectual categories. The body is regarded as the medium of identity and self-presentation, as well as an instrument of social and sexual appeal. However, its significance is understood merely in its physical and physiological essence, but undervalued and neglected in its role as the very ground of embodied existence and knowledge as well as the full understanding of the human condition.²

This division of body and mind has, of course, its solid foundation in the history of Western philosophy. Prevailing educational pedagogies and practices also regrettably continue to separate mental, intellectual and emotional capacities from the senses and the multifarious dimensions of human embodiment. Educational practices usually provide some degree of physical training for the body, but they do not acknowledge our fundamentally embodied and holistic essence. The body is addressed in sports and dance, for instance, and the senses are directly acknowledged in connection with art and music education, but our embodied existence is rarely identified as the very basis of our interaction and integration with the world, or of our consciousness and self-understanding. Training of the hand is provided in courses that teach elementary skills in the handicrafts, but the integral role of the hand in the evolution and different manifestations of human intelligence is not acknowledged. To put it simply, today's prevailing educational principles fail to grasp the indeterminate, dynamic and sensually integrated essence of human existence, thought and action.

It is, in fact, reasonable to assume that prior to our current industrial, mechanised and materialist consumer culture, situations in daily life as well as processes of maturation and education provided a more comprehensive experiential ground for human growth and learning due to their direct interaction with the natural world and its complex causalities. In earlier modes of life, the intimate contact with work, production, materials, climate and the ever-varying phenomena of nature provided ample sensory interaction with the world of physical causalities. I would also suggest that closer family and social ties, as well as the presence of domestic animals, provided more experiences for the development of a sense of empathy and compassion than today's individualised and molecular life world.

I spent my early childhood years at my grandfather's small farm in central Finland, and with age I have become increasingly aware of how indebted I am to the richness of this farmer's life sphere in the late 1930s and 1940s for providing an understanding of my own embodied existence, and of the essential interdependences of the mental and physical aspects of daily life. I believe now that even one's sense of beauty and ethical judgement are firmly grounded in the early experiences of the integrated nature of the human life world. Beauty is not a detached aesthetic quality; the experience of beauty arises from grasping the unquestionable causalities and interdependences of life.

In our age of massive industrial production, surreal consumption, euphoric communication and fictitious digital environments, we continue to live in our bodies in the same way that we inhabit our houses, because we have sadly forgotten that we do not live in our bodies but are ourselves embodied constitutions. Embodiment is not a secondary experience; the human existence is fundamentally an embodied condition. Today, our senses and bodies are objects of ceaseless commercial manipulation and exploitation. Physical beauty, strength, youth and virility are adored in the realms of social values, advertising and entertainment. In case we fail to possess ideal physical qualities, our bodies are turned against us as causes of deep disappointment and guilt. With ever-accelerating frequency, all our senses are exploited by consumer manipulation, yet at the same time these very same senses continue to be undervalued as prerequisites of our existential condition or as educational objectives. Intellectually, we may well have philosophically rejected the Cartesian duality of body and mind, but the separation continues to rule in cultural, educational and social practices.

It is tragic, indeed, that at the time in which our technologies offer a multi-dimensional perception of the world and ourselves, we should throw our consciousness and capacities back to a Euclidian world. I do not wish to dwell on nostalgic images of an Arcadian past, or to represent a conservative view of cultural development. I just want to remind myself as well as my readers of the very evident blind spots in our established understanding of our own historicity as biological and cultural beings.

Human consciousness is an embodied consciousness; the world is structured around a sensory and corporeal centre. 'I am my body,' Gabriel Marcel claims;³ 'I am what is around me,' Wallace Stevens argues;⁴ 'I am the space, where I am,' Noël Arnaud establishes;⁵ and finally, 'I am my world,' Ludwig Wittgenstein concludes.⁶

We are connected with the world through our senses. The senses are not merely passive receptors of stimuli, and the body is not only a point of viewing the world from a central perspective. Neither is the head the sole locus of cognitive thinking, as our senses and entire bodily being directly structure, produce and store silent existential knowledge. The human body is a knowing entity. Our entire being in the world is a sensuous and embodied mode of being, and this very sense of being is the ground of existential knowledge. '[U]nderstanding is not a quality coming to human reality from the outside; it is its characteristic way of existing,' as Jean-Paul Sartre claims.⁷



Human consciousness is an embodied consciousness and we are connected with the world through our senses. Our hands and entire body possess embodied skills and wisdom.
Gjon Mili, Drummer Gene Krupa performing in Gjon Mili's studio, 1941

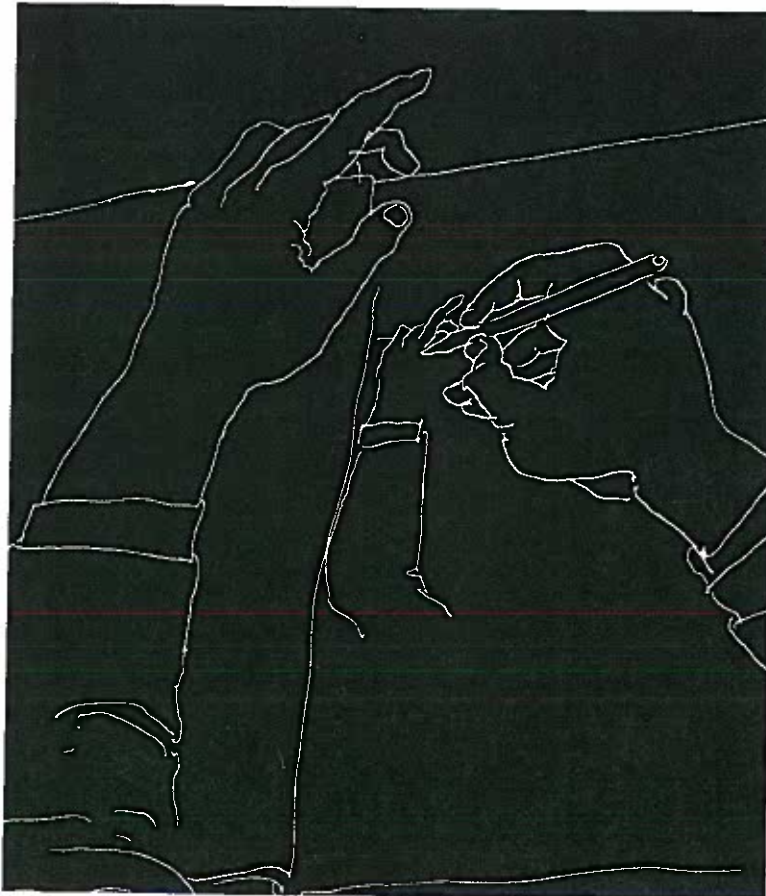
Existentially essential knowledge is not primarily a knowledge moulded into words, concepts and theories. In human interaction alone, 80 per cent of communication is estimated to take place outside the verbal and conceptual channel. Communication takes place even on a chemical level; the endocrine glands have been thought of as a closed system sealed in the body and only indirectly linked to the outside world. However, the experiments of AS Parker and HM Bruce show that chemical regulators, such as odoriferous substances, work directly on the body chemistry of other organisms, conditioning behaviour.⁸

The knowledge and skills of traditional societies reside directly in the senses and muscles, in the knowing and intelligent hands, and are directly embedded and encoded in the settings and situations of life. In accordance with Sartre's argument we are born into the world which in itself is the most important source of knowledge for us.⁹ In their thought-provoking book *Philosophy in the Flesh*, George Lakoff and Mark Johnson point out that even ordinary daily acts and choices call for a philosophical understanding; we must be able to make sense of our own lives in all the countless situations that we constantly face in life. The philosophers argue:

Living a human life is a philosophical endeavour. Every thought we have, every decision we make, and every act we perform is based upon philosophical assumptions so numerous we couldn't possibly list them all [...] Though we are only occasionally aware of it, we are all metaphysicians – not in some ivory-tower sense but as part of our everyday capacity to make sense of our experience. It is through our conceptual systems that we are able to make sense of everyday life, and our everyday metaphysics is embodied in those conceptual systems.¹⁰

Learning a skill is not primarily founded on verbal teaching but rather on the transference of the skill from the muscles of the teacher directly to the muscles of the apprentice through the act of sensory perception and bodily mimesis. This capacity of mimetic learning is currently attributed to human mirror neurons.¹¹ The same principle of embodying – or introjecting, to use a notion of psychoanalytic theory – knowledge and skill continues to be the core of artistic learning. The foremost skill of the architect is, likewise, to turn the multi-dimensional essence of the design task into embodied and lived sensations and images; eventually the entire personality and body of the designer becomes the site of the design task, and the task is lived rather than understood. Architectural ideas arise 'biologically' from unconceptualised and lived existential knowledge rather than from mere analyses and intellect. Architectural problems are, indeed, far too complex and deeply existential to be dealt with in a solely conceptualised and rational manner. Profound ideas or responses in architecture are not individual inventions *ex nihilo* either; they are embedded in the lived reality of the task itself and the age-old traditions of the craft. The role of this fundamental, unconscious, situational and tacit understanding of the body in the making of architecture is grossly undervalued in today's culture of quasi-rationality and arrogant self-consciousness.

Even masterful architects do not invent architectural realities; they rather reveal what exists and what are the natural potentials of the given condition, or what the given situation calls for. Álvaro Siza, one of the finest architects of our time in combining a sense of tradition with a unique personal expression, puts it sharply: 'Architects don't invent anything, they transform reality.'¹² Jean Renoir expresses the same idea of artistic humility in filmmaking somewhat differently: 'The director is not a creator, but a midwife. His task is to help the actor give birth to a child, that she has not realised having carried inside her,' he confesses in his humane memoirs.¹³ Architecture is also a product of the knowing hand. The hand grasps the physicality and materiality of thought and turns it into a concrete image.



The hand has its dreams and assumptions, as Gaston Bachelard notes
Álvaro Siza, *The Architect Sketching*

In the arduous processes of designing, the hand often takes the lead in probing for a vision, a vague inkling that it eventually turns into a sketch, a materialisation of an idea.

The pencil in the architect's hand is a bridge between the imagining mind and the image that appears on the sheet of paper; in the ecstasy of work, the draughtsman forgets both his hand and the pencil, and the image emerges as if it were an automatic projection of the imagining mind. Or, perhaps, it is the hand that really imagines as it exists in the flesh of the world, the reality of space, matter and time, the very physical condition of the imagined object.

Martin Heidegger connects the hand directly with the human thinking capacity: 'the hand's essence can never be determined, or explained, by its being an organ which can grasp [...] Every motion of the hand in every one of its works carries itself through the element of thinking, every bearing of the hand bears itself in that element [...].'¹⁴ Gaston Bachelard writes about the imagination of the hand: 'Even the hand has its dreams and assumptions. It helps us understand the innermost essence of matter. That is why it also helps us imagine [forms of] matter.'¹⁵ The capacity to imagine, to liberate oneself from the limits of matter, place and time, must be regarded as the most human of all our qualities. Creative capacity as well as ethical judgement call for imagination. It is evident, however, that the capacity of imagination does not hide in our brains alone, as our entire bodily constitution has its fantasies, desires and dreams.

All our senses 'think' and structure our relationship with the world, although we are not usually conscious of this perpetual activity. Knowledge is normally supposed to reside in verbalised concepts, but any grasp of a life situation and a meaningful reaction to it can, and indeed should, be regarded as knowledge. In my view, the sensory and embodied mode of thinking is particularly essential in all artistic phenomena and creative work. Albert Einstein's well-known description, in a letter to the French mathematician Jacques Hadamard, of the role of visual and muscular images in his thinking processes in the fields of mathematics and physics provides an authoritative example of embodied thinking:

The words or the language, as they are written or spoken, do not seem to play any role in my mechanism of thought. The psychological entities which seem to serve as elements in thought are certain signs and more or less clear images which can be 'voluntarily' reproduced and combined. [...] The above



The study of Albert Einstein
with a blackboard

mentioned elements are, in my case, of visual and some of muscular type. Conventional words or other signs have to be sought for laboriously only in a secondary stage, when the mentioned associative play is sufficiently established and can be reproduced at will.¹⁶

It is also evident that an emotional and aesthetic factor, as well as an embodied personal identification, is equally central in scientific creativity as in the making and experiencing of art. Henry Moore, one of the greatest sculptors of the modern era, emphasises the bodily identification and simultaneous grasp of several points of view in the sculptor's work:

[The sculptor] must strive continually to think of, and use form in its full spatial completeness. He gets the solid shape, as it were, inside his head – he thinks of it, whatever its size, as if he were holding it completely enclosed in the hollow of his hand. He mentally visualises a complex form from all round itself; he knows while he looks at one side what the other side is like; he identifies himself with its centre of gravity, its mass, its weight; he realises its volume, and the space that the shape displaces in the air.¹⁷

All art forms – such as sculpture, painting, music, cinema and architecture – are specific modes of thinking. They represent ways of sensory and embodied thought characteristic to the particular artistic medium. These modes of thinking are images of the hand and the body, and they exemplify essential existential knowledge. Instead of being mere visual aestheticisation, architecture, for instance, is a mode of existential and metaphysical philosophising through the means of space, structure, matter, gravity and light. Profound architecture does not merely beautify the settings of dwelling; great buildings articulate the experiences of our very existence.

Salman Rushdie points out that a distinct softening of the boundary between the world and the self takes place in an artistic experience: 'Literature is made at the boundary between self and the world, and during the creative act this borderline softens, turns penetrable and allows the world to flow into the artist and the artist to flow into the world.'¹⁸ This softening of the existential boundary, the fusion of the world and the self, object and subject, takes place in every meaningful artistic work and experience.

Creative work calls for a double perspective: one needs to focus simultaneously on the world and on oneself, the external space and one's inner mental space. All artworks articulate the boundary between the

The hands of a great sculptor:
Henry Moore in his studio in
the late 1970s



self and the world, both in the experience of the artist and in that of the viewer/listener/occupant. In this sense, the art form of architecture does not only provide a shelter for the body, it also redefines the contour of our consciousness, and it is a true externalisation of our mind. Architecture, as well as the entire world constructed by man with its cities, houses, tools and objects, has its mental ground and counterpart. As we construct our self-made world, we construct projections and metaphors of our own mindscapes. We dwell in the landscape and the landscape dwells in us. A landscape wounded by acts of man, the fragmentation of the cityscape, as well as insensitive buildings, are all external and materialised evidence of an alienation and shattering of the human inner space or *Weltinnenraum*, to use a beautiful notion of Rainer Maria Rilke.¹⁹

Even in the technological culture of today, the most important existential knowledge in our everyday life does not reside in detached theories and explanations, but it is a silent knowledge, beyond the threshold of consciousness, fused with daily environments and behavioural situations. The poet, too, speaks of encounters at the 'threshold of being', as Gaston Bachelard points out.²⁰ Art guides us to this 'threshold', and surveys the biological and unconscious realms of the body and mind. In so doing, it maintains vital connections with our biological and cultural past, the soil of genetic and mythical knowledge. Consequently, the essential time dimension of art points to the past rather than the future; significant art and architecture maintains roots and traditions instead of uprooting and inventing. However, today's obsession with uniqueness and novelty has misguided our judgement of artistic phenomena. Radical works of art and building surely appear as ruptures or discontinuities of convention but, at the same time on a deeper level, all profound artworks reinforce the perception and understanding of human bio-cultural historicity and continuity. Artistic revolutions always imply a reconnection with the invisible undercurrents of the universe of the human mind.

The duty of education is to cultivate and support the human abilities of imagination and empathy, but the prevailing values of culture today tend to discourage fantasy, suppress the senses, and petrify the boundary between the world and the self. Consequently, education in any creative field in our time has to begin with the questioning of the absoluteness of the lived world and with the re-sensitisation of the boundaries of self. The main objective of artistic education may not directly reside in the principles of artistic making, but in the emancipation and opening up of the personality of the student

and his/her self-awareness and self-image in relation to the immensely rich traditions of art, and to the lived world at large.

It is evident that an educational change concerning the significance of the sensory realm is urgently needed in order to enable us to rediscover ourselves as complete physical and mental beings, to fully utilise our capacities, and to make us less vulnerable to manipulation and exploitation. In the words of the philosopher Michel Serres, 'If a revolt is to come, it will have to come from the five senses.'²¹ The intelligence, thinking and skills of the hand also need to be rediscovered. Even more importantly, the unbiased and full understanding of human embodied existence is the prerequisite for a dignified life.

In 1995 I wrote the book *The Eyes of the Skin: Architecture and the Senses* (Academy Editions, London, 1996), which was republished in a new illustrated format with a preface by Steven Holl by John Wiley & Sons, Chichester, in 2005. The book is a critique of the dominance of the sense of vision in today's technological culture and contemporary architecture, and a general call for a multi-sensory approach in the arts and architecture.

A year ago, Helen Castle, Executive Commissioning Editor at John Wiley & Sons, asked me to write a book in their new book series *AD Primers* as a continuation of the ideas on human embodiment suggested in *The Eyes of the Skin*. I submitted half a dozen of my recent essays and lectures, and the publisher suggested a book that would be developed around the concept of *The Thinking Hand*, the title of a chapter in one of these essays.

The current book analyses the essence of the hand and its seminal role in the evolution of human skills, intelligence and conceptual capacities. As I argue – with the support of many other writers – the hand is not only a faithful, passive executor of the intentions of the brain; rather, the hand has its own intentionality, knowledge and skills. The study of the significance of the hand is expanded more generally to the significance of embodiment in human existence and creative work.

This book emphasises the relatively autonomous and unconscious processes of thinking and working in writing, craftsmanship and making art or architecture. The book turned out to be quite different from my initial idea as

a consequence of both the writing process itself and my subsequent literary research, which brought out chapters, concerns and ideas that I did not have any notion of before immersing myself in the project. In a way, the act of writing this little book directly proved the thesis that I present in the text.

Most of the examples and quotes in this book are drawn from painting, sculpture and literature, but due to my own professional background my focus is on issues of architecture. In my role as a teacher of architecture, I have always found it easier for myself and more efficient to explain phenomena of the art of architecture through other art forms. 'All painters and poets are born phenomenologists,' as JH van den Berg suggests.²² This observation implies that all artists look at the essence of things. Besides, all arts arise from a common soil; they are all expressions of the human existential condition.

The title of the book, *The Thinking Hand*, is a metaphor for the characteristic independent and active roles of all our senses as they constantly scan our life world. The subtitle – *Existential and Embodied Wisdom in Architecture* – refers to the other knowledge, the silent understanding that lies hidden in the human existential condition and our specific embodied mode of being and experiencing. Many of our existentially most crucial skills are internalised as automatic reactions beyond conscious awareness and intentionality. We are hardly aware of the fantastically complex and automated metabolic processes, for instance, without which we could not survive a split second. Even in the case of learning skills, the complex sequence of movements and spatial and temporal relationships in the execution of the task is unconsciously internalised and embodied rather than understood and remembered intellectually.

The prevailing educational philosophies regrettably continue to emphasise and value conceptual, intellectual and verbal knowledge over the tacit and non-conceptual wisdom of our embodied processes. This attitude continues against all the overwhelming evidence of this catastrophic bias provided today through philosophical arguments and recent developments and discoveries in neurology and cognitive science. The objective of this book is to help to shake the foundations of this hegemonic but erroneous and harmful paradigm in the realm of architecture.

References

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- 8 For the biochemistry of crowding see, Edward T Hall, *The Hidden Dimension*, Anchor Books/Doubleday (New York, London, Toronto, Sydney and Auckland), 1966, pp 32–40.
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A hand in the attitude of benediction embellished with charms to increase its power, late Roman period, bronze

The Mysterious Hand

'A hand is not simply part of the body, but the expression and continuation of a thought which **must** be captured and conveyed [...].'

Honoré de Balzac¹

'The hand is the window on to the mind.'

Immanuel Kant²

'If the body had been easier to understand, nobody would have thought that we had a mind.'

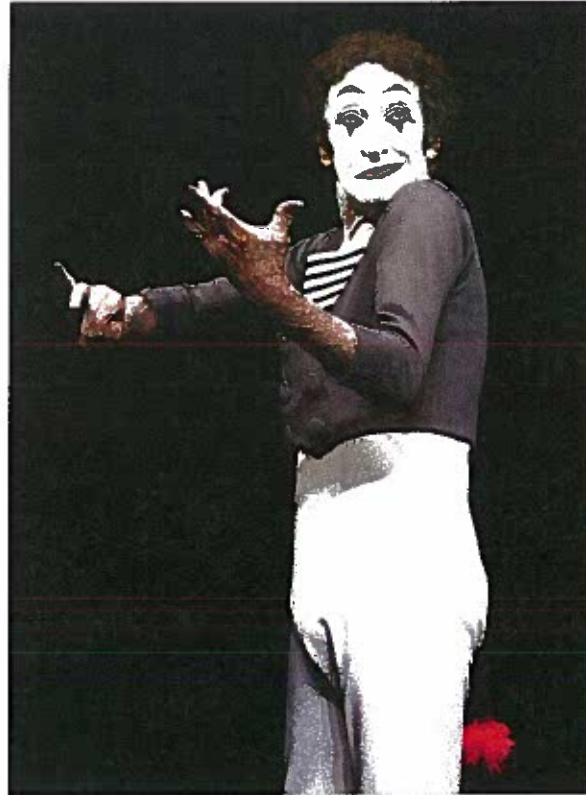
Richard Rorty³

The Multiple Essences of the Hand

We regard our own hand as a commonplace and self-evident member of the body, but in fact it is a prodigious precision instrument that seems to have its own understanding, will and desires. Often it even appears to be both the origin and the expression of pleasure and emotion. The hand, its motions and

gestures, are expressions of the person's character to the same degree as the face and body physique. Hands also have their unique appearances and features; they have their distinct personalities. They even reveal one's occupation and craft; just think of the robust hands of a steelworker or blacksmith, the often mutilated hands of a cabinet-maker, the hands of a shoemaker hardened and cracked by handling of the substances of the trade, the eloquently speaking hands of a pantomime artist, or the delicate, utterly precise and quick hands of a surgeon, pianist or magician. Hands are generic organs characteristic to *Homo sapiens*, but at the same time they are unique individuals. Imagine the hand of a child full of clumsy and innocent curiosity and excitement, and the nearly useless hand of an old person deformed by hard work and articular rheumatism. The vivid movement of the contours of Henri Matisse's colourful paper cut-outs acquire a special meaning after having seen a photograph of the aging artist warming his aching finger joints in the feathers of domesticated pigeons, or on his sickbed drawing on a sheet of paper on the wall with a charcoal attached to the end of a long bamboo stick. André Wogenscky, Le Corbusier's close assistant for 20 years, describes his master's hands poetically and suggestively:

Then I would let my eyes go from his face down to his hands. I would then discover Le Corbusier. It was his hands that revealed him. It was as if his hands betrayed him. They spoke all his feelings, all the vibrations of his inner life that his face tried to conceal [...] Hands that one might have thought Le Corbusier had drawn himself, with that trait made of a thousand small successive traces that seemed to look for one another but that in the end formed a precise and exact line, that unique contour that outlined the shape and defined it in space. Hands that seemed to hesitate but from which



The French mime artist Marcel Marceau during a performance at Sadler's Wells Theatre in London.

'The art of mime is the portrayal of the human being in its most secret yearnings. By identifying itself with the elements which surround us, the art of the mime makes visible the invisible and concrete the abstract.' Marcel Marceau.

A child's hand explores the world eagerly. The child's first impressions of the world are tactile images



precision came. Hands that always thought, just like he did in his thinking, and on his hands one could read his anxiety, his disappointments, his emotions and his hopes. Hands that had drawn, and were to draw, all his work.⁴

Le Corbusier appears as a somewhat enigmatic and distant person in literature written about him, his life and work, but his hands as observed by his assistant seem to reveal his inner character and intentions.

Hands can tell epic stories of entire lives; in fact, every epoch and culture has its characteristic hands; just look at the varying hands of the countless portraits through the history of painting. Moreover, every pair of hands is equipped with singular patterns of fingerprints, which do not change at all after five months prior to the person's birth; these engravings on the human skin are the secret prenatal hieroglyphs of individuality.

Our hands are our reliable and diligent servants, but now and then they seem to take command, lead their independent lives and demand their own liberties. But then, the integral completeness of the human figure is so powerful that we accept an armless statue as a valid and aesthetically pleasing representation of the human constitution, not as a deliberate depiction of mutilation. '[N]othing essential is missing. Standing before them, one has the sense of a profound wholeness, a completeness that allows for no addition,' Rainer Maria Rilke, the poet, writes of Auguste Rodin's vivid



Henri Matisse cutting painted paper in his studio in the Hôtel Régina in Nice, 1952

torsos.⁵ Or are we here rather seduced by the magic integrity of an artistic masterpiece?

The poet also describes the multiple roles and determinedly independent lives of the human hands:

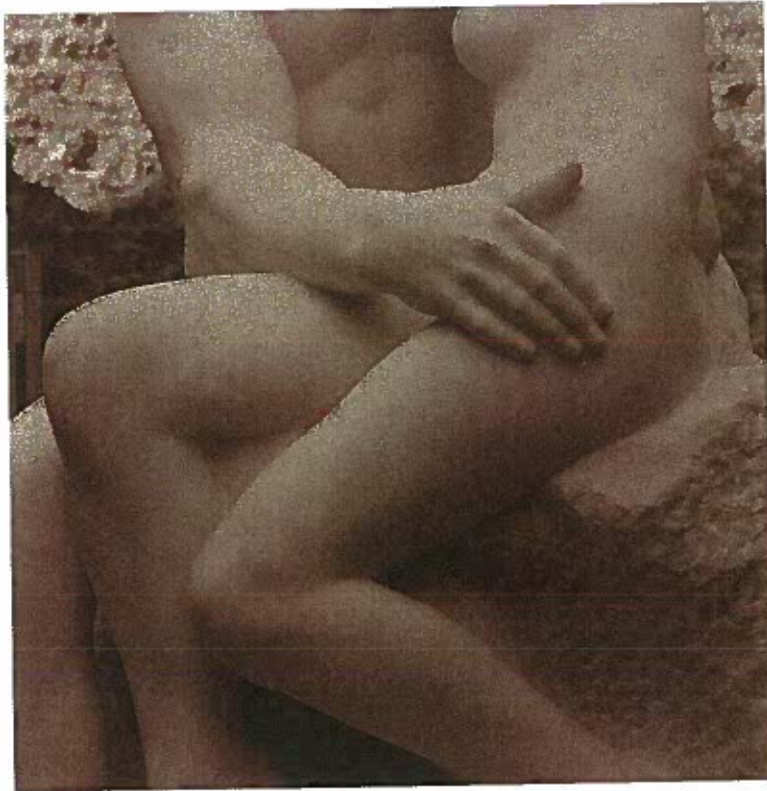
There are hands that walk, hands that sleep and hands that wake; criminal hands weighted with the past, and hands that are tired and want nothing more, hands that lie down in a corner like sick animals who know no one can help them. But then hands are a complicated organism, a delta in which life

from the most distant sources flows together, surging into the great current of action. Hands have stories; they even have their own culture and their own particular beauty. We grant them the right to have their own development, their own wishes, feelings, moods, and occupations [...].⁶

The hand has its social roles and behaviours, its amorous as well as hostile and aggressive acts, its gestures of welcome and rejection, friendship and animosity. The hands of God and Christ, as well as of the Pope, are hands of benevolence and blessing. The hand of Mucius Scaevola is the hand of bravery and heroic self-control, whereas the hands of Cain and Pontius Pilate are organs of crime and guilt. Regardless of its self-sufficiency, the hand may momentarily lose its independence and identity, and fuse with the body of the other. As Rilke observes: 'A hand lying on the shoulder or thigh of another body no longer belongs completely to the one it came from: a new thing arises out of it and the object it touches or grasps, a thing that has no name and belongs to no one, and it is this new thing, which has its own definite boundaries, that matters from that point on.'⁷ The hands of a mother and child or of two lovers turn into an umbilical cord that unites the two individuals.

Works of art and architecture extend the human hand through both space and time. When looking at the *Rondanini Pietà* (1555–64) in Castello Sforzesco in Milan, I can feel the passionate but already feeble hands of Michelangelo approaching the end of his life. The works of a great architect likewise invite the imagined presence of his figure and hand, as the architectural space, scale and detailing are unavoidably products and projections of the maker's body and hand. The greater the work, the more present the hand of the maker. I cannot look at a Vermeer painting at a close distance without thinking of the painter stooped over his painting with a thin, sharply shaped brush in his hand. No, I do not imagine the painter, I become him. My entire physique changes and my hand guides the brush to the still wet 'little patch of yellow wall' in his *View of Delft* (1660–1) that Marcel Proust admired and wrote about.⁸

As I am looking at a Suprematist painting by Kasimir Malevich, I do not see it as a geometric gestalt but as an icon meticulously painted by the artist's hand. The surface of cracked paint conveys a sense of materiality, work and time, and I find myself thinking of the inspired hand of the painter holding a brush.



'A hand lying on the shoulder or thigh of another body no longer belongs completely to the one it came from.' Rainer Maria Rilke. Auguste Rodin, *The Kiss* (detail), marble, complete sculpture 183 x 110 x 118 cm, 1886. Musée Rodin, Paris. Judging by the year in which the sculpture was created, it is likely that it was initially intended for *The Gates of Hell*.

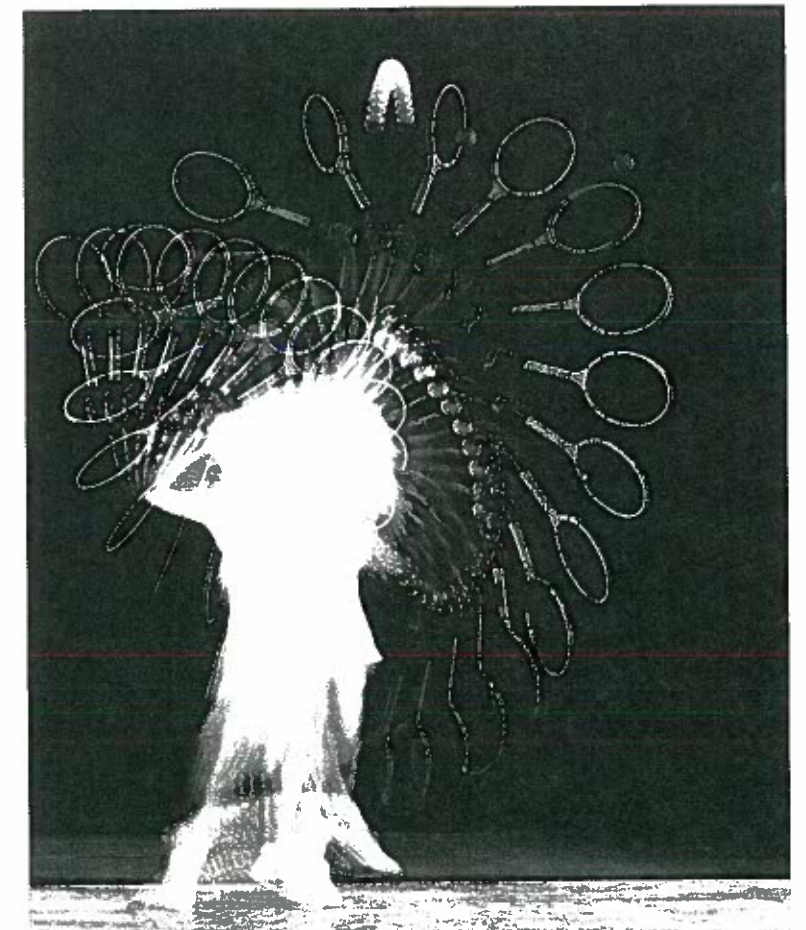
What is the Hand?

We use the notion of the 'hand' carelessly and without much thought, as if its essence were self-evident. 'The human hand is so beautifully formed, its actions so powerful, so free and yet so delicate that there is no thought of its complexity as an instrument; we use it as we draw our breath, unconsciously,' Sir Charles Bell wrote in 1833.⁹ But how should we really define the hand? When we say 'give me your hand', or 'I place this matter in his hands', or we speak of 'handwork' or 'handshake', what exactly do we mean?

Everyday use of the word as well as classical *surface anatomy* would probably argue that the hand is the human organ that extends from the wrist to the fingertips.¹⁰ From the viewpoint of *biomechanical anatomy*, the hand would be seen as an integral part of the entire arm. But the arm also functions in a

dynamic coordination with the muscles of the neck, back, and even the legs, and in fact with the rest of the body. Training in most sports aims exactly at this complete integration of the actions of the hands with the entire body. When I raise my hand for an oath or greeting, or give my fingerprints as evidence of my identity, the hand stands for my entire persona. *Physiological and functional anatomy* would even consider those parts of the brain that regulate hand functions as part of the hand. Altogether, we are bound to admit that the hand is everywhere in our body, as well as in all our actions and thoughts, and thus the hand is fundamentally beyond definability. As Frank R Wilson, neurologist and writer, argues in his seminal study of the

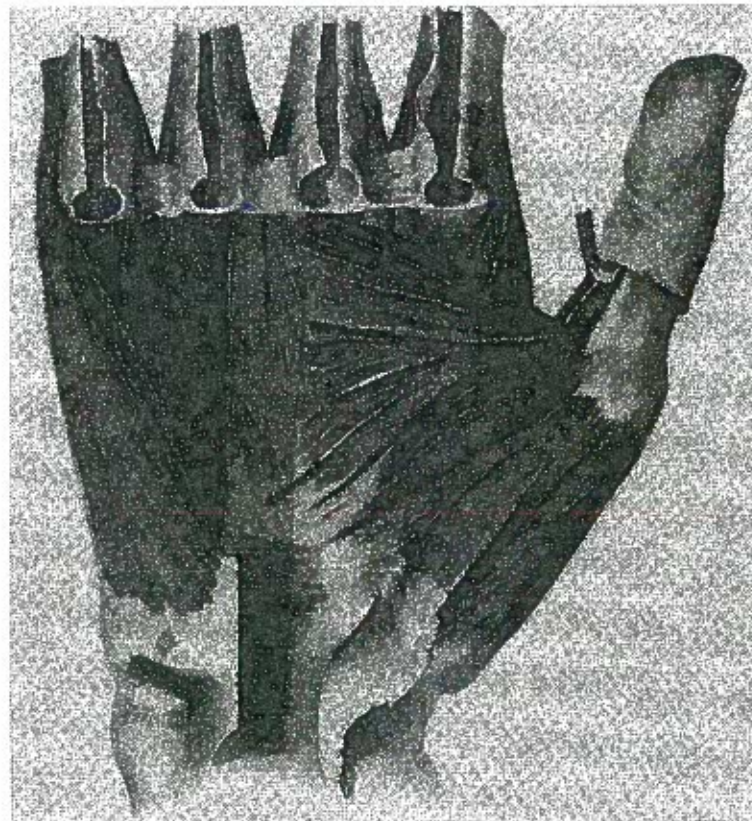
Hands act in collaboration and coordination with the entire body. Harold E Edgerton, *Tennis Player*, 1938, gelatin silver print



evolution and significance of the human hand entitled *The Hand: How Its Use Shapes the Brain, Language, and Human Culture*:

Bodily movement and brain activity are functionally interdependent, and their synergy is so powerfully formulated that no single science or discipline can independently explain human skill or behaviour [...]. The hand is so widely represented in the brain, the hand's neurological and biomechanical elements are so prone to spontaneous interaction and reorganization, and the motivations and efforts that give rise to individual use of the hand are so deeply and widely rooted, that we must admit we are trying to explain a basic imperative of human life.¹¹

Recent anthropological and medical research and theories even give the hand a seminal role in the evolution of human intelligence, language and symbolic



The mobility of the thumb results from the arrangement of nine muscles that enter the thumb from the forearm and from the hand

thought. The amazing mobile versatility, learning capacity, and apparently independent functions of the hand may not be a result of the development of the human brain capacity, as we tend to think, but the extraordinary evolution of the human brain may well have been a consequence of the evolution of the hand. 'Aristotle erred in asserting that humans had hands because they were intelligent; Anaxagoras was, perhaps, more correct in stating that humans were intelligent because they had hands,' as Marjorie O'Rourke Boyle notes.¹²

Wilson sees the brain as well as the interdependence of the hand and brain omnipresent in the body:

The brain does not live inside the head, even though it is its formal habitat. It reaches out to the body, and with the body it reaches out to the world. We can say that the brain 'ends' at the spinal cord, and that the spinal cord 'ends' at the peripheral nerve, and the peripheral nerve 'ends' at the neuromuscular junction, and on and on down to the quarks, but brain is hand and hand is brain, and their interdependence includes everything else right down to the quarks.¹³

We can certainly conclude that 'the hand speaks to the brain as surely as the brain speaks to the hand.'¹⁴ Even beyond its physical and neurological significances, Wilson regards the hand as an essential constituent of the story of human intelligence and its gradual evolution: '[A]ny theory of human intelligence which ignores the interdependence of hand and brain function, the historic origins of that relationship, the impact of that history on developmental dynamics in modern humans, is grossly misleading and sterile.'¹⁵ We usually think that our hands deal merely with the concrete, material world, but some theorists attribute to the hand a significant role even in the emergence of symbolic thought.¹⁶

Hand, Eye, Brain and Language

The human hand is the product of evolution. The extraordinary mobility of the arm and the hand, as well as the human eye-hand coordination and precise judgment of spatial positions and relations were already developed when the ancestors of hominids lived and moved up in trees. The earliest direct ancestors of humans were the australopithecines – misleadingly named 'southern apes' – who walked upright. The transition from moving on the branches of trees to walking on two legs on the flat floor of the savannah

changed the role of the hands and liberated them for new purposes and a new evolutionary development. In the 1960s the discovery of the remains of 'Lucy', who had lived 3.2 million years ago in Hadar in East Africa, was an anthropological sensation; our now famous female ancestor was named after the Beatles song 'Lucy in the sky with diamonds' that was played on a tape recorder in the anthropologists' camp.¹⁷ Prevailing theories already assumed that the human brain could have evolved as a consequence of the increase in tool use. 'The structure of modern man must be the result of the change in the terms of natural selection that came with the tool-using way of life [...] From the evolutionary point of view, behaviour and structure form an interacting complex, with each change in one affecting the other. Man began when populations of apes, about a million years ago, started the bipedal, tool-using way of life,' anthropologist Sherwood Washburn argues.¹⁸

The most seminal single aspect in the evolution of the hand was that the physical opposition between the thumb and fingers became increasingly articulate. At the same time this opposition combined with subtle changes occurring in the bones that support and strengthen the index finger.¹⁹ These anatomical changes enabled both the power and the precision grip in tool handling.²⁰

Current theories suggesting that language originates in early collective tool manufacture and tool use imply that even the development of language is linked with the co-evolution of the hand and the brain. Wilson argues assuredly: 'It is a virtual certainty that complex social structure – and language – developed gradually in association with the spread of more highly elaborated tool design, manufacture, and use.'²¹ The further refinement of the hand led to further development of the brain's circuitry:

There is growing evidence that *Homo sapiens* required in its new hand not simply the mechanical capacity of refined manipulative and tool-using skills but, as time passed and events unfolded, an impetus to the redesign, or reallocation, of the brain's circuitry. The

Hand tools from the Palaeolithic era, Czech Republic



The power grip preceded the precision grip in the evolution of the human hand. Modern wrenches



new way of mapping the world was an extension of ancient neural representations that satisfy the brain's need for gravitational and inertial control of locomotion.²²

The development and refinement of tool use is assumed to be related to the emergence of subjectivity and purposive thinking: 'It has been said that language is the prelude to the coming of man. That may be, but even before language comes *thinking in terms of tools*, i.e., the realization of mechanical connections and the invention of mechanical means for mechanical ends. To put it briefly, before the advent of speech, action comes to have a *subjective meaning*, i.e., it becomes consciously purposive.'²³ Psychologist Lev Vygotsky thought-provokingly suggests that speech and language, on the one hand, and thought, on the other, are of different biological origins: '[I]nitially thought is nonverbal and speech non-intellectual [...] [But, in humans] thought development is determined by language, i.e., by the linguistic tools of thought and by the sociocultural experience of the child.'²⁴

In fact, art and architecture guide us back to the origins of language, to the originary wonder and amazement when encountering the unforeseen. Artistic images expose us to images and encounters of things before they have been trapped by language. We touch things and grasp their essence before we are able to speak about them. Profound buildings place us at the central point of the lived world; even the tiny architectural space of the Kärtner Bar (1907) in Vienna by Adolf Loos turns into the nucleus of the world that seems to condense gravity and space, as well as all our existential knowledge, in its pre-verbal, compressed spatial and material configuration.

In his book *The Mind in the Cave: Consciousness and the Origins of Art*, David Lewis Williams proposes a convincing theory for the origins of image-making in the animals and symbols depicted on the walls of neolithic caves. He also provides an explanation for the mysterious fact that the Neanderthals, our nearest ancient relatives, who lived alongside our Cro-Magnon ancestors for over 10,000 years and borrowed their stone tool technology, never developed artistic expressions such as the cave paintings of the Cro-Magnons. In Williams's view, the explanation for this curious fact lies in the evolution of the human mind. The Cro-Magnons, unlike the Neanderthals, possessed a higher-order consciousness and a more advanced



In his book *The Ascent of Man*, Jacob Bronowski points out that the cave paintings reveal what dominated the minds of these early hunters: 'I think that the power we see expressed here for the first time is the power of anticipation, the forward-looking imagination.' Cave painting.

neurological structure which enabled them to experience shamanistic trances and vivid mental imagery. These mental images – the first recorded expressions of human imagination and artistic hand – were then painted on the cave walls, which were regarded as the membrane between the world of their pre-human occupants and their imaginary spirit world, in which the images originated.²⁵

The American psychologist Julian Jaynes argues that human consciousness did not emerge gradually in animal evolution but that it is a learned process which developed from an earlier hallucinatory mentality through cataclysmic and catastrophic events. He identifies the emergence of human consciousness in 'the breakdown of the bicameral mind' roughly at the time of the earliest Mesopotamian written records – which, at around 3,000 years ago, is surprisingly late.²⁶

The authors and researchers of the book *Gesture and the Nature of Language* suggest that actions of the hand directly moulded the development of language:

The very categories of language are created by intentional hand actions, so that verbs derive from hand movements, nouns hold things as names, and adverbs and adjectives, like hand tools, modify movements and objects. The focus here is particularly on how experiences of touch and grip [...] give language its directive power.²⁷

George Lakoff and Mark Johnson suggest another connection between language and the body. In their book *Metaphors We Live By*, they develop ideas of the fundamental grounding of language on metaphors originating in the human body and the ways the body is related with and positioned in space. '[M]etaphor is pervasive in everyday life, not just in language but in thought and action. Our ordinary conceptual system, in terms of which we both think and act, is fundamentally metaphorical in nature,' the two philosophers argue.²⁸ These metaphors of language, thought and action originate in the natural structures and aspects of the body and its relationship with space.

The decisive development of the human brain began about three million years ago with tool use, and in accordance with recent theoretical views, the modern human brain was completed 100,000 years ago, or perhaps even somewhat earlier.²⁹

Hand as Symbol

The fact that the hand is the part of the human body that appears most frequently in symbolisation³⁰ reflects the significance, subtlety, as well as expressiveness and multiple meanings of the human hand. Imprints and silhouettes of human hands appear already in Palaeolithic cave paintings. These early imprints of hands probably signify the individual whose hand left the mark in the same way that young children enjoy impressing their hand marks as expressions of their selves. The depictions of twisted finger joints and mutilated hands in the Gargas cave in France are assumed to commemorate acts of sacrifice.



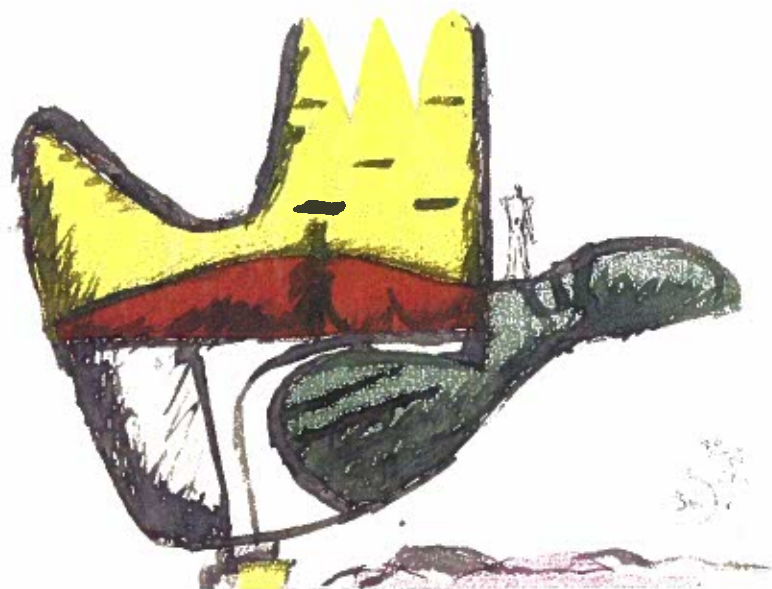
Palaeolithic hand marks and various symbols (according to Sigfried Giedion). El Castillo, Cantabria, Spain.

The hand can have multiple and even opposite symbolic meanings in social communication and art, as in the case of the hand gesture of taking hold or pushing away, that can express both positive and negative meanings. The image of the hand often appears in amulets, such as the Islamic Hand of Fatima. In Semitic cultures, 'hand' and 'might' are a single concept that refers to the power of a ruler. The hand contact is regulated by cultural and professional codes, such as the use of hand contact in the medical profession, or the social custom of greeting, but generally hand contact symbolises the magic touch. Laying on of hands signifies blessing and this sign transfers the powers of the touching person, or of a higher being, onto the person being touched. Raised or folded hands are a symbol of prayer, distinct gestures of the fingers signify oath or blessing, and a handshake is a general symbol of a friendly and accepting attitude.

In Christian iconography, Christ is referred to as the Right Hand of God; right hand altogether tends to have positive significations as opposed to the left hand. In many cultures the right hand is the 'clean' hand whereas the left hand is 'dirty'. Hands that are covered or hidden in sleeves refer to the age-old custom of covering one's hands as a sign of respect in the presence of rulers. The raised open hand of Byzantine rulers gave rise to the gesture of blessing in Christianity. The symbolism of both raised hands signifies a turning towards the heavens, the receptivity of the person in prayer, or the gesture of adoration. In Renaissance heraldry hands mean strength, loyalty, diligence, innocence and unity, whereas a hand with the fingers extended and spread apart means disunity, the closed hand or fist indicates strength and unity, and folded hands signify loyalty and union.

In artistic representation, a hand emerging from a cloud is an early form of representing the First Person of the Trinity. The hand that struck Christ is one of the instruments of the Passion, whereas a hand giving money refers to the payment to Judas, and the washing of hands referring to Pilate's hands after the trial of Christ represents innocence.³¹

The hand and fingers have varying connotations in different cultures; in the Islamic belief the five fingers signify proclaiming one's faith, prayer, pilgrimage, fasting and generosity. The traditional system of ritual hand postures, or mudras, of the Buddhists and Sivaists in India expresses a range of coded meanings for hand gestures and they can be an integral part of both secular and religious ritual performances. Each finger may be associated with its own hue, sound, element, and even its own celestial guardian.



A monument to the hand
Le Corbusier, *The Open Hand*,
sketch for a monument,
Chandigarh, 1954

Because of the characteristic Indian tendency to excessive classification, each mudra gesture may possess one meaning nested within another.

The intricate and individually unique lines of the palm are interpreted in palmistry, or chiromancy. The basic assumption of this practice is that symbolic analogies link the hand with its 'hieroglyphics', planetary forces and the potential life of the individual.

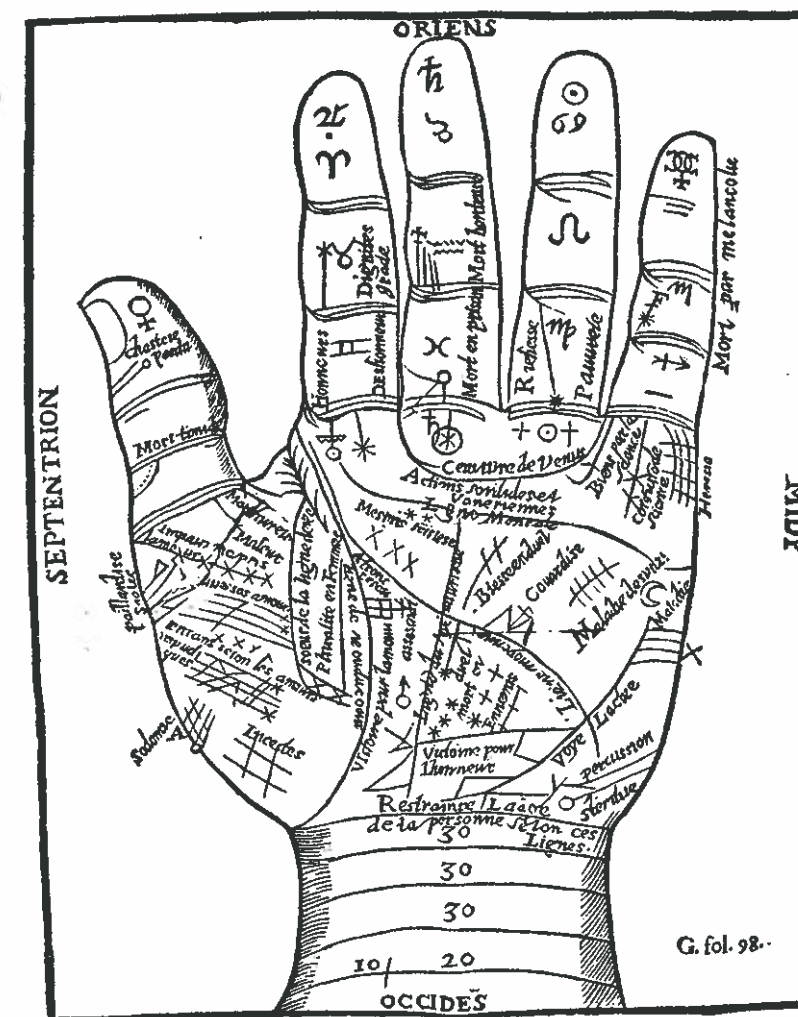
The hand is a signboard of personality; it expresses social class, wealth, allegiance, occupation and association. In many cultures the hand is decorated by tattoos or less permanent colorations and imagery. Hands are also bearers of rings and bracelets that communicate numerous coded meanings, such as marriage, profession, or membership in societies. Gestures, meanings and messages of the hand are also popular subjects in the arts.

Gestures of the Hand

There are theories that regard human gesture as the first evolutionary phase on the way to spoken and written language. The emotive power,

immediacy, universality and articulate nature of gestural expression certainly reflect the integrity of the human constitution as well as of the close connection between the mind and the hand. Also deficiencies and failures in the mind-hand unity are dramatically manifested: 'The hand is the mind's only perfect vassal, and when, through age or illness, the connection between them is interrupted, there are few more affecting tokens of human decay.'²²

A chiromantic hand from Jean Belot's *Oeuvres* (Lyons, 1649), reproduced in *A History of Magic*, published in the late 19th century Engraving by French School, 17th century





The hand as a work of art that reflects cultural restrictions and values
Shirin Eshai, *Untitled* (from 'Women of Allah' series), black and white photograph with ink, 23.8 x 16.5 cm, 1996

It is remarkable to realise that the meaning of numerous facial and hand gestures can be grasped quite independently of cultural background. 'The hand is the only speech that is natural to man [...] It may well be called the Tongue and general language of Humane Nature, which, without teaching, man in all regions of the habitable world doe at the sight most easily understand,' John Bulwer wrote in 1644 in his book *Chironomia*.³³ A more

recent scholar, Edward Sapir, the anthropologist and linguist, argues similarly: 'We respond to gestures with an extreme alertness and, one might almost say, in accordance with an elaborate and secret code that is written nowhere, known by none, and understood by all.'³⁴

Before they have learned even the foundations of linguistic communication, infants react correctly to basic gestures of threat or friendliness, for instance, and individuals born blind seem to be able instinctively to use basic facial and hand gestures.³⁵ Although they were divided linguistically, the various Native American peoples were able to communicate through sign language.³⁶ 'For purposes of trading, as much was done by the Hudson River Algonquins by signs with the thumb and fingers, as by speaking,' wrote Jonas Michaelius, a 17th-century clergyman.³⁷

Languages of the Hand

Besides the Native North Americans, sign languages were particularly highly developed among Australian aborigines and the Maoris. In addition to sign languages of indigenous cultures there are occult sign languages of secret societies and religious communities. Certain hand gestures that continue to be used today were already known in ancient Egypt and Babylon. Other gestural symbolic signs are used in art, theatre, heraldry and religion.³⁸ Even today, members of many trades and professions use private sign languages. A peculiar example of invisible hand communication is the bidding process at the markets of the delicious but highly poisonous globefish in Japan, performed secretly by the hands of the seller and bidder pushed inside a special sleeve.

Sir Richard Paget, who developed a universal sign language in 1939, estimated that by combining various postural movements of the upper arm, forearm, wrist and fingers, it is feasible to produce the staggering number of 700,000 distinct elementary signs; he assessed that 500 to 600 signs would suffice for the vocabulary of his New Sign Language. This estimation makes the human hand overwhelmingly more versatile than the mouth. This surprising realisation seems to open up immense possibilities for gestural communication.³⁹

Yet another, and the most common, realm of hand gestures are the mostly unconscious gestures of hands in conversation and public appearances. Hand gestures are, naturally, an important aspect of the arts of rhetoric and acting.

However, the working hand represents the true versatility of its actions as well as both its seamless unity with the intentioning mind and its cunning independence and capacity for autonomous thought.

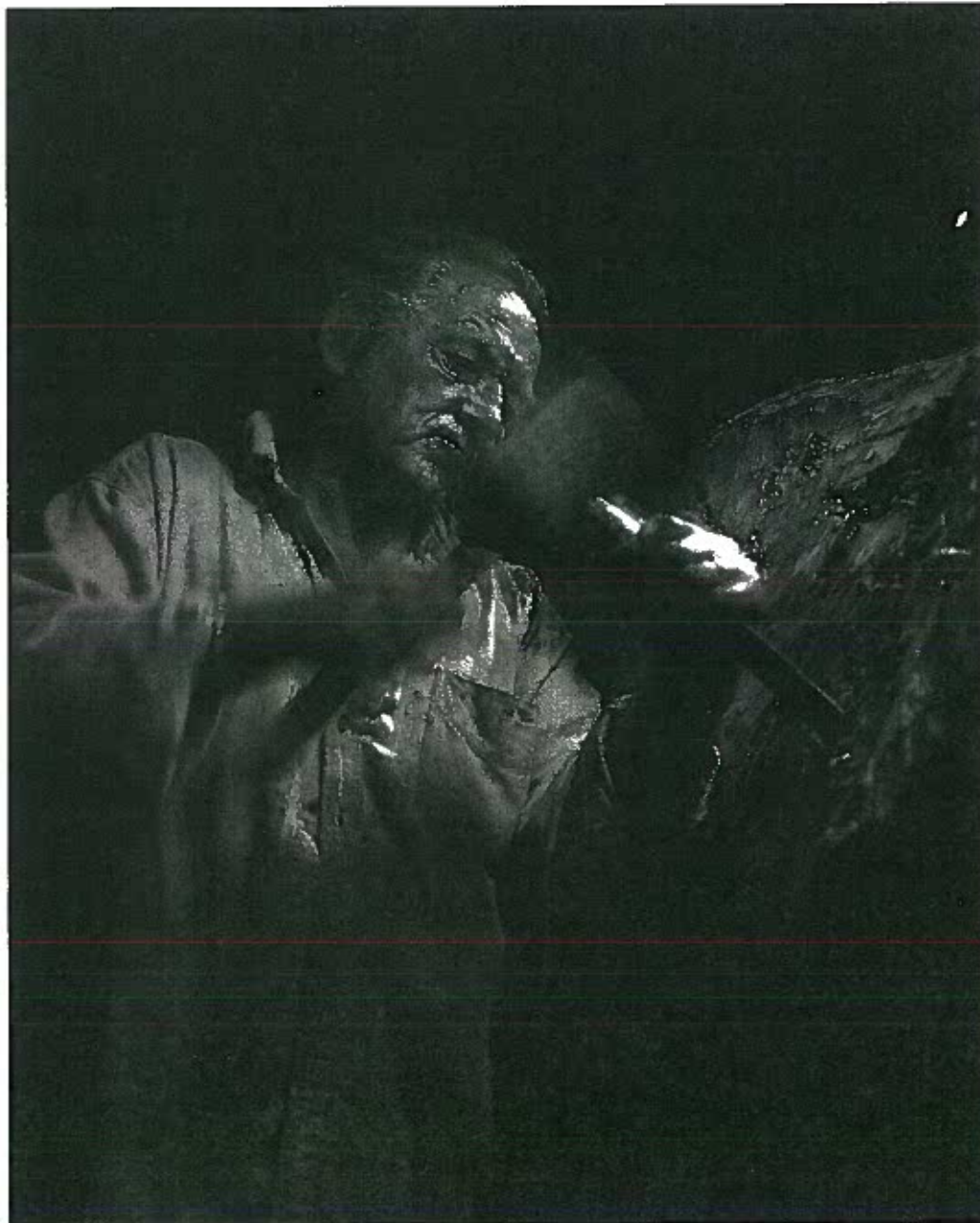


Gestures for miming from John Bulwer's *Chirologia, Natural Language of the Hand*, 1644. Engraving by English School, 17th century.

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- 39 Ibid, p. 220.

The Working Hand



The Finnish sculptor Kari Tapper
chiselling a wood sculpture,
1998

'But the craft of the hand is richer than we commonly imagine. [...] The hand reaches and extends, receives and welcomes – and not just things: the hand extends itself, and receives its own welcome in the hands of others. [...] But the hand's gestures run everywhere through language, in their most perfect purity precisely when man speaks by being silent. [...] Every motion of the hand in every one of its works carries itself through the element of thinking, every bearing of the hand bears itself in the element. All the work of the hand is rooted in thinking.'

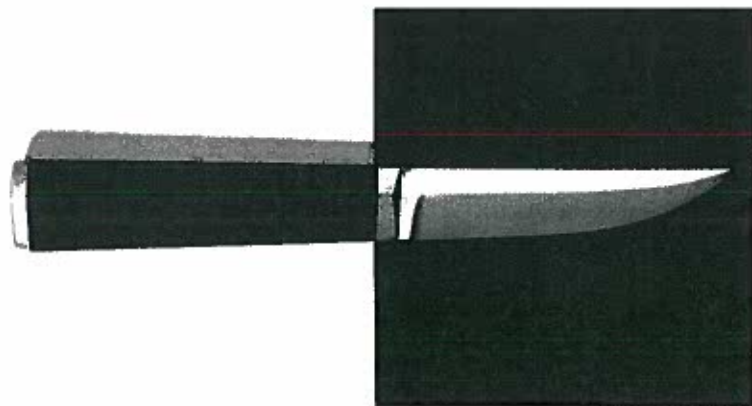
Martin Heidegger¹

The Hand and the Tool

The tool is an extension and specialisation of the hand that alters the hand's natural powers and capacities. When an axe or a sheath knife is being used, the skilled user does not think of the hand and the tool as

different and detached entities; the tool has grown to be a part of the hand, it has transformed into an entirely new species of organs, a tool-hand. Michel Serres, the philosopher, describes this perfect union of an animate and inanimate element eloquently: 'The hand is no longer a hand when it has taken hold of the hammer, it is the hammer itself, it is no longer a hammer, it flies transparent, between the hammer and the nail, it disappears and dissolves, my own hand has long since taken flight in writing. The hand and thought, like one's tongue, disappear in their determinations [...]'¹²

Tools evolve gradually through a process of small improvements, use and rejection. The finest tools are a result of a timeless anonymous evolution, and especially identifiable designer tools usually remain as momentary curiosities that do not become part of the real ancestry of the particular tool. Musical instruments, specifically conceived by designer professionals, exemplify these aestheticised curiosities. All great works of art similarly become an inseparable part of the tradition of the art form in question instead of being mere individualistic inventions. Great tools are moulded by the hand and its action directly. Centuries of continuous work have refined the basic tools – knife, hammer, axe, saw, plane – beyond improvement by an individual self-conscious designer, guided by intellectualised ideas of function and beauty. The development of tools in various cultures makes a mark on their specific 'DNA', as it were, that guides their evolution, resulting in a sense of relatedness. Like the human hand, the tool is generic and specific at the same time. It is possible to identify the genetic line of Japanese tools, for instance, as clearly distinct from the Scandinavian or North American family of tools;



Tools are usually anonymous products of the long tradition of use and successive improvement. A successful tool conceived by an individual designer is rare. Tapio Wirkkala, Finnish sheath knife (puukko), stainless steel and nylon, 1961

Tools possess an unarguable beauty brought about by the functional requirements and the anonymous tradition that have gradually perfected the object. Japanese carpenters' tools, late Edo or early Meiji period, private collection.



the performance and appearance of the tool unavoidably reflects the culture's particular attitude towards work and the social value placed on the work.

Tools possess a special and unarguable beauty. This is a beauty brought about by absolute causalities instead of being a materialisation of an aesthetic idea. Even the earliest stone tools express their use in the grip of the human hand and they convey the unarguable pleasure of perfect functionality and performance. The beauty of tools reflects the same pleasure of inevitability as living creatures; indeed, they possess the beauty of the human hand itself, the most perfect of all tools. Traditional tools, devices and vehicles developed in contexts where access to limited materials is limited, such as in the various Eskimo cultures, project a specially convincing and touching beauty that unites aesthetic pleasure with the pure joy of discovery.

Also buildings, such as the houses of Glenn Murcutt in Australia – perfectly adapted to their settings and functional requirements, and precisely expressive of climatic conditions and of their structural and material essence without any arbitrary aestheticising intentions – turn into architectural tools of sorts with

the same inevitability and beauty as the tools of crafts. Given the indefinable complexity of the hand, its actions and its relationship to the rest of the body as well as the brain, even simple hand tools are in essence body tools. Yet the complexity of performance in tools varies from one-hand tools to tools used by two hands, and tools, instruments and machines that function truly as extensions of the entire human bodily and neurological constitution, such as the bicycle, car or aircraft. In the same way that the boundary between the hammer and the hand disappears in the act of hammering, complicated tools such as musical instruments merge with the user's body; a great musician plays himself rather than a separate instrument. In drawing and painting, the pencil and the brush become inseparable extensions of the hand and the mind. A painter paints by means of the unconscious intentionality of the mind rather than the brush as a physical object.

Despite these magical integrations, tools are not innocent; they expand our faculties and guide our actions and thoughts in specific ways. To argue that for the purposes of drawing an architectural project the charcoal, pencil, ink pen and computer mouse are equal and exchangeable is to misunderstand completely the essence of the union of the hand, tool and mind.

The Hand of the Craftsman

Whenever I see the total correspondence and unexplainable affinity of a craftsman's persona, his/her hands and his/her workshop environment, I am deeply moved. The unity of a shoemaker's professional world and his hands, the dark workshop of a blacksmith covered by soot and the smell of burning coal, the fully integrated atmosphere of a cabinet-maker's persona, his tools, shop and the clean smell of wood, as well as the unity of the orderly and hygienic reception room of a dentist and his/her gloved hands, or the completeness of the highly technologised operation room of a microsurgeon and the masked doctor, all express the marriage of an individual and craft, responsibility and pride. This unity reflects dedication, determination and hope. Each one of these individuals has trained his/her hands for the highly specialised task and made a pact with the trade for the ultimate destiny of his/her life.

The studios of architects who have an uncompromising, exploratory and humbly ambitious attitude towards their architectural craft usually express the personality of the architect as well as the maker's devotion to and respect



The entire operation room with all its countless technical facilities, instruments and assistants are all extensions of the surgeon's hands. Hands of a surgeon.

for his/her work. These studios are epics of a lifetime of arduous work and faith in one's mission. A unique sense of purpose and order usually resides beneath the seemingly chaotic mess of sketches, working models, material samples, photographs, notes, memos and books.

In his recent book, *The Craftsman*, the cultural historian Richard Sennett narrates a concise history of craftsmanship, its characteristic ways of action and thinking, its relationship to tools and machines, the development of the required skills, and the craftsman's ethical stance. The tradition of craftsmanship is clearly gaining increasing value and appreciation in today's reality of the technological world, mechanical production, and the regrettable loss of the touch of the human hand in our mechanically mass-produced products and environments. In traditional cultures the entire life world is the product of human hands, and the daily sphere of work and life

means an endless passing of the hand skills and their products on to others; a traditional life world is a continuous meeting and joining of the hands of successive generations.

In my country, numerous traditional specialised crafts – such as the building of traditional church boats, basket making, burning of pine tar, restoration of buildings and objects, and painting of imitation materials in buildings – were almost lost in the period of euphoric industrialisation of the 1960s and

1970s. Fortunately, a new interest in traditions has followed the industrial rage and saved these and numerous other crafts, but there are still countless skills and an immense stock of unverbaised knowledge around the world, embedded in ageless modes of life and livelihoods, that need to be maintained and restored. These traditional cumulative practices of the human hand around the world form the true survival skills of humankind.

Craftsmanship arises from manual skill, training and experience – personal commitment as well as judgement. 'Every good craftsman conducts a dialogue between concrete practices and thinking; his dialogue evolves into sustaining habits, and these habits establish a rhythm between problem solving and problem finding,' Sennett points out.³ Even composers, poets and writers often consider themselves as craftsmen. Anton Chekhov used the Russian word *mastersvo* to describe his craft both as a medical doctor and as a writer.

Jorge Luis Borges likewise considered writing as a craft, and this attitude is reflected in the very title of his Harvard lectures of 1967–8, published in book form as *This Craft of Verse*.⁴

In addition to the tool, the skilled practice of a craft involves imagination with the hand; every masterful exercise of craft projects determined intentionality and an imagined vision of the completed task or object at hand. Richard Sennett makes two basic arguments about the interaction



A surprising similarity between an artist's own character and his work that underlines the interaction of the artist's sense of self and his work. Henri Cartier-Bresson, *Alberto Giacometti on rue d'Alésia*, Paris, 1961.

of the bodily actions of the hand and imagination:

First, that all skills, even the most abstract, begin as bodily practices; second, that technical understanding develops through the powers of imagination: The first argument focuses on knowledge gained in the hand through touch and movement. The argument about imagination begins by exploring language that attempts to direct and guide bodily skills.⁵

The craftsman needs to develop specific relationships between thought and making, idea and execution, action and matter, learning and performance, self-identity and work, pride and humility. The craftsman needs to embody the tool or instrument, internalise the nature of the material, and eventually turn him/herself into his/her own product, either material or immaterial. The physical likeness or resonance between the artist/maker and his/her work is often surprising; just think of the lean and melancholic figure of Alberto Giacometti, and his solitary, eroded walking sculptures clinging shyly to the surface of Mother Earth by their enormous feet.

In his book *Berger On Drawing*, John Berger points out this identification or fusion of the maker and his/her product in the craft of drawing: 'Each confirmation



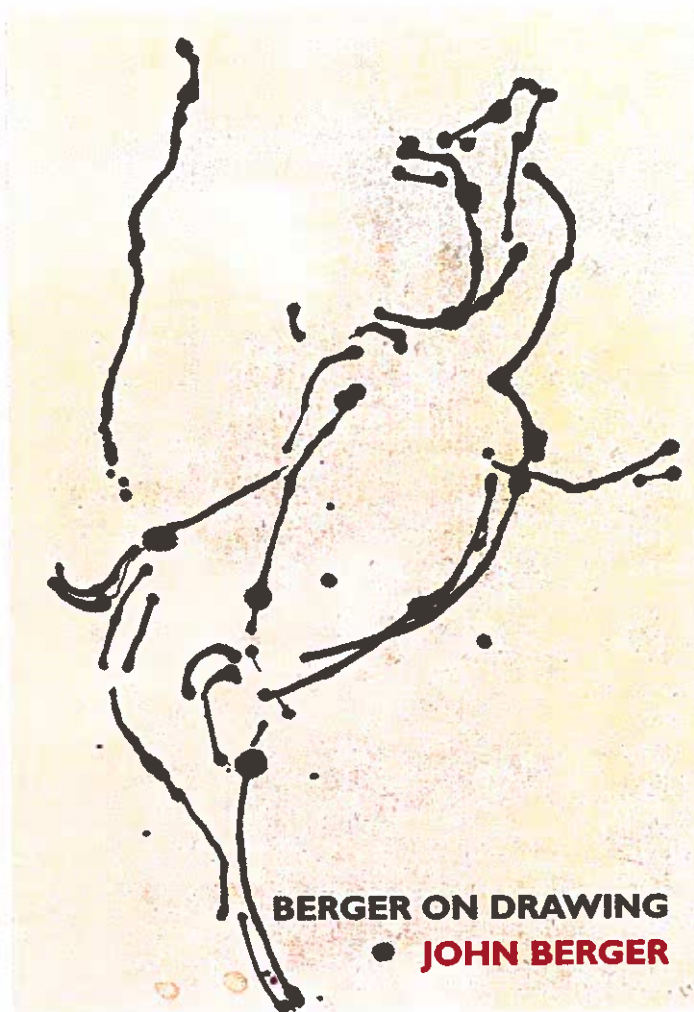
Alberto Giacometti, *Standing Woman*, 1948. Cast 1949

or denial brings you closer to the object, until finally you are, as it were, inside it: the contours you have drawn no longer marking the edge of what you have seen, but the edge of what you have become.⁶

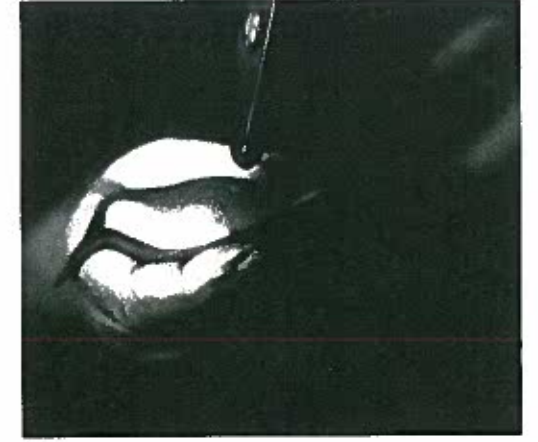
Tapio Wirkkala (1915–85), the legendary Finnish designer and master craftsman, worked in practically any material, and even in his role as industrial designer he carved the graphite moulds for the prototypes of his glass objects himself. His perfect manual control was expressed equally in cutting thin and even slices of Finnish rye bread or filleting a fish, carving a piece of sculpture or a prototype of an object, and drawing a perfect circle two-handedly on the blackboard. He could draw and write equally well with both hands, and even shift the pencil from one of his hands to the other in the middle of writing or drawing a line. His hands were equally determined and skilled in carving a huge birch plywood sculpture with power tools, and engraving a glass object with a diamond point, or drawing a tiny postage stamp.

Wirkkala explains his relationship with materials:

‘Making things with my hands means a lot to me. I could even say that when I sculpt or model nature’s materials it has an almost therapeutic effect. They inspire me and lead me on to new experiments. They transport me into another world. A world in which, if eyesight fails, my fingertips see the movement and the continuous emergence of geometrical forms.’⁷ He often used the expression ‘eyes at the fingertips’ referring to the subtlety and precision of the tactile sense of the hand.

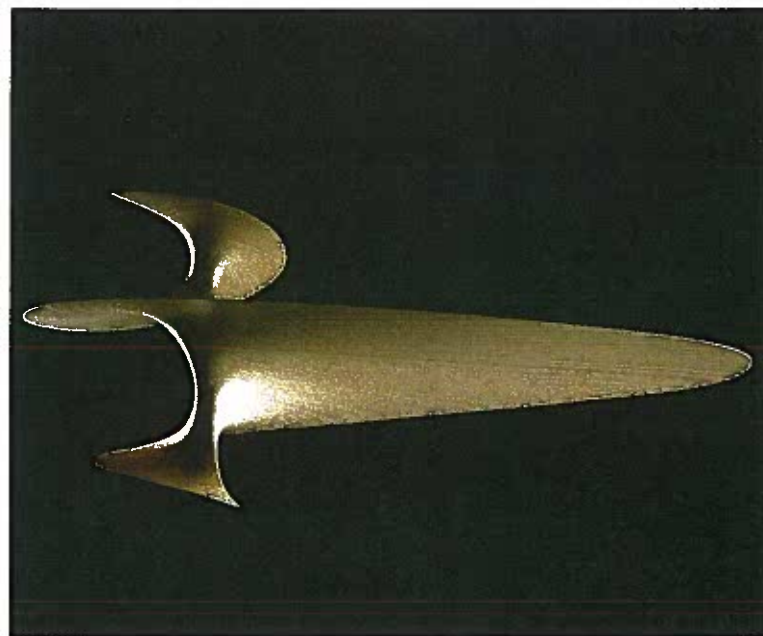


John Berger, *Drawings from Chauvet Cave*. Cover of Berger's book *Berger on Drawing* (Occasional Press (Aghabullogue, County Cork), 2007)



The maker's hands. Tapio Wirkkala, the designer, working on a drawing, finishing the surface pattern on a bird sculpture of cast bronze, carving a graphite mould for a glass object, and splitting wood with an axe of his own design

The work of the craftsman implies collaboration with his material. Instead of imposing a preconceived idea or shape, he needs to listen to his material. Brancusi was the magician of pure form, but he was also deeply concerned with the innate properties of materials: ‘[Y]ou cannot make what you want to make, but what the material permits you to make. You cannot make out of marble what you would make out of wood, or out of wood what you would make out of stone [...] Each material has its own life, and one cannot without punishment destroy a living material to make a dumb senseless thing. That is, we must not try to make materials speak our language, we must go with them to the point where others will understand their language.’⁸



Tapio Wirkkala, *Whirl*, laminated birch plywood, 140 x 159 cm, 1954.

Wirkkala, another master of form, expresses exactly the same concerns: 'All materials have their own unwritten laws. You should never violate the material you're working with. The designer's purpose is to be in harmony with the material. The craftsman has the advantage that at every stage of the work his material is in his hands to feel and command. In industry, the material is constantly subordinate to some preplanned law and machinery and once the job has begun it's difficult to make changes.'⁹

The Finnish sculptor Kain Tapper (1930–2004) relied on the feel of his palms rather than his eyes in finishing the fluidity of the shape or the rhythm of the surface texture in his wood and stone sculptures. He liked to polish his stone pieces at the shoreline of a lake because he felt that the horizontality of the water surface and the definitiveness of the horizon line sharpened his eyesight and tactile sense. His subtle wood reliefs could be called 'tactile paintings' as they address the hand and the skin as much as the eyes.

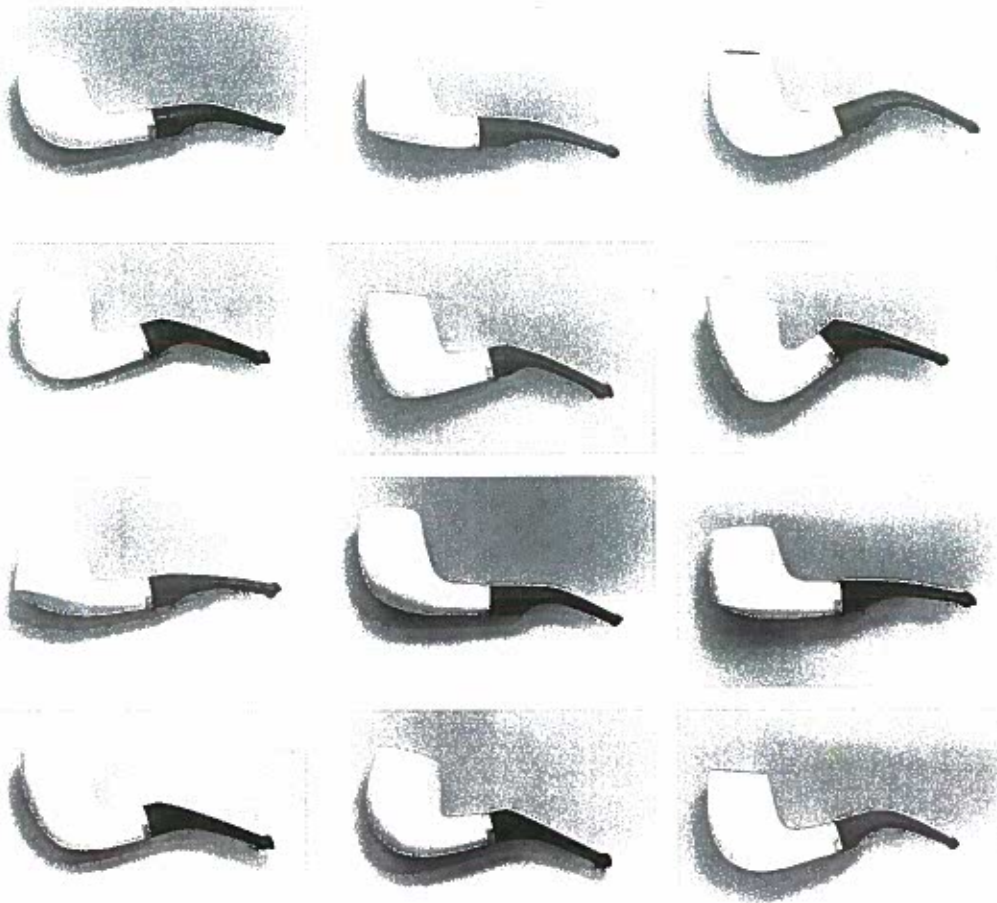
In another context, Wirkkala speaks about the interaction of two hand activities, drawing and model making:

Kain Tapper's sculptures and wood reliefs are products of the tactile sense of his hands and the muscular sense of his body as much as products of visual aspirations.
Kain Tapper, *Wind*, birch, 120 x 120 cm, 1964, Amos Andersson Art Museum, Helsinki.



A drawing or sketch is an idea which provides the basis to start work. I make dozens – sometimes hundreds – of sketches. From them I select those that offer some potential for development. For me it's important to see the object as a concrete thing before sending it on to the manufacturer. Making the model is an essential aspect of my work. I produce it from some solid material. I don't make just one, but several models which I can compare and then select one to continue working on. In this way the idea becomes clearer and the mistakes more apparent.¹⁰

Even in the age of computer-aided design and virtual modelling, physical models are incomparable aids in the design process of the architect and the designer. The three-dimensional material model speaks to the hand and the body as powerfully as to the eye, and the very process of constructing a model simulates the process of construction.



Models are used for a variety of purposes: they are a way of quickly sketching the essence of an idea; a medium of thinking and working, of concretising or clarifying one's own ideas; a means of presenting a project to the client or authorities; and a way of analysing and presenting the conceptual essence of the project. Models are also used to study specific aspects of architectural projects, such as illumination or acoustic qualities. The model both concretises and externalises ideas: the frequently diminutive scale of the model and the observer's externality invites and permits the identification and judgement of aspects that could otherwise be lost. The model helps the architect to think of the project in its full

A true craftsman is not bound to a single idea, as the formal idea often gives rise to a family of variations. Tapio Wirkkala, pipe models, 'meerscham' (sea foam) and nylon, 1974–6.



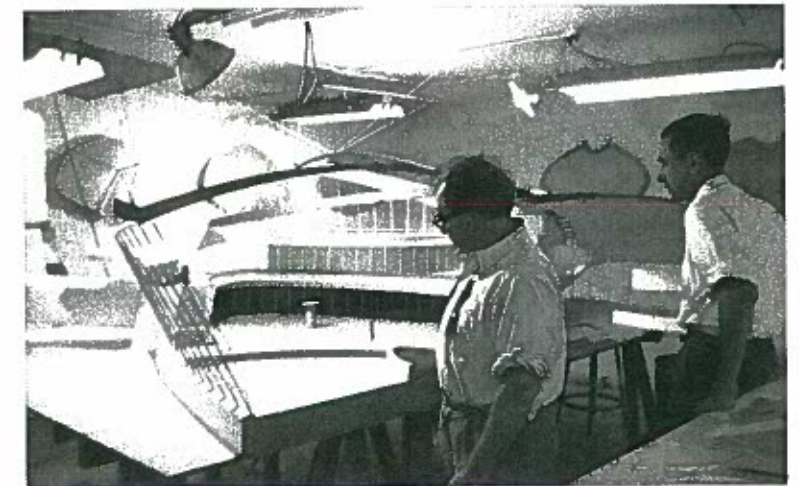
Architectural models may be tiny, yet serve as meaningful design tools in concretising ideas. Gerrit Thomas Rietveld, Sonisbeek Sculpture Pavilion, Arnhem, 1951. Tiny sketch model of the project held by the architect in his hand.

spatial completeness, as Henry Moore advises. In addition to their primary purpose of mediating and facilitating the design process itself, architectural models are often conceived and produced as semi-independent objects of art, or at least of aesthetic appreciation.¹¹

Collaborative Craftsmanship

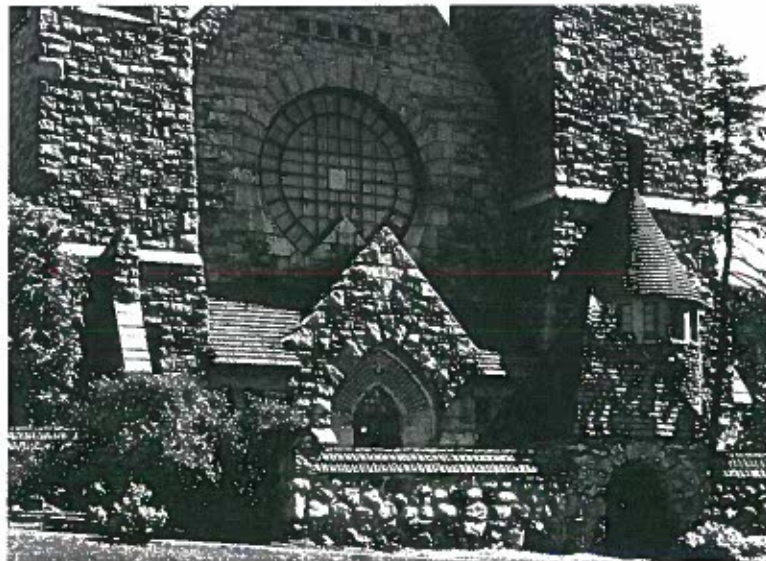
While drawing, a mature designer and architect is not focused on the lines of the drawing, as he is envisioning the object itself, and in his mind holding the object in his hand or occupying the space being designed. During the design process, the architect occupies the very structure that the lines of the drawing represent. As a consequence of the mental transfer from the actuality of the drawing or the model to the material reality of the project, the images with which the designer advances are not mere visual renderings; they constitute a fully haptic and multi-sensory reality of imagination. The architect moves about freely in the imagined structure, however large and complex it may be, as if walking in a building and touching all its surfaces and sensing their materiality and texture. This is an intimacy that is surely difficult, if not impossible, to simulate through computer-aided means of modelling and simulation.

Architectural models are sometimes enormous constructions that one can enter, particularly in the case of acoustical test models for concert halls. Eero Saarinen and Cesar Pelli with the model of the Trans World Airlines Terminal at the John F Kennedy Airport, New York, USA, 1956–62.



While working on a drawing you concretely touch all the edges and surfaces of the designed object with the tip of your pencil that has become an extension of your fingertips. The hand–eye–mind connection in drawing is natural and fluent, as if the pencil were a bridge that mediates between two realities, and the focus can constantly be shifted between the physical drawing and the non-existent object in the mental space that the drawing depicts.

Drawings and models have the double purpose of facilitating the design process itself and mediating ideas to others. Working drawings, finally, transfer instructions of the conceived design to the craftsmen and builders for the purposes of execution. An element of magic resides even in this final phase of communicating instructions for execution, usually regarded as a mere necessity that only requires precision and clarity of logic in the communication. I have often marvelled at the natural stone walls of the Tampere Cathedral by Lars Sonck (1870–1956) that seem to radiate an extraordinary sense of attention and care as if every individual stone had been selected and put in its place in a state of inspiration or ecstasy. Vibrating in light and expressing the entire history of brick building traditions, the red brick walls of Alvar Aalto's Säynätsalo Town Hall (1948–52), Jyväskylä University (1952–7) and other buildings of his 'red period' project similarly a powerful tactile invitation that speaks of the act of bricklaying and the touch

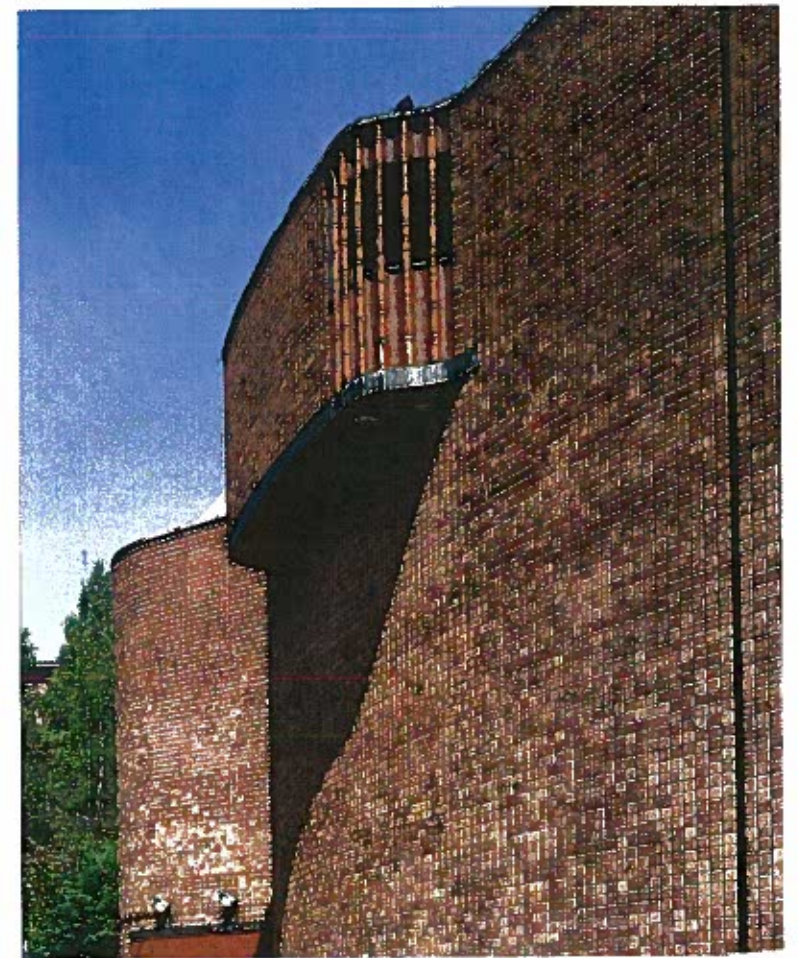


Even the material execution of great buildings often radiates a sense of inspiration. Lars Sonck, Tampere Cathedral, Finland, 1902–7.

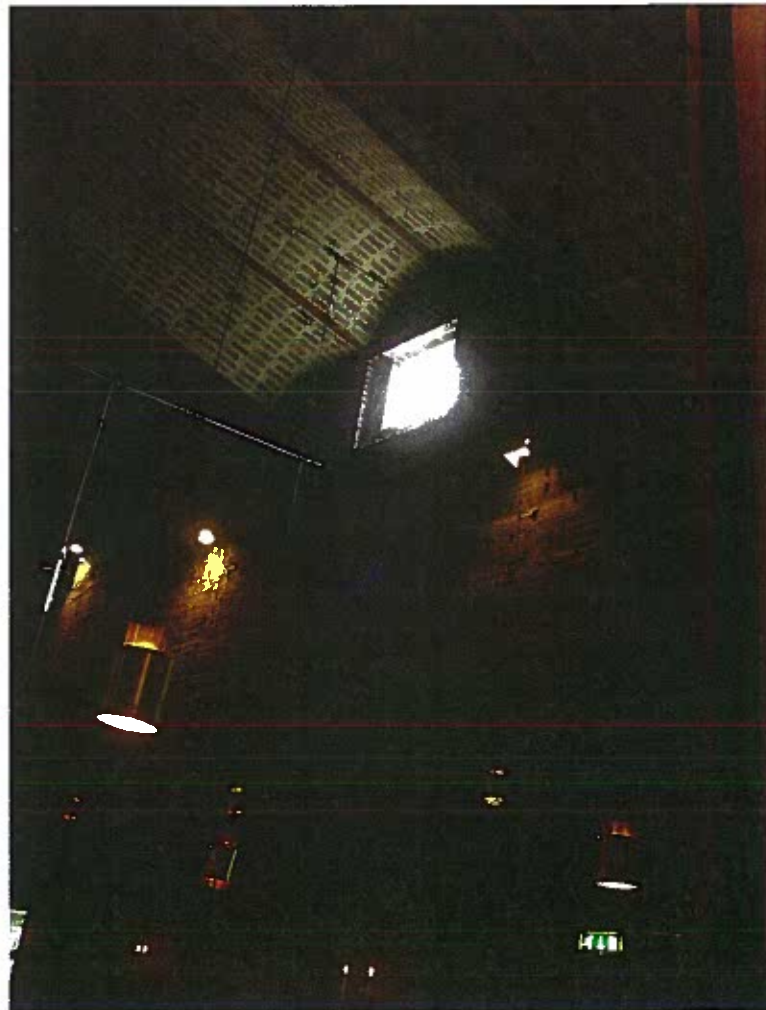
of the hand, whereas the very same bricks in a building nearby, designed by a less gifted architect, appear as lifeless surfaces made of industrially mass-produced construction elements.

The Swedish master architect Sigurd Lewerentz (1885–1975) is told to have arrived at the construction sites of his Churches of St Mark in Björkhagen (1956–60) and St Peter in Klippan (1963–6) early in the morning, at the time his bricklayers began their daily work, and, sitting in a chair, pointed with his umbrella at one brick at a time in the stack and then at

Wall made of specially designed wedge-formed brick to permit both convex and concave curved surfaces. The brick wall projects a rare sense of authority and timelessness. The projection room bulging out of the main volume is the architect's late improvisation during the course of design. Alvar Aalto, House of Culture, Helsinki, Finland, 1952–8.



its destined place on the brick wall under construction. In the Lewerentz walls and vaults, laid with wide plaster seams, each brick maintains its individuality, and the coarseness of bricklaying expresses the physicality of work; one can almost sense the smell of sweat and hear the chatting of the bricklayers. Lewerentz is reported to have told the bricklayers the white lie that the brick surfaces would finally be plastered; otherwise the bricklayers would not have agreed to execute such brutal bricklaying work



The coarse brickwork in the late Sigurd Lewerentz churches speaks of the physicality of manual labour
Sigurd Lewerentz, St Mark's Church, Björkhamnen (Stockholm), Sweden, 1956–60.

under today's professional practices of strict quality standards. Whether this is true or not, the story emphasises the importance of intentionality and intimate human communication even in work that seems to be mechanical.

Most designers – such as glass artists or furniture designers, not to mention architects – rarely make the objects they design themselves. Consequently, they need to understand the possibilities and limits of the materials and crafts, and communicate their ideas and intentions to the specialist craftsman, whose hands become the designer's surrogate hands in the execution of the work. The architect often needs an entire army of surrogate hands, both in the studio and on the construction site, to execute his work. In the case of the unique glass objects that Tapio Wirkkala designed for Venini in Venice, the Finnish designer collaborated with the knowledge and skills accumulated at the Murano factories through several generations of Venetian master glass-blowers. The shared knowledge of the material, the shared ambition to perform at the limit of the capacity of the craft and one's personal skill, and the logic of the work itself provided the syntax for the unspoken language between the Finnish designer and the Venetian maker. In fact, the designer's collaborator was the ageless tradition of his art, rather than any single individual craftsman.

I used to think that the architect's duty was to design structures and details that are as easy to execute as possible. Having realised that every serious professional has his ambition and pride, I have changed my view entirely. Skilled craftsmen and builders like to face challenges, and consequently the work needs to meet the full potential of the maker in order to provide the desired inspiration and satisfaction. Work that is too simple and repetitious kills ambition, self-esteem, pride and, finally, the craft itself. Most importantly, collaborative craftsmanship requires mutual respect. Alvar Aalto was a master in his communication with the various professionals and craftsmen of his varied productions; the highly respected academician spoke to a carpenter and bricklayer as his equals, and inspired them to internalise their work and perform supremely at the limit of their professional capabilities.

Mastering one craft personally helps the designer and architect to grasp the nuances of other crafts and, before all, to respect the special skill and experience of the craftsman executing his design. Besides, learning any skill intimately teaches one welcome humility. Arrogance does not go with true skill.



Tapio Wirkkala collaborated with the master glass-blowers of the Venini Glass Factory in Murano, and this interaction brought the age-old mastership of Venetian glass art to new heights. Wirkkala's glass objects designed for Venini pay homage to the long traditions of Venetian glass-blowing. Tapio Wirkkala, *Coreano* dish, 40 cm diameter. Free-blown, in swirled overlapping colours, turquoise and apple green.

Architecture as Workmanship

The architectural profession was traditionally regarded as a craft, or close to the notion of craft. Architectural ideas were created in close interaction with the actual physical construction at the site, and drawings did not emerge as means of conceiving architecture until the Renaissance period.¹² Prior to that, architecture was seen as a manual occupation along with painting and sculpture. In order to raise these manual and mechanical arts to the level of the 'liberal arts' of arithmetic, geometry, astronomy and music that formed the *quadrivium* of the mathematical arts, these practices had to be given a firm theoretical – ie mathematical – foundation, something that was at the time to be found in musical theory.¹³ The essence of architecture was largely in technical practicalities as conveyed, for instance, by Vitruvius's (84–14 bc) seminal treatise *De architectura libri decem* (*The Ten Books on Architecture*). In addition to conceiving the novel structural principle for the elliptical dome

of the cathedral of S Maria del Fiore in Florence (1417–46) – which, 43 metres in diameter and rising to a height of 115 metres, was erected in two ribbed shells without centring – Filippo Brunelleschi, a clocksmith by initial training, also had to invent all the devices for transporting huge blocks of stone and lifting them to the soaring heights of the dome. It should also be remembered that the master architects of the Renaissance were usually painters and sculptors as well.

In some countries, such as Denmark, an alternative path to the profession of the architect has traditionally existed through some of the building crafts, such as bricklaying, carpentry, or cabinet-making. The connection with the construction site and processes of construction has also traditionally been intimate in the architectural profession until the modern era that has, however, emphasised specialisation and the consequent separation of the architect from the physical work of construction. The fact that being an apprentice at the construction site used to be a mandatory part of architectural education, and that architects often practised a craft, drawing, painting or sculpture as a hobby or a means of acquiring manual skill and carrying out formal experiments, reinforced the connection between professional architectural practice and the realities of making – between idea and matter, form and its execution.

During the post-war decades, the intellectual emphasis in architectural education, and the growing practical as well as mental distance between the architect's studio and the construction site have, however, decisively weakened the craft essence of the architect's work. Today the architect usually works from the distance of the architectural studio through drawings and verbal specifications, much like a lawyer, instead of being directly immersed in the material and physical processes of making. In addition, the increasing specialisation and division of labour within the architectural practice itself has fragmented the traditional entity of the architect's self-identity, working process, and end result. Finally, the use of the computer has broken the sensual and tactile connection between imagination and the object of design.

Design-build practices particularly in the United States – such as Sam Mockbee's Rural Studio in Alabama, Rick Joy's in Arizona, and Dan Rockhill's in Kansas – have reintroduced the intimate connection between designing and building, thinking and making. The design-build practice also brings the execution and detailing back into the full control of the designer,

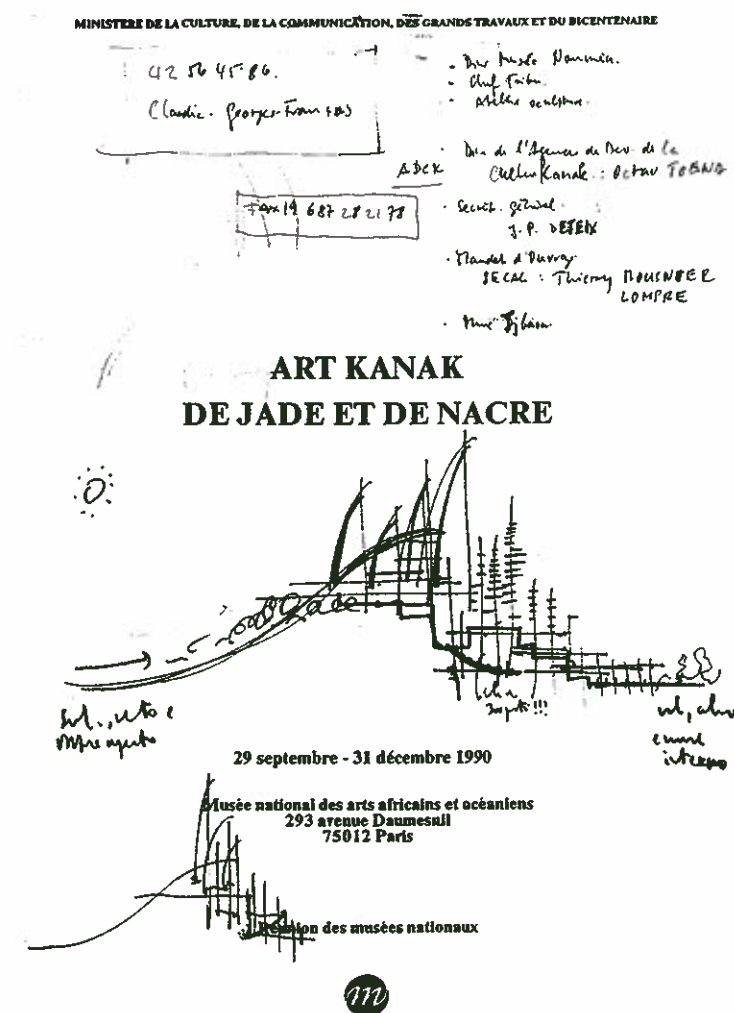


and potentially eliminates the conservatism and carelessness of today's construction companies. Rockhill's Studio 804 and Brian MacKay-Lyons's Ghost International Architectural Laboratory in Nova Scotia exemplify architectural courses that aim at linking architectural education back to practices and processes of physical making.

Small studio practices around the world often embrace a craftsman-like ethos and maintain an intimate, tactile connection with the work. Renzo Piano is surely one of the most sophisticated High-Tech architects today, but he has deliberately maintained a craftsman's approach to the process of architectural design, experimentation, and execution of the work. Piano explains his craftsmanlike working methods as follows: 'You start by sketching, then you do a drawing, then you make a model, and then you go to reality – you go to the site – and then you go back to the drawing. You build up a kind of circularity between drawing and making and back again.'¹⁴ The architect's

Renzo Piano emphasises the 'circularity' of his design process – the repeated shifting from sketching to drawing to model building and testing, and back again. Renzo Piano, sketch for the Jivi Tjibaou Cultural Centre in Hounéa, New Caledonia, Canada, 1990.

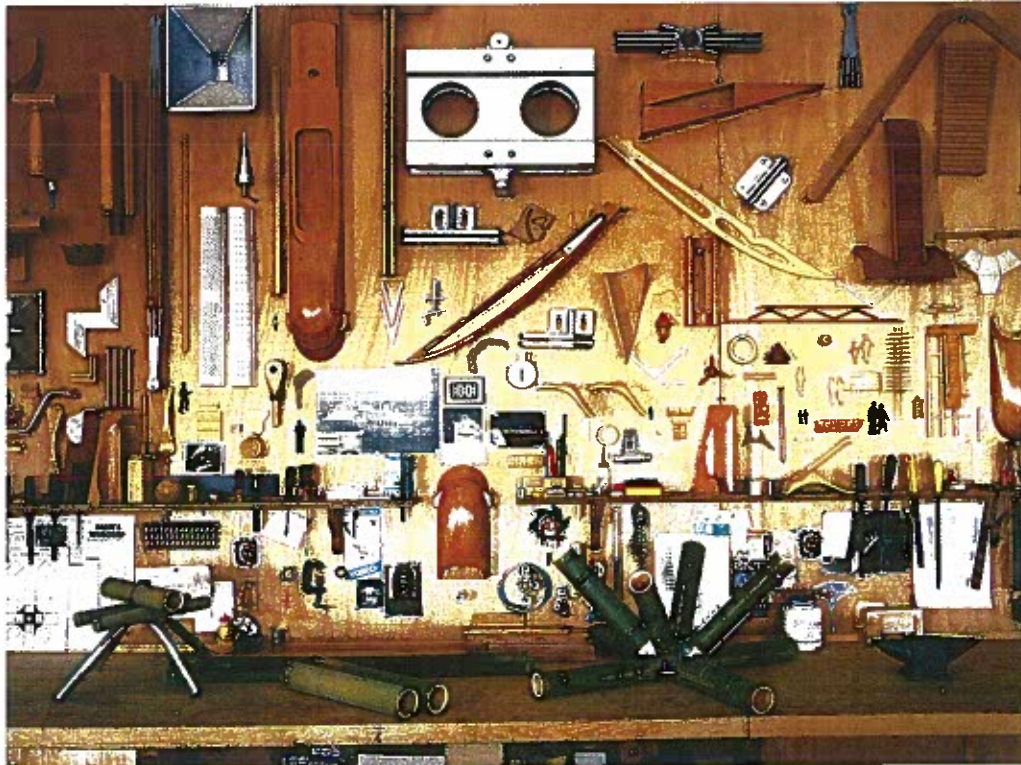
approach here seems to be close to the working method of the craftsman-designer as exemplified by Tapio Wirkkala. The important aspect of the process is its 'circularity', the constant shifting of viewpoints from the idea to the sketch, the model, a full-scale test, and back again. As a consequence of this arduous and complex process, the building exists as a full immaterial



mental construction long before the actual construction work begins. In fact, the building has often been built and tested as a mental construction in several alternatives before the final concept is chosen.

Tireless repetition is an essential feature of Renzo Piano's way of working. 'This is very typical of the craftsman's approach. You think and you do at the same time. You draw and you make. Drawing [...] is revisited. You do it, you redo it, and you redo it again.'¹⁵ Piano has appropriately named his studio 'Renzo Piano Building Workshop' to reflect the idea of teamwork, and to suggest the long traditions of craftsmen's and artists' workshops since the Middle Ages with their intimate relationship between the master, apprentice and work. The feeling of the workshop of a medieval guild of sorts separates Piano's workshop, reflecting the materiality and physicality of things as well as physical labour, from the neatness and sterility of a businesslike architect's office of today.

Models and prototypes in the Renzo Piano Building Workshop



In my view, the connection with the processes of making continues to be seminal, and a wise architect today searches deep personal friendships with craftsmen, artisans and artists in order to reconnect his/her intellectualised world and thinking with the source of all true knowledge: the real world of materiality and gravity, and the sensory and embodied understanding of these physical phenomena.

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Eye- Hand-Mind Fusion

'This image [of the artist's model] is revealed to me as though each stroke of charcoal erased from the glass some of the mist which until then had prevented me from seeing it [...] beyond the haze of this uncertain image I can sense a structure of solid lines. This structure releases my imagination, which works, at the next sitting, in accordance with the inspiration which comes both from the structure and directly from the model. [...] Drawings containing all the subtle observations made during the work arise from a fermentation within, like bubbles in a pond.'

Henri Matisse¹

Experimentation and the Art of Play

David Pye divides workmanship into two categories in his book *The Nature and Art of Workmanship*: 'workmanship of risk' and 'workmanship of certainty'. The first attitude to workmanship 'means that at any moment,

In creative exploration the actions of the hand, eye and mind fuse into a singular process of semi-unconscious scanning. Here, the architect's attention keeps shifting from floor plan to section and various details, back and forth.
Alvar Aalto, early sketches for the Church of the Three Crosses at Vuoksenniska, Imatra, Finland, presumably 1955. Pencil on tracing paper.





'Workmanship at risk', in which a single slip of the tool would destroy the entire piece. David Pye, English walnut bowl.

whether through inattention, or inexperience, or accident, the workman is liable to ruin the job'. In the second approach 'the quality of the result is predetermined and beyond the control of the operative'. David Pye, a master craftsman of skilled wood objects himself, concludes: 'All the works of men which have been most admired since the beginning of our history have been made by the workmanship at risk, the last three or four generations only excepted.'²

This thought-provoking separation of craft practices into two categories with their distinct ethical connotations also applies to architectural practices today. Most practices apply rather established and tested standard methods and solutions throughout their work, while more ambitious and courageous studios tend to experiment with novel structures, forms, materials, details, and their combinations. These practices are willing to employ a 'workmanship of risk'. The 'risk' usually implies the mental uncertainty of advancing on untrodden paths, as the actual risks in relation to safety, durability, appearance and suchlike can usually be minimised by working experience, careful calculations, research, experimentation, and laboratory or prototype tests. The risk is directed to the architect's own persona, values, beliefs and ambitions – one's self-identity as an architect and professional. The creative state is a condition of haptic immersion where the hand explores, searches and touches semi-independently. Reima Pietilä (1923–93), the Finnish architect, compared the design process with the acts



Facile probing through sketching. Two early sketches for the non-geometric shape of the architectural project Raiti and Reima Pietilä, Kaleva Church, Tampere, Finland, competition 1959, execution 1964–6. Two early charcoal sketches.

of hunting and fishing; you cannot be certain what you are going to catch, or whether you will catch anything at all. Pietilä's working method was a curious fusion of linguistic and visual explorations; his sketching appears as probing by means of invented words, whereas his spoken and written language often projects characteristics of visual sketching. Both his lines and words probe and mould the contours of an unknown territory. Studies in the morphology of characteristic Finnish landscapes often revealed to him the formal language, structure, texture and rhythms of his projects.³

Alvar Aalto provides a rare and intimate insight into the associative and experimenting creative process of a great mind, indicating the seminal role of the absent-minded hand and its seemingly unconscious and aimless play in sketching.

This is what I do – sometimes quite instinctively. I forget the whole maze of problems for a while, as soon as the feel of the assignment and the innumerable demands it involves have sunk into my subconscious. I then move on to a method of working that is very much like abstract art. I simply draw by instinct, not architectural syntheses, but what are sometimes quite childlike compositions, and in this way, on an abstract basis, the main idea gradually takes shape, a kind of universal substance that helps me to bring the numerous contradictory components into harmony.⁴



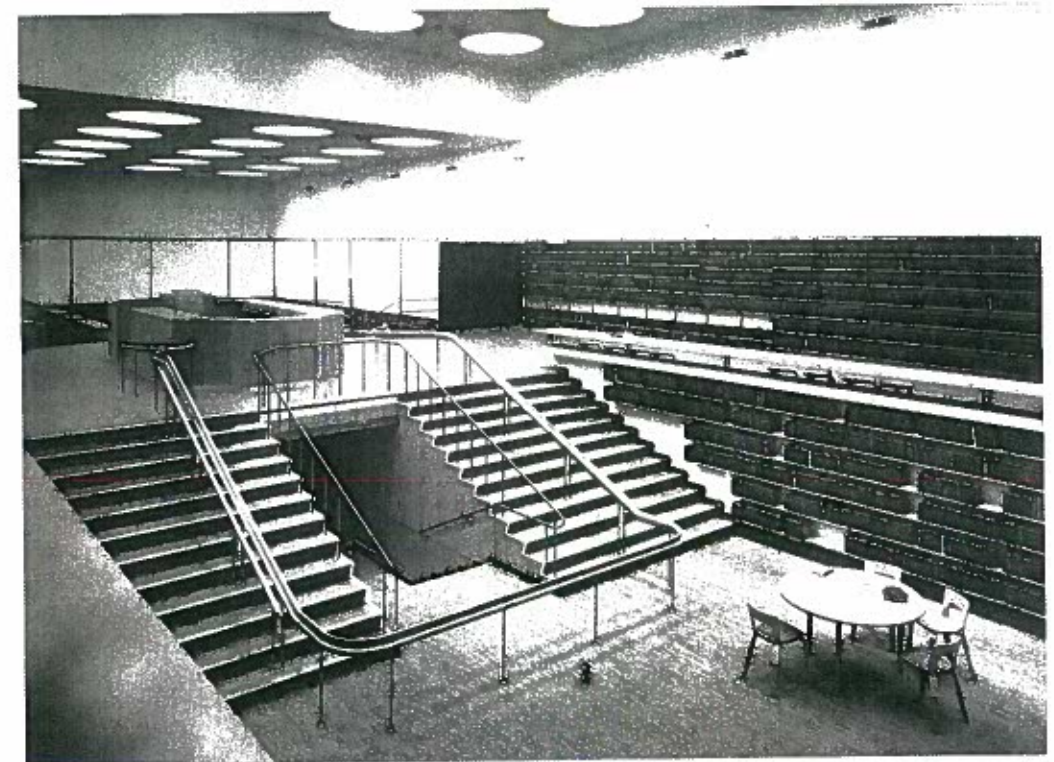
Alvar Aalto, early sketches of 'a fantastic mountain landscape with slopes lit by many suns' during the design process of the Viipuri City Library, Viipuri, Finland, 1927–35

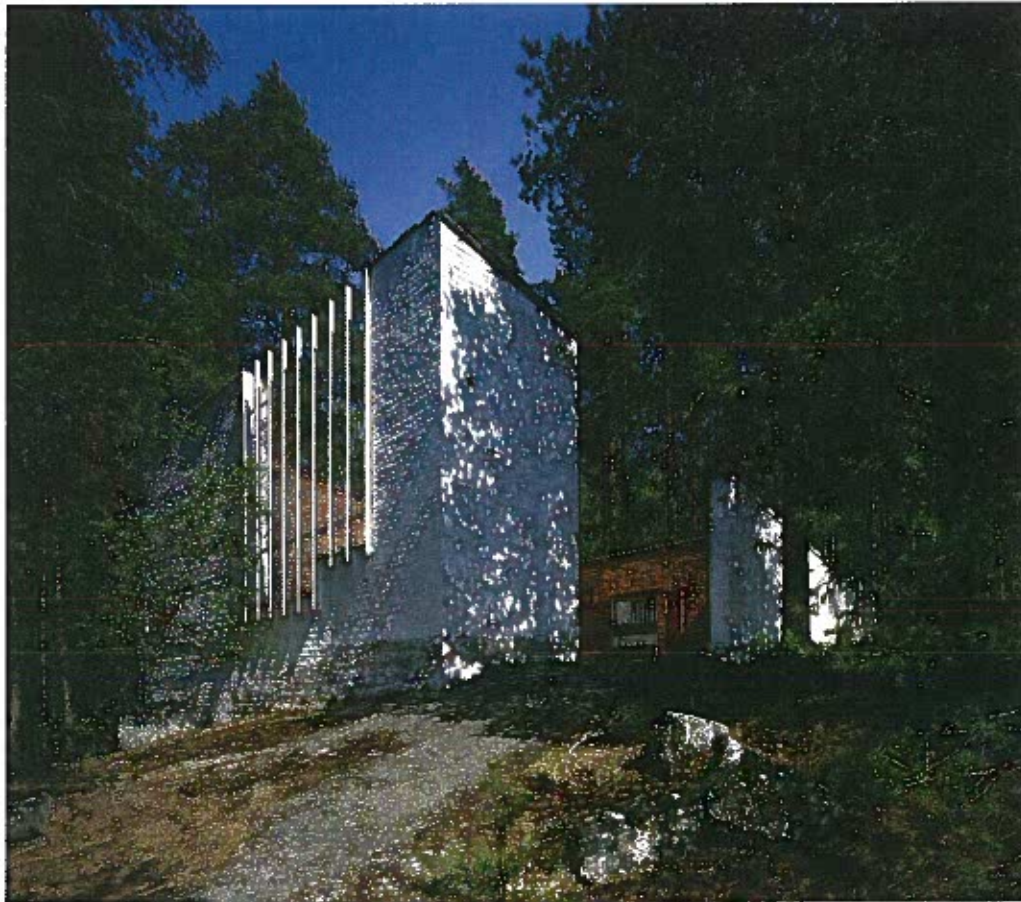
Aalto's design approach points out that in creative work a focused consciousness needs to be momentarily relaxed and replaced by an embodied and unconscious mode of mental scanning. The eye and the external world are dimmed for an instant, as consciousness and vision are internalised and embodied.

When I designed the Viipuri City Library (and I had plenty of time, a whole five years), I spent long periods getting my range, as it were, with naïve drawings. I drew all kinds of fantastic mountain landscapes, with slopes lit by many suns in different positions, which gradually gave rise to the main idea of the building. [...] My childlike drawings were only indirectly linked with architectural thinking, but they eventually led to an interweaving of the section and ground plan, and to a kind of unity of horizontal and vertical construction.⁵

Alvar Aalto, Viipuri City Library, Viipuri, Finland, 1927–35
Reading room with lower entry area, elevated zone for the circulation desk, and circular skylights that emerged through his doodling of imaginary mountain landscapes.

Aalto's subconscious sketches of 'mountain landscapes' and 'many suns' eventually led him to the solution of the library consisting of stepped floor levels and in total 57 conical skylights with a 1.8-metre diameter that prevent direct sunlight at the maximum sun angle at 52 degrees or less. The project that arose from absent-minded doodles turned into one of the seminal architectural projects of Functionalism. Aalto used to sketch on thin rolled tracing paper that he could pull out in endless strips and keep on sketching in the manner akin to the 'train-of-thought' or 'automatic writing' method. These strips of tracing paper open up a view into Aalto's working mind that keeps shifting from the whole to the parts, from plan and sectional ideas to details, basic calculations of measurements and areas or verbal notes, and back again. Sometimes in the middle of working on a distinct project, his mind seems momentarily to waver into a completely different project – or, perhaps, a piece of furniture or light fitting. Aalto's sketches show concretely the non-linearity of the design process and the essential aspect of zooming





back and forth between various scales and aspects of a project similarly, in fact, to Renzo Piano's confession. In addition to the fluidity of his creative process, Aalto's soft sketches also demonstrate the essential seamless eye-hand-mind collaboration.

In the context of his Experimental House at Muuratsalo (1952–3), Aalto points out the importance of experimentation and play in his design method, while at the same time emphasising the sense of responsibility:

[I have] a firm conviction and instinctive feeling that in the midst of our labouring, calculating, utilitarian age, we must continue to believe in the

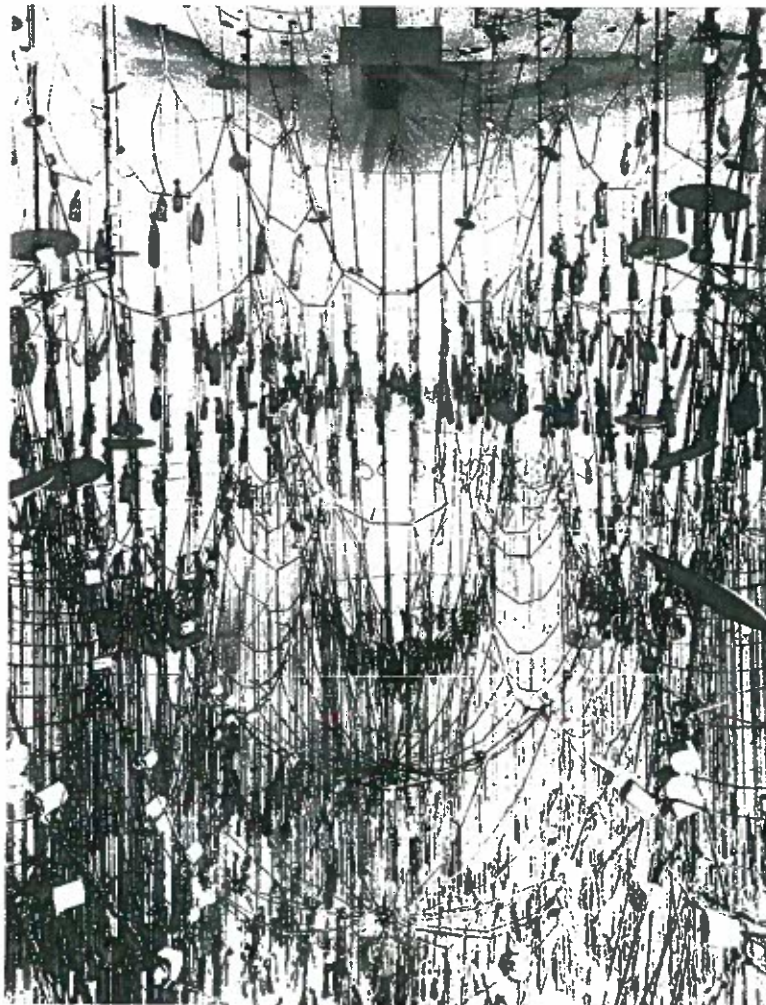
Alvar Aalto conceived his own summerhouse as a deliberate architectural experiment and consequently named it 'The Experimental House'.
Alvar Aalto, Experimental House, Muuratsalo, Finland, 1952–3.

Alvar Aalto carried out numerous exercises in various methods of bending wood, in addition to being technical experiments they were conceived as artistic objects. Wood bending experiment that juxtaposes a piece of tree and a 'man-made tree', birch, 47.5 x 37.5 cm, 1947.



crucial significance of play when building a society for human beings, those grown-up children. The same idea, in one form or another, surely lies at the back of every responsible architect's mind. A one-sided concentration on play, however, would lead us to play with forms, structures and, eventually, the body and soul of other people; that would mean treating play as a jest [...] we must combine serious laboratory work with the mentality of play, or vice versa. Only when the constructive parts of a building, the forms derived from them logically, and our empirical knowledge is [sic] coloured with what we might seriously call the art of play; only then are we on the right path. Technology and economics must always be combined with a life-enhancing charm.⁶

The Experimental House is an experiment on a conceptual or philosophical level as well as in the use of materials and detailing. The summerhouse built of brick in the Finnish lake landscape, in a region and building typology of prevailing wood construction, reflects the imagery of a Mediterranean atrium house, but the project also contains experiments in various uses of brick and ceramic tiles, free-form foundations on natural rocks, a free-form support system, solar heating, and the 'aesthetic effect of plants'.



Antoni Gaudí studied his architectural structures through suspended models. Photographs of the models were turned upside down to examine actual structures in compression. Study model for the church of Colonia Güell. Mas Archive, Barcelona.

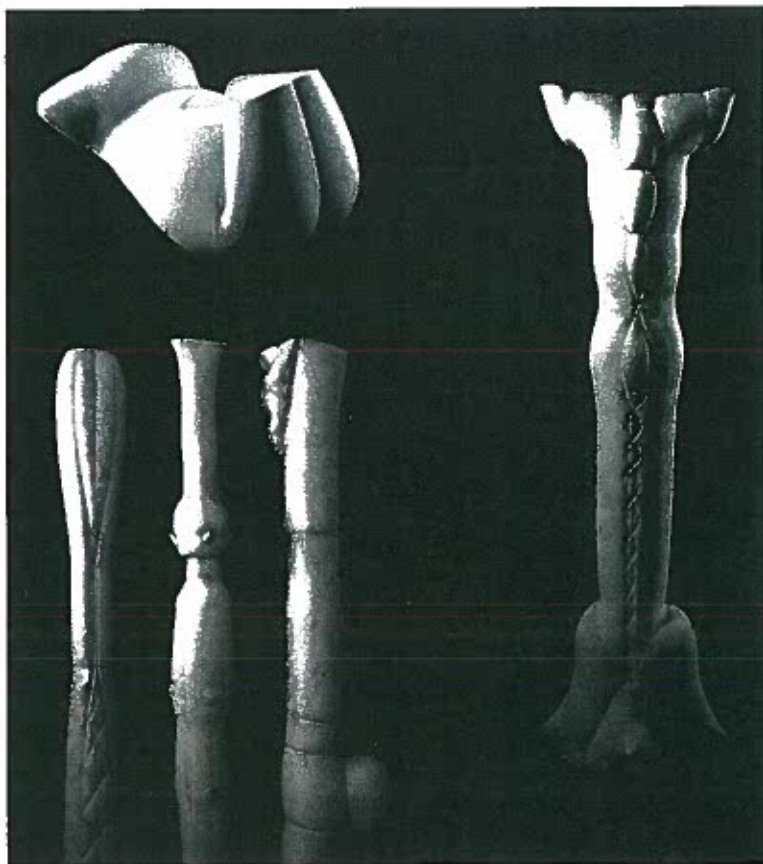
Alvar Aalto's sculptural experiments in various methods of bending wood that were conducted while he was developing his bentwood furniture during the 1930s and early 1950s demonstrate the role of semi-independent artistic experimentation in design. On the other hand, Antoni Gaudí's famous inverted structural experiments exemplify the use of physical models devised to define the performance and shape of an architectural structure. Mark West's current experiments at the University of Manitoba's Faculty of Architecture on concrete structures poured in canvas moulds continue this line of devising novel structural ideas through direct material experimentation; the processes of making give rise to theoretical formulations rather than vice versa.

Skill and Boredom

A craft is based on learned specialised skill; Sennett defines skill as a trained practice.⁸ Any skill calls for tireless practising: 'When I was a student at Juilliard, we all practised fourteen hours a day, and we knew that any time spent away from the piano was a waste of time,' pianist Misha Dichter confesses.⁹ This confession makes clear that any special manual skill – be it that of a pianist, puppeteer or juggler – requires endless practice based on obsessive dedication and commitment. The established estimate based on research concludes that any demanding specialised manual or physical skill requires about ten thousand hours of practice. 'In studies of composers, basketball players, fiction writers, ice skaters, ... and master criminals, this number comes up again and again,' the psychologist Daniel Levitin points out.¹⁰

Excessive exercise as well as thought can also stifle the performance. Anton Ehrenzweig (1908–66) – pianist, singer and scholar on the psychoanalytic view of art and artistic making – points out the necessary balance between the precision of the act, and the simultaneous pulse of life as its necessary counterpoint: 'The dutiful pianist wishes to acquire first the necessary craftsmanship for regularizing and equalizing the action of his fingers. If he ignores the spontaneous inflections of his playing he will kill the spirit of living music. He will neither listen to what his own body tells him nor respect the independent life of his work.'¹¹

Joseph Brodsky warns similarly of the negative impact of expertise: 'In reality (in art and, I would think, science), experience and the



In the CAST Building Workshop at the University of Manitoba Faculty of Architecture, Black West and his students study concrete structures cast in inexpensive canvas moulds. Fabric-cast concrete sculpture.

accompanying expertise are the maker's worst enemies.¹² The poet's warning here concerns the false preparedness and sense of certainty easily adopted by established and acknowledged expertise. The profound creative individual and craftsman approaches each task anew, and this attitude is the opposite of that of the expert.

Brodsky also emphasises the importance of the process of working over the end result in the maker's consciousness: '[N]o honest craftsman or maker knows in the process of working whether he is making or creating ... [T]he first, the second, and the last reality for him is the work itself, the very process of working. The process takes precedence over its result, if only because the latter is impossible without the former.'¹³ The poet seems

to say that although the perfection of the end result has, of course, a seminal significance for the artist and maker, it arises from and is refined by the process, instead of being a mere preconception. In the very same way, beauty or simplicity cannot be preconceived, conscious targets in artistic work; one arrives at these qualities by struggling for other ends. Brodsky, in fact, criticises Ezra Pound for this mistake of aiming at beauty directly '[H]e hadn't realized that beauty can't be targeted, that it is always a by-product of other, often very ordinary pursuits.'¹⁴ Constantin Brancusi, the master of reduction, makes exactly the same argument concerning the search for purity or simplicity: 'Simplicity is not an end of art, but one arrives at simplicity in spite of oneself, in approaching the real essence of things, simplicity is at bottom complexity and one must be nourished on its essence to understand its significance.'¹⁵ In another context the sculptor confesses: 'I never seek to make what they call a pure or abstract form. Pureness, simplicity is never in my mind; to arrive at the real sense of things is the one aim.'¹⁶

Training for a skill implies endless practice and repetition that borders upon boredom.¹⁷ However, the gradual improvement of performance, combined with dedication, keeps the negative sense of boredom at bay. In fact, the experience of slow time and boredom initiates meditative mental activity. I have personally learned to be grateful for my painfully endless days as a kid during the war years at my grandfather's small farm, and the gnawing experience of boredom resulting from the lack of external stimuli that could have been provided by friends, hobbies, entertainment, or books that, however, were not available in solitary Finnish farm life almost seven decades ago. I have become thankful for the sense of curiosity and hunger for observation evoked by the absence of early stimuli deliberately programmed and provided for me by others. As Odo Marquard notes, in today's world we have largely lost 'the art of solitude'.¹⁸

The experience of boredom in early childhood ignites imagination and initiates independent and self-motivated observation, play and imagination. This condition also guides one to realise essential causalities between things. The current tendency of parents and teachers to over-stimulate children may have catastrophic consequences for the children's capacity of imagination, invention and self-identity. In today's ordinary life, the mechanised, automated and electronic equipment and gadgets, with their invisible workings and functions, may well weaken the sense of physical causation even in adults, not to speak of the eventual impact of standard

entertainment and digital games on human and social interaction and sense of compassion.

Boredom and repetition are related, yet learning of any specialised skill calls for repetition *ad absurdum*. It seems to me that the current over-stimulated youth tends to take repetition as a mere pain. Even a slowed-down pace of events, such as watching the slow progress of Andrey Tarkovsky's films, is experienced as physically intolerable by many of today's students conditioned by the accelerated stimulation of action-cinema.

Eye, Hand and Mind

For the sportsman, craftsman, magician and artist alike, the seamless and unconscious collaboration of the eye, hand and mind is crucial. As the performance is gradually perfected, perception, action of the hand and thought lose their independence and turn into a singular and subliminally coordinated system of reaction and response. Finally it is the maker's sense of self that seems to be performing the task as if his/her existential sense exuded the work, or performance. The maker's identification with the work is complete. At its best, the mental and material flow between the maker and the work is so tantalising that the work seems to be producing itself. This is actually the essence of the ecstatic experience of a creative outburst; artists repeatedly report that they feel that they are merely recording what is revealed to them and what emerges involuntarily beyond their conscious intellectual control. 'The landscape thinks itself in me, and I am its consciousness,' Paul Cézanne confesses.¹⁹ William Thackeray pointed out the independence of his characters: 'I do not control my characters; I am in their hands and they take me wherever they want.'²⁰ As Honoré de Balzac was criticised for creating a hero that only moves from one tragic mishap to the next, he responded: 'Do not disturb me ... these people do not have any backbone at all. What happens to them is inevitable.'²¹

The union of the eye, hand and mind creates an image that is not only a visual recording or representation of the object, it *is* the object. As Jean-Paul Sartre observes: 'He [the painter] makes them [houses], that is, he creates an imaginary house on the canvas and not a sign of a house. And the house, which thus appears preserves all the ambiguity of real houses.'²²



and columns by Mark West.
'The landscape thinks itself in
me, and I am its consciousness'
Paul Cézanne, *Mont Sainte-
Victoire*, oil on canvas, 60 x 70
cm, 1904–5.

At the moment when the player of a ball game hits or catches the ball, the eye–hand–mind complex has already gone through instantaneous and unconscious computations of relative spatial positions, speeds and movements, as well as a series of strategic planning. This demanding task of fusing dimensions of time – perception, objective and response – in a split-second action, is only conceivable through assiduous practice that has succeeded in embodying the task, through making it an ingredient of the athlete's sense of self instead of confronting the situation as a detached, externalised task. In the same way the musician and the painter must fuse the actions of the eye, hand and mind into a unified and singular response. 'A universe has been narrowed into what lies at each end of a paintbrush,' as the poet Randall Jarrell remarks of the work of a mature painter.²³

When a painter, say Vincent van Gogh or Claude Monet, paints a scene, the hand does not attempt to duplicate or mimic what the eye sees or the mind conceives. Painting is a singular and integrated act in which the hand sees, the eye paints, and the mind touches. 'The hands want to see, the eyes want to caress,' as Johann Wolfgang von Goethe remarked;²⁴ or: 'Nothing escapes the great Thinker – he knows all, he sees all, he hears all. His eyes are in his ears, his ears in his eyes,' as Brancusi describes the totally focused state of his sculpture *Socrates* (1922).²⁵

Intention, perception and the work of the hand do not exist as separate entities. The sole act of painting and its very physicality and materiality is both the means and the end. 'In Monet as in other very different painters part of the object is to work until it is no longer possible to tell what paint is on top and what is underneath [...] The meadow is no longer a green card scattered with cutout plants, but a rich loam matted with plant life and moving with living shadows. Monet's texture strokes help that happen by raising glints of light that sparkle randomly among the painted stalks and leaves, confusing the eye and mimicking the hopeless chaos of an actual field,' James Elkins describes the experiential alchemy of a Monet painting.²⁶ The process, the product and the maker are fully merged: 'Like poetry or any other creative enterprise, painting is something that is worked out in the making, and the work and its maker exchange ideas and change one another ... Thoughts at the moment of beginning are only guideposts, and the actual substance of the work is entirely inchoate.'²⁷

The same fusion of the external and the internal, material and mental, thought and execution takes place in the designer's and architect's work, although their work is usually painfully prolonged, and interrupted by less creative and intimate phases. One of the most demanding requirements of an architect is the capacity to sustain a sense of inspiration and freshness of approach for several years, and sometimes through several successive alternative projects.²⁸

Elkins writes about the *materia prima* of the painter and its relationship with concepts and practices in alchemy, arguing that, 'it has to be both nothing (nothing yet, nothing that has been formed) and everything (everything in *potentia*, all the things that wait to exist) ... *Materia prima* is a name for the state of mind that sees everything in nothing.'²⁹

The painter's hand does not only reproduce the visual appearance of the object, person or event – observed, remembered or imagined – the hand perfects the impossible task of recreating the object's very essence, its sense of life, in all its sensory and sensual manifestations. The individual brushstrokes in a portrait of Rembrandt or in an Impressionist landscape do not only depict the form, colour and illumination of the object; the spots of colour, texture and light awaken the object back to full life. 'Art must give suddenly. All at once, the shock of life, the sensation of breathing,' as Brancusi states.³⁰

In addition to breathing life into the scene, a profound work projects the object's metaphysical essence, and in fact, it creates a world. 'If a painter presents us with a field or a vase of flowers, his paintings are windows which are open on the whole world,' as Jean-Paul Sartre states.³¹ This world evoked by a profound piece of art is an experientially real world. Merleau-Ponty points out the multi-dimensional and multi-sensory nature of an artistic work:

We see the depth, the smoothness, the softness, the hardness of objects; Cézanne even claimed that we see their odour. If the painter is to express the world, the arrangement of his colours must carry with it this indivisible

When looked at from a close distance, a painting turns into a meaningless chaos of colours and textures. Rembrandt van Rijn, *Self-portrait* (detail), Oil on canvas, 114 x 94 cm, 1661–2.





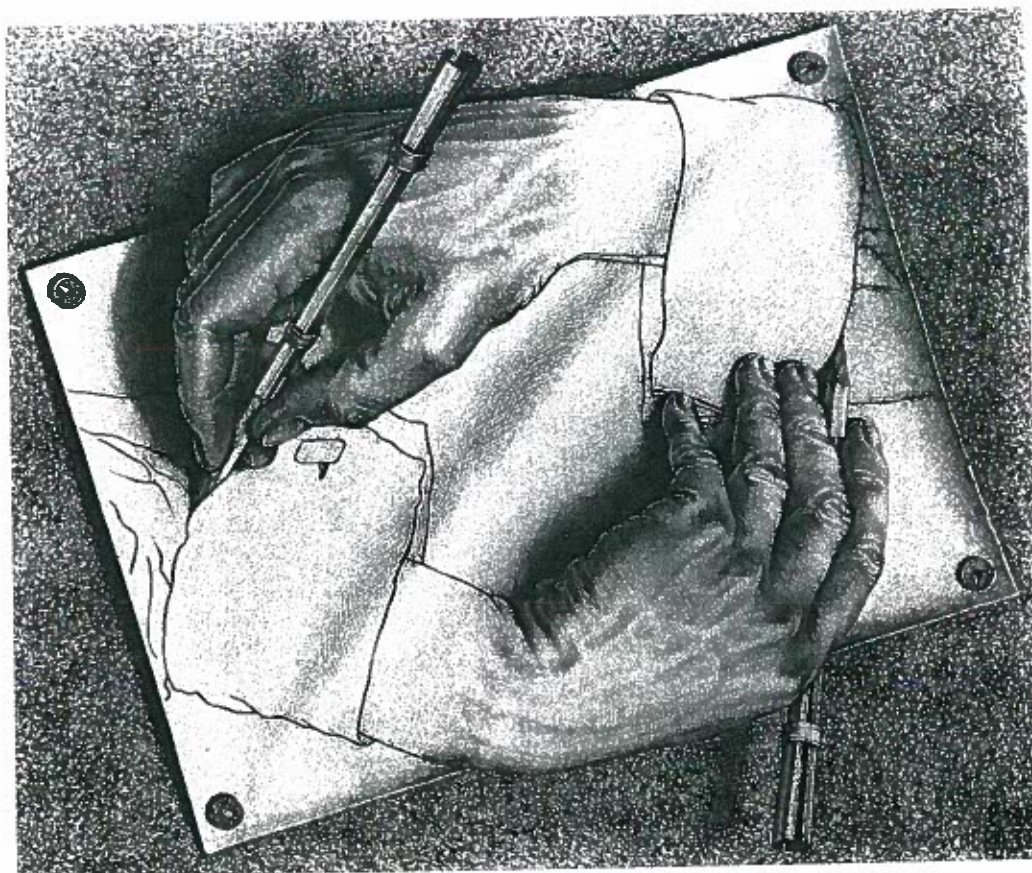
whole, or else his picture will only hint at things and will not give them in the imperious unity, the presence, the unsurpassable plenitude, which is for us the definition of the real. That is why each brushstroke must satisfy an intimate number of conditions [...] as Bernard said, each stroke must 'contain the air, the light, the object, the composition, the character, the outline, and the style'. Expressing what exists is an endless task.³²

Yet, regardless of the endlessness or logical impossibility of the artist's task, the masterpieces succeed in re-creating, not only the existence of a singular object, but the very essence of our lived world.

The Gestalt quality of Monet's paintings became increasingly vague in the paintings of his old age. Claude Monet, *The Path with Rose Trellises, Giverny*, oil on canvas, 90 x 92 cm, c. 1922. Collection of the Musée Marmottan, Paris (5088)

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- 15 Catalogue of Brancusi Exhibition, Brummer Gallery, New York, 1926. As republished in Eric Shanes, *Constantin Brancusi*, Abbeville Press (New York), 1989, p 106.
- 16 Quoted by Dorothy Dudley in 'Brancusi', *Dial* 82 (February 1927). As republished in Shanes, *Constantin Brancusi*, op cit, p 107.
- 17 The Norwegian philosopher Lars Fr H Svendsen maps the history and constantly changing meaning of boredom in his enjoyable book *Philosophy of Boredom*, Tammi (Helsinki), 2005. Basing his argument on Brodsky, the writer proposes the notion, for instance, that 'our existence is for the greater part boring, and that is why we give so much weight to originality and innovation. Nowadays we emphasize more what is "interesting" than what possesses some "value". When we regard something only on the basis of whether it is "interesting" or not, we regard it purely in an aesthetic perspective [...] . The aesthetic eye needs to be excited by ever greater intensity or, preferably, by something new; the ideology of the eye is superlative' (p 31) (translated by Juhani Pallasmaa).
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M.C. Escher, *Drawing Hands*,
lithograph, 1948. M.C. Escher
Foundation, Gemeentemuseum,
The Hague, Netherlands

4

The Drawing Hand

'[T]he art of portraiture is one of the most remarkable. It demands especial gifts of the artist, and the possibility of an almost total identification of the painter with his model. The painter should come to his model with no preconceived ideas. Everything should come to him in the same way that in a landscape all the scents of the countryside come to him: the smells of the earth, the flowers linked with the play of clouds, the movement of trees, and the different sounds of the countryside.'

Henri Matisse¹

Drawing and the Self

Sketching and drawing are spatial and haptic exercises that fuse the external reality of space and matter, and the internal reality of perception, thought and mental imagery into singular and dialectic entities. As I sketch a contour of an object, human figure or landscape, I actually touch and feel the surface of the subject of my attention, and unconsciously I sense and internalise its character. In addition to the mere correspondence of

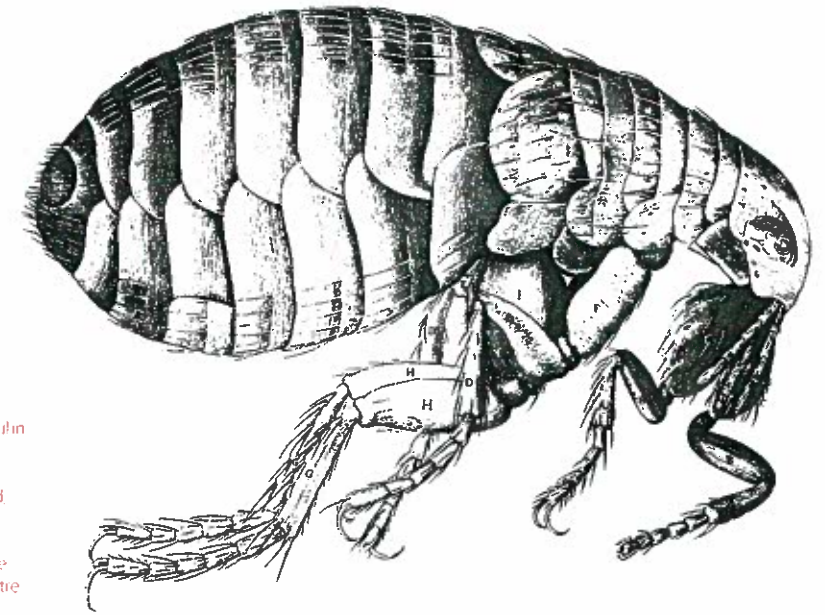
the observed and depicted outline, I also mimic the line rhythm with my muscles, and eventually the image becomes recorded in the muscular memory. In fact, every act of sketching and drawing produces three different sets of images: the drawing that appears on the paper, the visual image recorded in my cerebral memory, and a muscular memory of the act of drawing itself. All three images are not mere momentary snapshots, as they are recordings of a temporal process of successive perception, measuring, evaluation, correction and re-evaluation. A drawing is an image that compresses an entire process fusing a distinct duration into that image. A sketch is in fact a temporal image, a piece of cinematic action recorded as a graphic image.

This multiple nature of the sketch, its layered exposure, as it were, makes me remember vividly each one of the hundreds of scenes that I have sketched during fifty years of my travels around the world, whereas I can hardly recall any of the places that I have photographed as a result of the weaker embodied recording of taking a photograph. This argument does not, of course, reduce the value of the photograph as an art form in its own right, but it underlines the corporeal limitations of photography as an act of recording experiences.

In the last decades of the 19th century, at the time when photography emerged as the technique of recording and interpreting the physical and biological world, Santiago Ramón y Cajal, the father of modern neurobiology, insisted that all his students take lessons in watercolour painting and reasoned:

If our study is concerned with an object related to anatomy or natural history, etc. observations will be accompanied by sketching, for, aside from other advantages, the act of depicting something disciplines and strengthens the attention, obliging us to cover the whole of the phenomenon studied and preventing, therefore, details from escaping our attention which are frequently unnoticed in ordinary observation [...] The great Cuvier [Georges Léopold Cuvier, French naturalist and zoologist (1769–1832)] had reason to affirm that without the art of drawing, natural history and anatomy would have been impossible. It is not without reason that all great observers are skilful in sketching.²

Drawing is a process of observation and expression, receiving and giving, at the same time. It is always a result of yet another kind of double perspective;



"All great observers are skilful in sketching"
Flea, wingless bloodsucking parasitic insect, 1665. Oxford Science Archive.
The original engraving was reproduced as a folded plate measuring almost half a metre in length

a drawing looks simultaneously outwards and inwards, to the observed or imagined world, and into the draughtsman's own persona and mental world. Each sketch and drawing contains a part of the maker and his/her mental world, at the same time that it represents an object or vista in the real world, or in an imagined universe. Every drawing is also an excavation into the drawer's past and memory. John Berger describes this seminal merging of the object and the drawer him/herself: 'It is the actual act of drawing that forces the artist to look at the object in front of him, to dissect it in his mind's eye and put it together again; or, if he is drawing from memory, that forces him to dredge his own mind, to discover the content of his own store of past observations.'³

Tactility of Drawing

When sketching an imagined space, or an object being designed, the hand is in a direct and delicate collaboration and interplay with mental imagery. The image arises simultaneously with an internal mental image and the sketch mediated by the hand. It is impossible to know which appeared first, the line

on the paper or the thought, or a consciousness of an intention. In a way, the image seems to draw itself through the human hand.

John Berger points out this dialectic interaction of external and internal reality: 'Every line I draw reforms the figure on the paper, and at the same time it redraws the image in my mind. And what is more, the drawn line redraws the *model*, because it changes my capacity to perceive.'⁴ Henri Matisse makes a similar remark: 'When I paint a portrait, I come back again and again to my sketch, and every time it is a new portrait that I am painting: not one that I am improving, but a quite different one that I am beginning over again; and every time I extract from the same person a different being.'⁵ It is evident that the act of drawing mingles perception, memory and one's sense of self and life: a drawing always represents more than its actual subject matter. Every drawing is a testimony. 'A drawing of a tree shows, not a tree, but a tree-being-looked-at [. . .] Within the instant of the sight of a tree is established a life-experience.'⁶ A drawing does not reproduce the tree as it manifests itself in the objective reality; the drawing records the way the tree is seen or experienced.

The initial mental image may emerge as a visual entity, but it can as well be a tactile, muscular or bodily impression, or a shapeless feeling that the hand concretises in a set of lines projecting a shape or structure. One cannot know whether the image first arose in one's mind, and was then recorded by hand, or whether the image was produced by the hand independently, or whether it emerged as a result of a seamless collaboration of the hand and the drawer's mental space. It is often the act of drawing itself, the deep engagement in the act of unconscious thinking through making, that gives rise to an image or an idea. The second meaning of the word 'drawing' – to pull – points to this essential meaning of the drawing as a means of pulling out, revealing and concretising internal mental images and feelings as much as recording an external world. The hand feels the invisible and formless stimulus, pulls it into the world of space and matter, and gives it a shape. 'Everything his eye sees, he fingers,' John Berger comments on the tactility of a Vincent van Gogh drawing.⁷ This very act of fingering the objects of observation or dreams, intimate or remote, gives rise to the creative process.

Similarly in the act of writing, it is frequently – perhaps even most often – the process of writing itself that gives birth to unexpected ideas and an especially fluent and inspired mental flow. It is beyond doubt that the hand has a central role also in writing. But not only the hand, as even writing poetry or

Vincent van Gogh, *Olive Trees at Montmajour*, pencil, reed pen, brown and black ink on Whatman paper, 48 x 60 cm, 1888. Musée des Beaux-Arts de Tournai, Belgium.



music is an embodied and existential act. Charles Tomlinson, the poet, points out the bodily basis in the practice of painting and poetry: 'Painting wakes up the hand, draws-in your sense of muscular coordination, your sense of the body, if you like. Poetry also, as it pivots on its stresses, as it rides forward over the line-endings, or comes to rest at pauses in the line, poetry also brings the whole man into play and his bodily sense of himself.'⁸

John Berger gives a poetic description of the embodied acts, internalisations and projections that he envisions taking place in van Gogh's drawing process:

The gestures come from his hand, his wrist, his arm, shoulder, perhaps even the muscles in his neck, yet the strokes he makes on paper are following currents of energy which are not physically his and which only become visible when he draws them. Currents of energy? The energy of a tree's growth, of a plant's search for light, of a branch's need for accommodation with its neighbouring branches, of the roots of thistles and shrubs, of the weight of rocks lodged on a slope, of the sunlight, of the attraction of the shade for whatever is alive and suffers from the heat, of the Mistral from the north which has fashioned the rock strata.⁹

In Berger's description, the muscles of the artist's entire body seem to participate in the physical act of drawing, yet the act draws its energy from the subject itself. It is evident that the common understanding of drawing and painting as purely visual endeavours is completely erroneous. Due to the innate and concrete spatiality of architecture, and its irrefutable embodied and existential essence, a visual understanding of this art form is also grossly misleading.

Modernity at large has been obsessed by vision and has suppressed tactility, but many visual artists have been concerned with the tactile sense. Brancusi, for instance, exhibited his *Sculpture for the Blind* (1916) in 1917 in New York concealed in a cloth bag, so that it could only be experienced through the sense of touch.¹⁰

Echoing Brancusi's idea, Sanda Iliescu at the University of Virginia teaches drawing to students of architecture through the sense of touch. The objects that the students draw are placed inside a cubic volume made of black cloth and provided with sleeves through which they can be studied by touching

Professor Aulis Blomstedt drawing on the blackboard at the Helsinki University of Technology with his eyes blinded by a bag, presumably in the early 1960s



them with one's hand. It is remarkable that students pay attention to entirely different characteristics and qualities of objects in their drawings when observing them through their hands instead of the eyes. The drawings of touch are also distinctly different from the drawings of visual observation in their overall ambience.

The eye-hand-mind union is normally the mode of artistic making, but there have also been serious attempts to weaken or eliminate this closed circuit. My professor and mentor Aulis Blomstedt liked to draw with his eyes closed in order to eliminate the close coordination of the eye and hand. Some artists today, such as Brice Marden, draw with long sticks to emancipate the line from the strict control of the hand. Abstract Expressionists, such as Jackson Pollock and Morris Louis, had their colours spread on the canvas by means of gravity and various processes of pouring and throwing the paint rather than the visual guidance of the eye and the muscular control of the hand. Cy Twombly experimented with sketching in the dark, and for a time he also forced himself to draw with his left hand.¹¹

The Computerised Hand

It would surely be an ignorant and prejudicially Luddite view to deny the benefits of the computer. In a very short time computational technology has completely changed countless aspects of research, production and everyday life. It has also changed the architectural practice beyond return. However, at the same time that we acknowledge the benefits of the computer and associated digital technologies, we need to identify the ways in which they differ from previous instruments of design. We must consider the limitations and problems that they may pose, for instance, in the mental and sensuous aspects of the work of the architect.

The computer undoubtedly expedites most aspects of architectural production decisively and in addition to being a tool of precise and rapid drawing, it has been put to good use in analysis, testing and virtual prototyping prior to building construction. Furthermore, the computer is used to directly generate artistic, architectural and urban forms. The problems of fully computerised design are evident particularly in the most sensitive and vulnerable early phases of the design process when the architectural essence of the building is conceived and determined. The hand with a charcoal, pencil or pen creates a direct haptic connection between the object, its

representation and the designer's mind; the manual sketch, drawing or physical model is moulded in the same flesh of physical materiality that the material object being designed and the architect himself embody, whereas computer operations and imagery take place in a mathematicised and abstracted immaterial world.

My particular suspicion concerns the false precision and apparent finiteness of the computer image as compared with the natural vagueness and innate hesitancy of hand drawing that only through repetition, trial and error, and a gradually achieved assurance and precision, arrives at a satisfactory resolution. This is the innate structure of all creative endeavour as described convincingly nearly half a century ago by Anton Ehrenzweig in his two seminal studies of the creative process, *The Psycho-Analysis of Artistic Vision and Hearing* (1953)¹² and *The Hidden Order of Art* (1967).¹³ The William James sentence that Ehrenzweig quotes on the title page of his first book makes his intention very clear: 'It is, in short, the reinstatement of the vague to its proper place in mental life which I am so anxious to press on the attention.'¹⁴ Ehrenzweig refers to Jacques Hadamard's interesting suggestion 'that Greek geometry lost its creative impetus in Hellenistic times because of too precise visualization. It produced generations of clever computers and geometers, but no true geometricians. Development in geometric theory stopped altogether.'¹⁵

Precision of thought and performance, as well as emotion, is crucial, but only in a counterpoint and dialogue with embracing and vague, all-encompassing and oceanic creative imagery. The seminal role of vagueness is totally ignored in today's pedagogic philosophies and methods.

Ehrenzweig argues:

Creativity is always linked with the happy moment when all conscious control can be forgotten. What is not sufficiently realized is the genuine conflict between two kinds of sensibility, conscious intellect and unconscious intuition [...] It is not an advantage if the creative thinker has to handle elements that are precise in themselves such as geometric or architectural diagrams.¹⁶

His reluctance must surely apply to the false precision of the computer as a means of developing an idea, although this novel tool did not exist in Ehrenzweig's day. Does the computerised hand allow 'the happy moment

when all conscious control can be forgotten'? Does it permit a multi-sensory imagery and an embodied identification?

Ehrenzweig defines the cause of his suspicion for excessive precision in architectural design further:

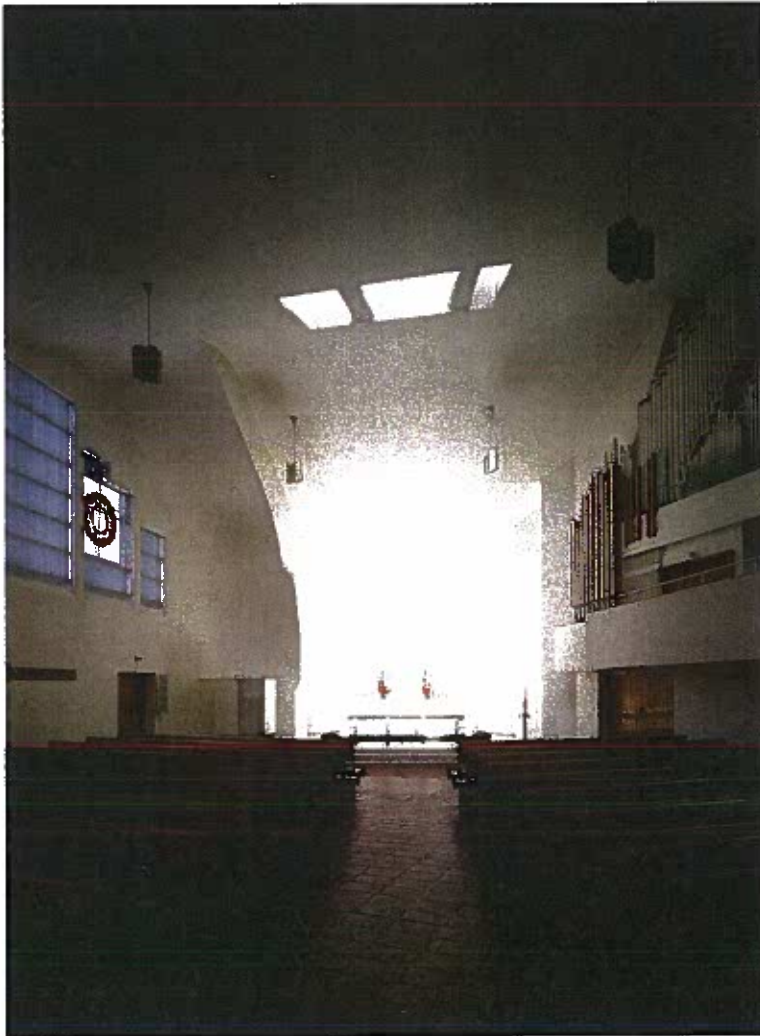
Motifs preserve their fertility only if their connection with the final result remains obscure. Otherwise they turn into mechanical assembly devices. I have mentioned how architectural design is hampered by its tendency to visualize too precisely and by its abuse of diagrammatic aids (ground plan, elevation, etc.). These visual aids seem to allow a precise presentation of the architectural problem, but in fact obscure it. It is vital for good design to break down the design process into stages that have no obvious connection with the final result.¹⁷

The computer is usually enthusiastically presented as a solely beneficial invention that liberates human fantasy. In my view, however, computer imaging tends to flatten our magnificent multi-sensory and synchronic capacity of imagination by turning the design process into a passive visual manipulation, a retinal survey. The computer creates a distance between the maker and the object, whereas drawing by hand or building a model puts the designer in skin-contact with the object or space. In our imagination we touch the designed object or space from the inside outwards, as it were. More precisely, in imagination the object is simultaneously held in the palm of the hand and inside the brain: we are inside and outside of the object at the same time. Ultimately the object becomes an extension and part of the designer's body.

In drawing by hand and pencil/pen, the hand follows the outlines, shapes and patterns of the object, whereas when drawing by mouse and computer, the hand usually selects the lines from a given set of symbols that have no analogical – or, consequently, haptic or emotional – relation to the object of drawing. Whereas the hand drawing is a mimetic moulding of lines, shades and tones, the computer drawing is a mediated construction.

My further reservation concerns the relationship between the whole and the parts that creates a natural two-way relationship and dialectic continuum in the processes of manual drawing and model making, whereas the computerised process, in its very perfection, tends to create a sense of fragmentation and discontinuity. Entities can only be grasped by suppressing and dimming details and exactitude.

Visual properties and proportions can be grasped through drawings at any scale, whereas a tactile imagination usually calls for a full-scale drawing. Although the computer drawing factually takes place in a one-to-one reality, the experiential scalelessness of the drawing and the lack of a tactile connection through the hand with imagination tend to weaken the haptic feeling of the designed entity in the computer-generated drawing.



The sensuality and plasticity of a formally exceptionally complex architectural project fully conceived through hand drawings and models
Alvar Aalto, Church of the Three Crosses, Vuoksenniska, Imatra, Finland, 1955–8

Fully computer-generated designs may well project a seductive surface appeal, but in fact they take place in a world in which the observer has no skin, hands or body. The designer himself remains an outsider in relation to his/her own design and body. Computer drawings are devices for a bodiless observer.

I for one strongly support hand drawing and working with physical models in the early phases of both design education and the work on an architectural project. In many discussions on the relationships of manual work and computerised design in various schools around the world, I have presented the view that all students of design and architecture should first be taught to work with their internalised mental imagery and their hands before they are allowed to use the computer. In my opinion, the computer probably cannot do much harm after the student has learned to use his/her imagination, and has internalised the crucial process of embodying a design task. Without this mental internalisation, however, the computerised design process tends to turn into a purely retinal journey in which the student him/herself remains an outsider and observer without having built a vivid mental model of the conceived reality. In my view, each student should pass a test that assures his/her capacity of mental imagery before being permitted the use of the computer.

Computer-aided design is usually supported by the argument that it enables the design of complex spatial, topological and formal situations that would otherwise be impossible to conceive and execute. Alvar Aalto's Church of the Three Crosses at Vuoksenniska in Imatra (1955–8) is arguably as complex in its fully moulded three-dimensional spatiality as any of today's CAD-designed buildings. However, the church is an extraordinarily plastic and sensuous building with a convincing sense of material and structural reality. It is powerfully real, and due to its strong sense of materiality and construction it addresses and awakens our bodies and imagination so gracefully. The architectural structure exists fully and comfortably in the same reality of life that our bodies occupy, not in a weightless and scaleless mathematical space. Yet the building was designed before the computer era, and in order to prepare his office for this exceptionally demanding design task Aalto sent his chief assistant in the job – Kaarlo Leppänen, a talented Finnish architect – to refresh his knowledge of trigonometry at the University of Helsinki for a couple of months.

I feel that I need to re-emphasise my point: I am not speaking against the computer. I am merely arguing that the computer is a fundamentally different tool from the traditional instruments of drawing and methods of making physical models. The line of charcoal, pencil and pen is an expressive and emotional line and so is a model crafted by human hand. They can express hesitation and assurance, judgement and passion, boredom and excitement, affection and repulsion. Every move, weight, shade, thickness and velocity of the hand-drawn line carries a particular meaning. The line traced by the hand is a spatial one: it is placed in a distinct perceptual or imagined space. In comparison to the expressive richness and emotive life of the hand-drawn line, the computer line is a laconic and uniform connection of two points (computer lines can, of course, be articulated to simulate lines drawn by hand, but their essence is the emotionless factuality of mathematicised space).

Primacy of Touch: Hapticity of Self-image

The boundary line between the self and the world is identified by our senses. Our contact with the world takes place through the skin of the self by means of specialised parts of our enveloping membrane. All the senses, including vision, are extensions of the tactile sense; the senses are specialisations of skin tissue, and all sensory experiences are modes of touching, and thus related to tactility. 'Through vision we touch the sun and the stars,' as Martin Jay remarks poetically in reference to Merleau-Ponty's philosophy.¹⁸ This fundamental hapticity of the human life world heightens the significance of the hand.

In their book *Body, Memory and Architecture*, one of the early studies of the embodied essence of architectural experience, Kent C Bloomer and Charles W Moore point out the primacy of the haptic realm: 'The body image [...] is informed fundamentally from haptic and orienting experiences early in life. Our visual images are developed later on, and depend for their meaning on primal experiences that were acquired haptically.'¹⁹ The view of the anthropologist Ashley Montagu, based on medical evidence, confirms the primacy of this haptic realm:

[The skin] is the oldest and the most sensitive of our organs, our first medium of communication, and our most efficient protector [...] Even the transparent cornea of the eye is overlain by a layer of modified skin [...] Touch is the

parent of our eyes, ears, nose, and mouth. It is the sense which became differentiated into the others, a fact that seems to be recognized in the age-old evaluation of touch as 'the mother of the senses'.²⁰

Touch is the sensory mode that integrates our experiences of the world and of ourselves. Even visual perceptions are fused and integrated into the haptic continuum of the self; my body remembers who I am and how I am situated in the world. In the first volume of Marcel Proust's *In Search of Lost Time*, the protagonist, waking up in his bed, reconstructs his identity and location through the memory of his body:

My body, still too heavy with sleep to move, would endeavour to construe from the pattern of its tiredness the position of its various limbs, in order to deduce therefrom the direction of the wall, the location of the furniture, to piece together and give a name to the house in which it lay. Its memory, the composite memory of its ribs, its knees, its shoulder-blades, offered it a whole series of rooms in which it had at one time or another slept, while the unseen walls, shifting and adapting themselves to the shape of each successive room that it remembered, whirled around it in the dark. And even before my brain, hesitating at the threshold of times and shapes, had reassembled the circumstances sufficiently to identify the room, it – my body – would recall from each room in succession the style of the bed, the position of the doors, the angle at which the sunlight came in at the windows, whether there was a passage outside, what I had had in mind when I went to sleep and found there when I awoke.²¹

We are here encountering a composite experience that brings to mind a fragmented and re-composed Cubist composition.

My body is truly the navel of my world, not in the sense of the viewing point of a central perspective, but as the sole locus of reference, memory, imagination and integration.

Unconscious Touch in Artistic Experience

We are not usually aware that an unconscious experience of touch is unavoidably concealed in vision. As we look, the eye touches, and before we see an object, we have already touched it and judged its weight, temperature and surface texture. The eye and the hand constantly



collaborate; the eye carries the hand to great distances, and the hand informs the eye at the intimate scale. Touch is the unconsciousness of vision, and this hidden tactile experience determines the sensuous qualities of the perceived object. This is the hidden element in touching and the activation of tactile judgement and memory that is involved in drawing. The sense of touch mediates messages of invitation or rejection, nearness or distance, pleasure or repulsion. It is exactly this unconscious dimension of touch in vision that is disastrously neglected in today's visually biased hard-edged architecture and design. Our architecture may entice and amuse the eye, but it does not provide a domicile for the touch of our bodies, memories and dreams.

Drawing, and especially painting, is not solely a matter of recording the visual essence of the scene; the apparently visual percept conveys the entire sensual essence of the thing. In the 1890s the American art critic and writer Bernard Berenson developed further Goethe's notion of 'life-enhancing', and he suggested that when experiencing an artistic work,

Louis I Kahn, Salk Institute for Biological Studies, La Jolla, California, USA, 1959–65

we actually imagine a genuine physical encounter through 'ideated sensations'. The most important of these Berenson called 'tactile values'.²² In his view, the work of authentic art stimulates our ideated sensations of touch, and this stimulation is life-enhancing.

A fine architectural work generates similarly an indivisible complex of impressions, or ideated sensations, such as experiences of movement, weight, tension, structural dynamics, formal counterpoint and rhythm, which become the measure of the real for us. When entering the extraordinary space of the marble-paved courtyard at the Salk Institute (1959–65) in La Jolla, California, designed by Louis Kahn, delineated by two rows of buildings, with the sky as its sublime ceiling and the horizon of the Pacific Ocean as its hypnotising back wall, I felt immediately compelled to walk to the nearest concrete wall surface and sense its temperature, the suggestion of silk and live skin was overpowering. Louis Kahn actually sought the grey softness of 'the wings of a moth' and added volcanic ash to the concrete mix in order to achieve this extraordinary inviting matt softness.²³ The architectural ensemble succeeds in fusing the immensity of the setting and the intimacy of the touch of the hand into a single experience.

The true mastership of Kahn was to turn architectural schemes that appear simplistic or even dull – such as Kimbell Art Museum (1966–72) in Fort

The door handle or pull is the handshake of the building
Juhani Pallasmaa, cabinet door pulls generated by the grip of three fingers (thumb, index and middle finger), 1991. Cast bronze.



Worth, Texas, and the Yale Center for British Art (1969–74) in New Haven, Connecticut – into magical pieces of complexity and subtlety, materiality and light, gravitas and levitation.

Pleasurable objects and buildings mediate an experience of the processes by which the object or structure was made; in a way, they invite the viewer/user to touch the hand of the maker. The door pull is one of the details of any building that call for close ergonomic attention and provide an opportunity for a nearly physical contact between the architect's hand and the hand of the occupant through the mediation of this object. The door pull or handle of the main door is the handshake of the building, and the pulling of the door with one's body weight is often the most intimate encounter with an architectural structure.

True architectural quality is manifested in the fullness and unquestioned dignity of the experience. A resonance and interaction takes place between space and the experiencing person; I set myself in the space and the space settles in me. This is the 'aura' of artistic work observed by Walter Benjamin.²⁴

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- 2 As quoted in J Allan Hobson, *The Dreaming Brain*, Basic Books (New York), 1988, pp 95–7. Republished in William Irwin Thompson, 'An introduction to "What Am I Doing in Österfärnebo?" by Cornelia Hesse-Honegger', *Cornelia Hesse-Honegger, After Chernobyl*, Verlag Hans Müller (Baden), 1992, p 16.
- 3 John Berger, *Berger On Drawing* (edited by Jim Savage), Occasional Press (Aghabullogue, County Cork), 2007, p 3.
- 4 *Ibid*, p 112.
- 5 Henri Matisse, 'Looking at life with the eyes of a child' (1953), in Flam (ed), *Matisse on Art*, op cit, p 149.
- 6 Berger, *Berger on Drawing*, op cit, p 71.
- 7 *Ibid*, p 16.
- 8 Charles Tomlinson, 'The poet as painter', in JD McClatchy (ed), *Poets on Painters*, University of California Press (Berkeley, Los Angeles and London), 1990, p 280.
- 9 Berger, *Berger on Drawing*, op cit, p 14.
- 10 The sculptor's friend Henri-Pierre Roché reports 'it was exhibited [...] enclosed in a bag that had two sleeves through which the hands could enter.' Eric Shanes, *Constantin Brancusi*, Abbeville Press (New York), 1989, p 74.
- 11 Richard Lacayo, 'Radically retro', *Time*, Vol 172, No 6, 2008, p 47.
- 12 Anton Ehrenzweig, *The Psycho-Analysis of Artistic Vision and Hearing: An Introduction to a Theory of Unconscious Perception* (first published 1953), Sheldon Press (London), third edition, 1975.
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- 15 Ehrenzweig, *The Hidden Order of Art*, op cit, p 58.
- 16 *Ibid*, p 57.
- 17 *Ibid*, p 66.
- 18 As quoted in David Michael Levin (ed), *Modernity and the Hegemony of Vision*, University of California Press (Berkeley and Los Angeles), 1993, p 14.
- 19 Kent C Bloomer and Charles W Moore, *Body, Memory and Architecture*, Yale University Press (New Haven, Connecticut and London), 1977, p 44.
- 20 Ashley Montagu, *Touching: The Human Significance of the Skin*, Harper & Row (New York), 1971, p 3.
- 21 Marcel Proust, *In Search of Lost Time: Swann's Way* (translated by CK Scott Moncrieff and Terence Kilmartin), Random House (London), 1992, pp 4–5.
- 22 Bernard Berenson, *Aesthetics and History*, Pantheon Books (New York), 1948, pp 66–70. Somewhat surprisingly, Merleau-Ponty objects strongly to Berenson's view: 'Berenson spoke of an evocation of tactile values, he could hardly have been more mistaken: painting evokes nothing, least of all the tactile.
- What it does is much different, almost the inverse; thanks to it we do not need a "muscular sense" in order to possess the voluminosity of the world [...] The eye lives in this texture as a man lives in his house' (Maurice Merleau-Ponty, 'Eye and Mind', *The Primacy of Perception*, Northwestern University Press (Evanston, Illinois), 1964, p 166). I can only understand this argument of the philosopher and agree with it when I assume that he is critical of the idea of a mediated or secondary sensation instead of regarding all sense perceptions as equally real. Experiencing the temperature and moisture of air and hearing the noises of carefree daily life in the erotically sensuous paintings of Matisse or Bonnard, one appreciates the reality of the multiplicity of sensations conveyed by a profound painting.
- 23 As quoted in Scott Poole, *Pumping Up: Digital Steroids and the Design Studio*, unpublished manuscript, 2005.
- 24 'The aura of an artwork is an effect of its tradition: this presents the artwork as present and absent, unique to the here and now, and yet distant, and elsewhere' Howard Caygill, 'Benjamin, Heidegger and the destruction of tradition', in Andrew Benjamin and Peter Osborne (eds), *Walter Benjamin's Philosophy: Destruction and Experience*, Routledge (London and New York), 1994, p 23.



The rhythmic organisation of stepping stones addresses directly the body and muscular sense without an intellectual content or mediation. The sawatari-ishi, 'steps across the marsh', in the garden of the Heian Shrine in Kyoto, Japan.

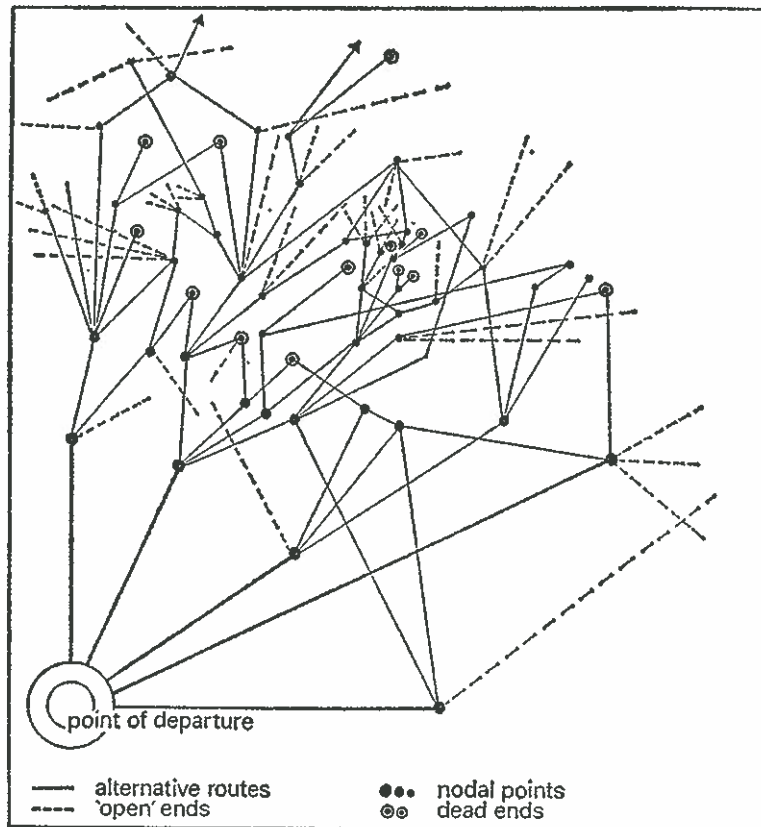
Embodied Thinking

'Rodin's hands were his principal tools, and with them he plopped and punched and gouged and smoothed, making both curves and straight lines wavy, allowing shoulders to flow into torsos and torsos to emerge from blocks (even when they hadn't), encouraging elbows to establish their own identity, his fingers everywhere busy at fostering the impressions of life, giving strength and will to plaster, ethereality and spirit to stone.'

William H Gass'

Creative Fusion

A creative insight in architecture is rarely an instantaneous intellectual discovery that could reveal a complex entity in its complete and finite resolution in a moment; neither is it a linear process of logical deduction. Most often the process begins with an initial idea that is developed for a while, but soon the concept branches out to new paths, and this pattern of criss-crossing trajectories grows ever denser through the process itself. Design is a process of going back and forth among hundreds of



Anton Ehrenzweig's chart for creative scanning. 'The maze (serial structure) of a creative search. The creative thinker has to advance on a broad front keeping open many options. He must gain a comprehensive view of the entire structure of the way ahead without being able to focus on any single possibility.' reads his caption for the chart in *The Hidden Order of Art* (1973). In my view, the creative maze is even more complex than Ehrenzweig's chart due to repeated turning back to already passed phases or rejected ideas, and entirely new beginnings.

ideas, where partial solutions and details are repeatedly tested in order to gradually reveal and fuse a complete rendition of the thousands of demands and criteria, as well as the architect's personal ideals of coordination and harmonisation, into a complete architectural or artistic entity. An architectural project is not only a result of a problem-solving process, as it is also a metaphysical proposition that expresses the maker's mental world and his/her understanding of the human life world. The design process simultaneously scans the inner and the outer worlds and intertwines the two universes.

More often than not, the initial idea and first elaboration of the scheme have to be abandoned and the entire process started anew. This is a search in the obscurity and darkness of uncertainty, in which a subjective certainty

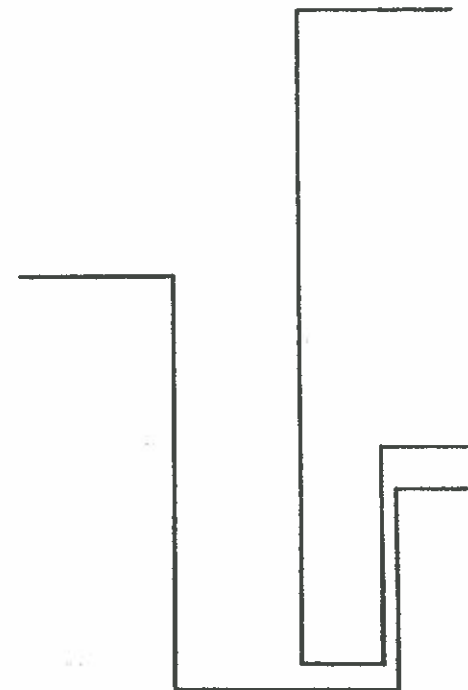
is gradually achieved through the laborious process of the search itself. This search is as much an embodied and tactile journey, guided by the hand and feelings of the body, as it is a visual and intellectual enterprise. An architectural task is not a simple logistical or rational problem to be solved. In architectural design both the appropriate end and the means have to be identified and concretised. In addition to resolving rational problems and fulfilling functional, technical and other demands, profound architecture is always expected to evoke human, experiential and existential values that cannot be prescribed. Every true piece of architecture relocates man in the world and casts some new light on man's existential enigma. Every architectural task that is taken seriously also calls for a distinct idealisation of the situation, the client and the future use of the building. Architecture needs to build a better world, and this projection of an idealised human dimension calls for an existential wisdom rather than professional expertise, skill and experience. In fact a design task is an existential exploration in which the architect's professional knowledge, life experiences, ethical and aesthetic

sensibilities, mind and body, eye and hand, as well as his/her entire persona and existential wisdom eventually merge.

The Work of Thinking: The Value of Uncertainty

Creative thinking is work, labour, in the proper meaning of the word, rather than merely an unexpected and effortless flash of insight. Such miracles may only happen to a true genius, but even in such a case the genius has laboriously worked his/her way to the threshold of realisation. Work is usually a sweaty and messy business. I personally want to see the traces, stains and dirt of my work, the layering of erased lines, errors and failures, the repeated re-tracings on the drawing, and the collage of corrections, additions and eliminations on the page that I am writing, for as long as I am developing an idea. These traces help me to feel the continuity and purposefulness of the work, to dwell in the

Erased lines of sketching are part of the final drawing, and they reveal the sequence of trial and error and suggest a dimension of time and spatial depth.
Juhani Blomstedt: *Drawing*, charcoal on paper, 106 x 76 cm, 1985.





The dirt of work on a draughtsman's hand

work, and to grasp the multiplicity, the plasticity as it were, of the task. They also help me to maintain the mental state of uncertainty, hesitation and undecidedness needed in the process for long enough. A sense of certainty, satisfaction and finality that arises too early can be catastrophic. The hesitancy of the drawing itself expresses and maintains my own inner uncertainty. Most importantly, the sense of uncertainty maintains and stimulates curiosity. As long as uncertainty is not permitted to escalate into hopelessness and depression, it is a driving force and source of motivation in the creative process. Design is always a search for something that is unknown

in advance, or an exploration into an alien territory, and the design process itself, the actions of the searching hands, need to express the essence of this mental journey.

Joseph Brodsky points out the value of insecurity and uncertainty for the creative endeavour. His perceptive and ethically uncompromising views of the poet's task have taught me a great deal concerning the architect's mission. 'In the business of writing, what one accumulates is not expertise but uncertainties,'² the poet confesses, and I feel that a sincere architect likewise ends up accumulating uncertainties. Brodsky connects uncertainty with a sense of humility: 'Poetry is a tremendous school of insecurity and uncertainty [...] [P]oetry – writing it as well as reading it – will teach you humility, and rather quickly at that. Especially if you are both writing and reading it.'³ This observation surely applies to architecture as well, as it can be particularly humbling if you are both making it and theorising about it. But the poet suggests that these mental states which are usually considered detrimental can actually be turned into a creative advantage: 'If this [uncertainty or insecurity] does not destroy you, insecurity and uncertainty in the end become your intimate friends and you almost attribute to them an intelligence all their own,' the poet advises.⁴

Billy Collins, another poet, explains why he insists on writing with pen or pencil rather than keyboard: 'I always compose by either pen or pencil, only because the keyboard, to me, makes everything kind of look done, look frozen, and writing on a page gives me a feeling of fluidity, that what I'm writing is provisional for the moment. And also since I don't know where the poem is going and I don't want to know until I get there, I always feel like the poem, as I'm writing, it is working toward some kind of understanding of itself.'⁵

I personally share the poets' views. In both writing and drawing, the text and image need to be emancipated from a preconceived sense of purpose, goal and path. When one is young and narrow-minded, one wants the text and drawing to concretise a preconceived idea, to give the idea an instant and precise shape. Through a growing capacity to tolerate uncertainty, vagueness, lack of definition and precision, momentary illogic and open-endedness, one gradually learns the skill of cooperating with one's work, and allowing the work to make its suggestions and take its own unexpected turns and moves. Instead of dictating a thought, the thinking process turns into an act of waiting, listening, collaboration and dialogue. The work becomes a journey

that may take one to places and continents which one has never visited before, or whose existence has been unknown prior to having been guided there by the work of one's own hand and imagination, and one's combined attitude of hesitation and curiosity.

There is an inherent opposition between the definite and the indefinite in art. An artistic phenomenon wants to escape definition until it has reached its self-sufficient existence – and even beyond that point, I believe. Simply expressed, true creative fusion always achieves more than can be projected by any theory, and profound design always achieves more than the brief or anyone participating in the process could anticipate.

I have to confess personally that ever since the foolishly self-assured days of my youth (that – of course disguised – genuine uncertainty, narrowness of understanding, and short-sightedness), my sense of uncertainty has grown constantly to the degree that it has become nearly intolerable. Every issue, every question, each detail, is so deeply embedded in the mysteries of human existence that often there does not seem to exist a satisfactory answer or response at all. In a fundamental sense, I can say that by age and experience one becomes increasingly more an amateur, rather than turning into a professional possessing immediate and assured responses. An established and successful professional would hardly stop to ponder questions such as, what is the floor, the window, or the door. But can anyone really tell me what are the fundamental metaphysical essences of these architectural events, and their human significance, outside and before a specific design task?

Resistance, Tradition and Freedom

A word that one hears rather often in the studios of schools of architecture, and in juries of architectural competitions, is 'Freedom'. The word seems to describe an artistic independence of the project. Independence from tradition and precedents, structural or material constraints, or sheer reason, is usually seen as a dimension of artistic freedom. Yet already Leonardo da Vinci taught us that 'strength is born from constraints and it dies in freedom'.⁶

It is thought-provoking, indeed, that great artists of any era rarely speak of the dimension of freedom in their work. They emphasise the role of restrictions and constraints in their materials and artistic medium, the cultural and social situation, and the shaping of their personality and style. The

greatness of an artist arises from the identification of his/her own territory and personal limits, rather than an indeterminate desire for freedom. Instead of longing for freedom, they emphasise the disciplined, tradition-bound character of their art form. In his memoirs *My Life and My Films*, Jean Renoir writes about the 'resistance of technique'⁷ in filmmaking, while Igor Stravinsky speaks of 'the resistance of material and technique'⁸ as important counter-forces in his work as a composer. Stravinsky scorns any yearning for freedom: 'The ones who try to avoid subordination support unanimously the opposite counter-traditional view. They reject constraint and they nourish hope – always doomed to failure – of finding the secret of strength in freedom. They do not find anything but the arbitrariness of freaks and disorder, they lose all control, they go astray [...]'.⁹ Stravinsky, the arch-modernist of music, argues forcefully that artistic strength and meaning can only derive from tradition. In his view, an artist who deliberately seeks novelty is trapped in his very aspiration: 'His art becomes unique, indeed, in the sense that its world is totally closed and it does not contain any possibility for communication.'¹⁰ The composer holds the concept of tradition so central an ingredient to art that he concludes with the Catalan philosopher Eugeni d'Ors's enigmatic statement, 'Everything that remains outside of tradition is plagiarism.'¹¹

Limits and restrictions are equally important in all arts. Paul Valéry, the poet, states unambiguously: 'The greatest liberty is born of the greatest rigour.'¹² In his book *The Power of Limits* – which studies proportional harmony in nature, art and architecture, and especially the repeated occurrence of the Golden Section in these phenomena – György Doczi notes: '[I]n our fascination with our powers of invention and achievement we have lost sight of the power of limits.'¹³ This is a seminal wisdom in our age – an age that seems to be neglecting the importance of limits.

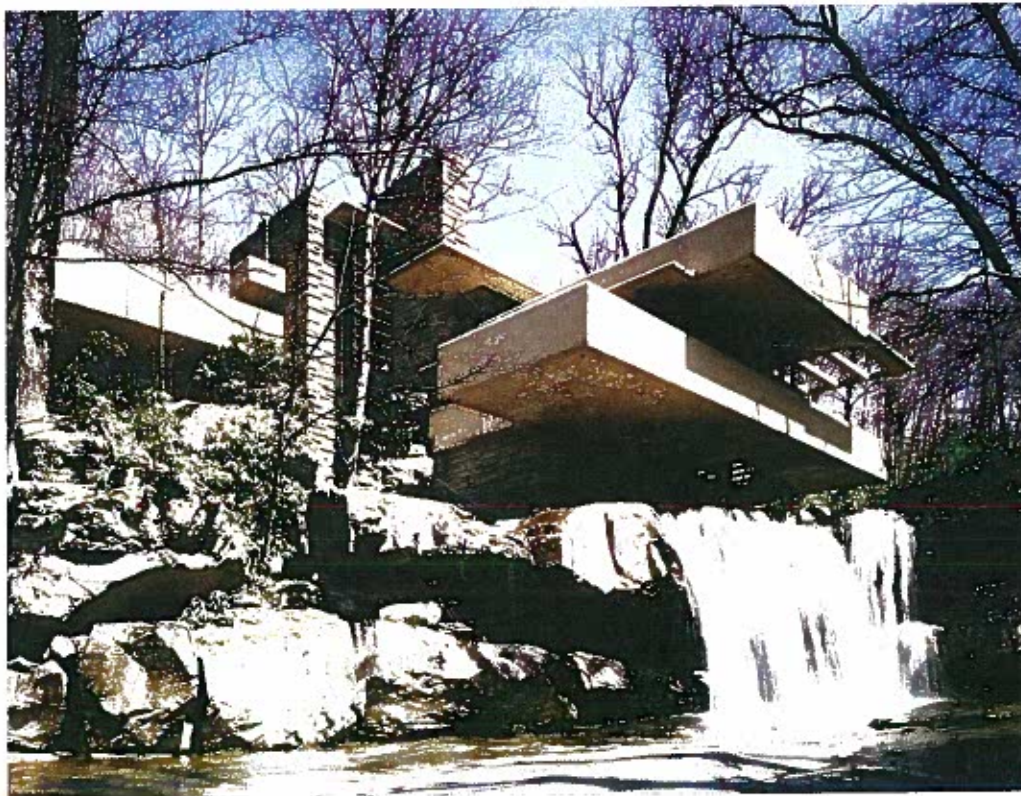
By rejecting the wisdom and resistance of tradition, architecture also drifts towards a deadening uniformity on the one hand, and towards a rootless anarchy of expression on the other. Every art form has its ontology as well as its characteristic field of expression, and limits are posed by its very essence, inner structures and materials. Generating architectural expression from the unquestionable realities of construction is the long tradition of the art of architecture. The tectonic language of architecture, the inner logic of construction itself, expresses gravity and structure, the language of materials as well as processes of construction and details of joining units and materials to one another. In my view, architecture arises from the identification and articulation of the realities of the task in question, rather than from individual

fantasy. Aulis Blomstedt used to advise his students at the Helsinki University of Technology wisely: 'The capacity to imagine situations of life is a more important talent for an architect than the gift of fantasising space.'¹⁴

Thinking Through the Senses

All significant architecture is the result of serious thinking – or, more precisely, of a distinct way of thinking through the medium of architecture. In the same way that cinema is a mode of cinematic thinking, painting a means of articulating painterly ideas, and sculpture a way of elaborating and expressing sculptural thoughts, architecture is a means of philosophising about the world and human existence through the embodied material act of constructing. Architecture develops existential

Architecture creates existential metaphors through space, structure, matter, gravity and light. Great buildings are also icons of life and metaphysical mandalas.
Frank Lloyd Wright, Fallingwater, Edgar J Kaufmann House, Mill Run, Pennsylvania, USA 1934–7.



and lived metaphors through space, structure, matter, gravity and light. Consequently, architecture does not illustrate or mimic ideas of philosophy, literature, painting or any other art forms; it is a mode of thinking in its own right. The ideas articulated by the arts are painterly, musical, cinematic or architectural thoughts, conceived and expressed through the inherent medium and artistic logic of the particular art form in a dialectical process with its tradition. Artistic ideas are not necessarily ideational and translatable into verbal terms, as they are embodied metaphors of the world and of the particular ways we exist in this world. Also architecture is an artistic expression as far as it transcends its purely utilitarian, technical and rational realm, and turns into a metaphoric expression of the lived world and the human condition.

There is a rather widely accepted view that wants to get rid of the boundaries between various art forms entirely. I for one feel that the ontological differences between various arts are as significant to acknowledge as the commonalities or shared ground of the arts. Every art form has its origins and traditions, and when this ontological backbone of a discipline is lost, the art form weakens – at least if we believe the testimony of Ezra Pound, the arch-modernist poet, in his book *ABC of Reading*: '[M]usic begins to atrophy when it departs too far from dance [...] poetry begins to atrophy when it gets too far from music [...]'.¹⁵

In my view, architecture turns similarly into mere aesthetics when it departs from its originary motives of domesticating space and time, an animistic understanding of the world, and the metaphoric representation of the act of construction. Every art form needs to be reconnected with its ontological essence, particularly at periods when the art form tends to turn into an empty aestheticised mannerism. The architectural works of the modern era, as well as of our time, that echo the tremors of origins – such as the works of Sigurd Lewerentz, Louis Kahn, Aldo van Eyck and Peter Zumthor, for example – project an authoritative radiance and depth of feeling. Such works are not always necessarily aesthetically polished as they pose a deep and disturbing emotive power and open up questions rather than provide well-formulated answers. Frank Lloyd Wright's Fallingwater (1934–7) in Mill Run, Pennsylvania, or any other masterpiece of architecture, opens up a new horizon to human existence instead of providing an answer to any question.

Louis Kahn preached the importance of beginnings: 'The spirit of the start is the most marvellous moment at any time for anything. Because in the start

lies the seed for all things that must follow. A thing is unable to start unless it can contain all that ever can come from it. That is the characteristic of a beginning, otherwise it is no beginning – it is a false beginning.¹⁶

Our entire bodily constitution and senses ‘think’ in the fundamental sense of identifying and processing information about our situation in the world, and mediating sensible behavioural responses. In the view of Henry Plotkin, professor of psychobiology, knowledge signifies more than consciously known words or facts: ‘Knowledge is any state in an organism that bears a relationship to the world.’¹⁷ The dancer and the soccer player ‘think’ with their body and legs, the craftsman and sculptor with their hands, and composers with their ears. In fact, our entire body and existential sense participate in all processes of thinking. ‘The dancer has his ear in his toes,’ as Nietzsche argues.¹⁸

In his essay ‘What calls for thinking?’, Martin Heidegger relates thinking with the art of cabinet-making. The philosopher gives the hand an essential role in the processes of thinking and he links the hand with the capacity to speak, a theme that was already discussed in earlier chapters of this book:

Perhaps thinking, too, is just something like building a cabinet. At any rate, it is a craft, a ‘handicraft’, and therefore has a special relationship to the hand. In the common view, the hand is part of our bodily organism. But the hand’s essence can never be determined, or explained, by its being an organ which can grasp. [...] The hand is infinitely different from all the grasping organs, [...] different by an abyss of essence. Only a being who can speak, that is, think, can have hands and can handily achieve works of handicraft.¹⁹

An artistic thought is not merely a conceptual or logical deduction, it implies an existential understanding and a synthesis of lived experience that fuses perception, memory and desire. Perception fuses memory with the actual percept, and consequently, even ordinary sense perceptions are complex processes of comparison and evaluation.

Embodied Memory and Thought

Merleau-Ponty extends the idea of the processes of embodied thinking to include the entire body as he argues: ‘The painter “takes his body with him” (says Paul Valéry). Indeed we cannot imagine how a mind could paint.’²⁰ It is

surely equally unthinkable that a mind could conceive architecture because of the irreplaceable role of the body in the very constitution of architecture. Buildings are not abstract, meaningless constructions, or aesthetic compositions, they are extensions and shelters of our bodies, memories, identities and minds. Consequently, architecture arises from existentially true confrontations, experiences, recollections and aspirations.

The most abstract of tasks would become nonsensical when detached from its ground in human embodiment. Even abstract art articulates the ‘flesh of the world’,²¹ and we share that very flesh, as well as the gravitational reality of the world, with our bodies. ‘The mind is not merely embodied, but embodied in such a way that our conceptual system draws largely upon the commonalities of our bodies and of the environments we live in,’ as the authors of *Philosophy in the Flesh* argue.²² We are occupants of this world with its physical realities and mental mysteries, not outside observers or theoreticians of the world.

The body is also part of our system of memory. Philosopher Edward S Casey, who has written seminal phenomenological studies on place, memory and imagination, points out the role of the body in the act of memorising: ‘Body memory is [...] the natural center of any sensitive account of remembering.’²³ In another context he elaborates his view: ‘There is no memory without body memory [...] In claiming this, I do not mean to say that whenever we remember, we are in fact directly engaging in body memory. [...] Rather, I am saying that we could not remember [...] without having the capacity for body memory.’²⁴ Furthermore, there are recent philosophical studies, such as *The Body in the Mind* by Mark Johnson, and *Philosophy in the Flesh* by Johnson and George Lakoff, that argue emphatically for the embodied nature of thinking itself.²⁵

In my own collaboration with painters, sculptors and craftsmen extending over four decades, I have learnt to admire their capacity to grasp essences of things through their hands and bodies, and through their non-conceptualised existential understanding rather than through intellectual and verbal analyses. They rely on the silent wisdom of the body and the hand. I have also had the chance to observe that the hand and the body produce distinctly different ideas than the head. The latter tend to be conceptual, intellectual, and geometricised ideas whereas the former usually project a spontaneity, sensuality and tactility. The hand registers and measures the pulse of lived reality.

An embodied manner of learning and maintaining skills as well as responding to life situations is the dominant mode of knowledge also in traditional societies. Learning a skill is primarily a matter of embodied muscular mimesis acquired through practice rather than conceptual or verbalised instruction. I cannot personally recall much talking in my youth at my grandfather's farm; everyday life and work took place 'in the flesh' of farm life, everyone knew his or her place in the family and the cycles of daily work, and everyone learnt and remembered countless practical skills as embodied patterns of life and work itself. I cannot recall anyone ever asking another whether he/she could do a certain thing; it was naturally assumed that everyone could do everything required in the tasks of daily farm life. The farmer's knowledge was constituted of crucial embodied skills that were coded into the seasons and cycles of the year and the concrete situations of daily life rather than books and notes.

Existential Knowledge

The prevailing view in our culture makes a fundamental distinction between the worlds of science and art. Science is understood to represent the realm of normality, and of rational and objective knowledge, whereas art stands for the world of subjective, emotional and essentially irrational sensations. The first is understood to possess an instrumental and operational value, whereas the second is seen as a form of exclusive cultural entertainment.

In an interview in 1990 concerning complexities and mysteries of new physics, Steven Weinberg, who won the Nobel Prize in Physics in 1979 for his discovery of the relationship between electromagnetism and the weak nuclear force, was asked: 'Whom would you ask about the complexity of life. Shakespeare or Einstein?' The physician answered immediately: 'Oh, for the complexity of life, there's no question – Shakespeare.' And the interviewer continued: 'And you would go to Einstein for simplicity?' 'Yes, for a sense of why things are the way they are – not why people are the way they are, because that's the end of such a long chain of inference ...'²⁶

Art articulates our existentially essential experiences, but as has been pointed out in the previous chapter, it also represents particular modes of thinking. Reactions to the world and processing of information take place directly as an embodied and sensory activity without being turned into concepts or even entering the sphere of consciousness.

It is evident that we need to rethink some of the very foundations of architectural experience and making in the light of these arguments. In addition to balancing the visual bias in architectural thinking, we need to be critical of approaching architecture with an intellectual and logistical emphasis. A wise and mature architect works with his/her entire body and sense of self. While working on a building or an object, the architect is simultaneously engaged in the reverse perspective of his/her self-image in relation to the world, and his/her existential knowledge. In addition to *operative and instrumental knowledge* and skills, the designer and the artist need *existential knowledge* moulded by their experiences of life. Existential knowledge arises from the way the person experiences and expresses his/her existence, and this knowledge provides the most important context for ethical judgment. In design work, these two categories of knowledge merge, and as a consequence, the building is a rational object of utility and an artistic/existential metaphor at the same time.

All professions and disciplines contain both categories of knowledge in varying degrees and configurations. The instrumental dimensions of a craft can be theorised, researched, taught and incorporated in the practice fairly rationally, whereas the existential dimensions are integrated within one's own self-identity, life experience and ethical sense as well as one's personal sense of mission. The category of existential wisdom is also much more difficult to teach, if not outright impossible. Yet, it is the irreplaceable condition for creative work. It is thought-provoking, indeed, to recognise that in most countries there is hardly any formal academic education for poets and novelists; their work is so strongly based on existential knowledge that these artists are expected to emerge and grow without explicit pedagogically formalised education.

The teaching of existential wisdom in education takes place primarily through the growth of one's personality, which is often a reflection of the teacher's persona and character on the self-identity of the student. This life wisdom is a slow accumulation of experience, a gradual maturation of personality and an internalisation – I would again use the word embodiment – of a sense of responsibility and ambition. By ambition I am not referring to social aspirations or goals, but to one's internal sense of responsibility and honour, and the willingness to cross limits of one's prior skills and knowledge.

Heidegger considers teaching even more difficult than learning: 'Teaching is even more difficult than learning [...] Not because the teacher must have a larger store of information, and have it always ready. Teaching is more difficult than learning because what teaching calls for is this: to let learn. The real teacher, in fact, lets nothing else be learned than – learning.'²⁷ The difficulty of teaching concerns especially the task of teaching existential wisdom.

Rainer Maria Rilke provides a moving and poetic description of the existential knowledge required for the writing of a single line of verse:

For verses are not, as people imagine, simply feelings [...] they are experiences. For the sake of a single verse, one must see many cities, men and things, one must know the animals, one must feel how the birds fly and know the gesture with which the little flowers open in the morning. [...] And still it is not yet enough to have memories. One must be able to forget them when they are many and one must have the great patience to wait until they come again. For it is not yet the memories themselves. Not till they have turned to blood within us, to glance and gesture, nameless and no longer to be distinguished from ourselves – not till then can it happen that in a most rare hour the first word of a verse arises in their midst and goes forth from them.²⁸

The qualifications for writing a line of verse listed by one of the finest poets of all time should surely humble anyone seeking to become a poet, artist or architect.

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A skilled soccer player internalises the entire complex situation of other players and the ball as well as their current and expected movements on the field into a single unconscious percept, and responds to that instinctively

Body, Self and Mind

‘During the sittings his eye sees far more than he can record at the time. He forgets none of it, and often the real work begins, drawn from the rich store of his memory, only after the model has left. His memory is wide and spacious, impressions are not changed within it, but they adjust to their surroundings, and when they pass into his hands it is as if they were the entirely natural gestures of these hands.’

Rainer Maria Rilke¹

The Body as Site

A powerful identification and projection takes place in creative work; the entire bodily and mental constitution of the maker becomes the site of the work. Even Ludwig Wittgenstein, whose philosophy is rather detached from body imagery, acknowledges the interaction of both philosophical and architectural work and the image of self: ‘Work on philosophy – like work in architecture in many respects – is really more work on oneself. On one’s own conception. On how one sees things. (And what one expects of them.)’²

The architect and artist alike are directly engaged with their sense of self rather than being focused intellectually on an external and objectifiable problem. A great musician plays himself rather than his/her instrument, in the same way that a masterful soccer player plays the entity of himself and the internalised and embodied field. 'The player understands where the goal is in a way, which is lived rather than known. The mind does not inhabit the playing field, but the field is inhabited by a "knowing body",' Richard Lang writes when commenting on Merleau-Ponty's views on the skill of playing soccer.³ The work is identified with and introjected in one's body. I become my work. I cannot perhaps intellectually analyse or know what is wrong with my work during the design or writing process, but my body knows it as feelings of uneasiness, distortion, asymmetry, pain, and a curious sensation of incompleteness and shame. I know that I have arrived at an acceptable rendition of the work only when my body feels relaxed and balanced; the body gives its signal of approval. The sensation of shame turns into a feeling of calmness and satisfaction.

The World and the Self

During its emergence, creative work calls for two simultaneous foci – the world and the self; and as a consequence of this double focus every profound work is essentially a microcosmic representation of the world and an unconscious self-portrait at the same time. Jorge Luis Borges gives a memorable expression of this double perspective: 'A man sets himself the task of portraying the world. Over the years he fills a given surface with images of provinces and kingdoms, mountains, bays, ships, islands, fish, rooms, instruments, heavenly bodies, horses, and people. Shortly before he dies he discovers that this patient labyrinth of lines is a drawing of his own face.'⁴

In our current understanding of architecture, both as architects and as occupants, we tend to close ourselves outside of the world and the phenomenon of architecture itself, and be mere observers. Yet it is the very boundary line between the self and the world that is opened and articulated in an artistic and architectural experience. It is also the boundary line between myself and the other. As Salman Rushdie argues in his essay in memory of Herbert Read (already quoted in the Introduction to this book): 'Literature is made at the boundary between self and the world, and during the creative act this borderline softens, turns penetrable and allows the world to flow into the artist and the artist to flow into the world.'⁵

Architecture is surely likewise conceived and experienced at this existential boundary line, and there is no artistic or architectural experience without the merging of the space and the viewer's/listener's/occupant's sense of self. When arguing that the architect's or artist's sense of self is a seminal focus in the work, I am not suggesting a narcissistic element in the process of design or making art. Simply, all the ideas have to be tested through the designer's/artist's own imagination, the wisdom of the body, and one's empathic capacity; there is no other authority or testing ground. The artist is the sole authority of his/her work; only weak artists seek external acceptance and authorisation. Only shallow artists seek recognition, as the confrontation with one's own existential boundary does not call for external or social recognition.

Besides, when designing a house, for instance, the architect cannot mentally design the house for the other as an outsider occupant. The architect needs to internalise the client, the other, and develop the design for his altered self. I lend my body, hands and mind to the service of the other, as if I in my role as architect were the surrogate mother for the birth of the house. In the very end of the process the product is given to the other, the client and the occupant. Without this deep personal internalisation and identification with the client/occupant/user, the architect only resolves the explicit requirements of the brief and satisfies the explicit intentions and desires of the other. Thus, a profound house is always a double portrait; it is an image of the client (or a specific cultural condition) and the architect at the same time. A profound piece of architecture is also essentially a gift.

In a way, the architect's relationship with the client is somewhat antagonistic. The designer does not think of the requests, attitudes or preferences of the client directly as he works with the internalised client as him/herself. 'Thinking about the reader is a deadly error for the writer,' JM Coetzee, the South African Nobel Laureate writer argues.⁶ In the Postscript to his book *The Name of the Rose*, Umberto Eco argues similarly that there are two categories of writers: the first type writes what he expects the reader to want to read, whereas the second creates his ideal reader as he writes. In Eco's view, the first writer will be able to produce mere kiosk literature whereas the second type is capable of writing literature that timelessly touches and elevates the human soul.⁷

I do not accept narcissism or self-centredness in architecture; but, in my view, the architect needs to create his/her ideal client in the process of design.

Meaningful architecture is conceived for a 'glorified' client, and it aspires to an idealised world, a form of life that is at least slightly more cultured, human and understanding than the concurrent actuality. This is why profound architecture always transcends its given conditions and achieves more than it is consciously commissioned to do. This is the true political dimension of architecture.

The World and the Mind

'How would the painter or the poet express anything other than his encounter with the world,'⁸ writes Maurice Merleau-Ponty, whose writings analyse the intertwining of the senses, the mind and the world, providing a stimulating ground for the understanding of artistic intention and impact.

Art and architecture structure and articulate our experiences of being-in-the-world. A work of art does not mediate conceptually structured knowledge

Genuine artistic works cannot be intellectually and logically explained, and they create a momentary short-circuit in the viewer's consciousness. Meret Oppenheimer, *My Nurse*, 1936, high-heeled shoes with paper ruffles on an oval platter, 14 x 21 x 33 cm, National Museum, SKM, Stockholm.



about the objective state of the world, but it renders possible an intense experiential and existential encounter. Without presenting any precise propositions concerning the world or its condition, art directs our view on the boundary surface between our sense of self and the world. 'It is bewildering that while grasping what surrounds him, what he is observing, and giving shape to these perceptions, the artist does not, in fact, say anything else about the world or himself, but that they touch each other,'⁹ writes the Finnish painter Juhana Blomstedt, echoing Merleau-Ponty's argument. The artist touches the skin of his world with the same sense of wonder as a child scratches the surface of a frosted window.

An artistic work is not an intellectual riddle seeking an interpretation or explanation. It is a complex of images, experiences and emotions which enters directly our consciousness. It has an impact on our mind before it becomes understood, or without ever being intellectually understood. The artist finds a way behind words, concepts and rational explanations in his/her repeated search for an innocent re-encounter with the world. Rational constructions provide little help for artistic search as the artist has to rediscover the boundary of his own existence repeatedly, time after time.

The artist's exploration focuses on lived experiential essences, and this aim defines his/her approach and method. As Jean-Paul Sartre states: 'Essences and facts are incommensurable, and one who begins his inquiry with facts will never arrive at essences [...] understanding is not a quality coming to human reality from the outside; it is its characteristic way of existing.'¹⁰ All artistic works approach this natural mode of understanding entwined with our own experience of being.

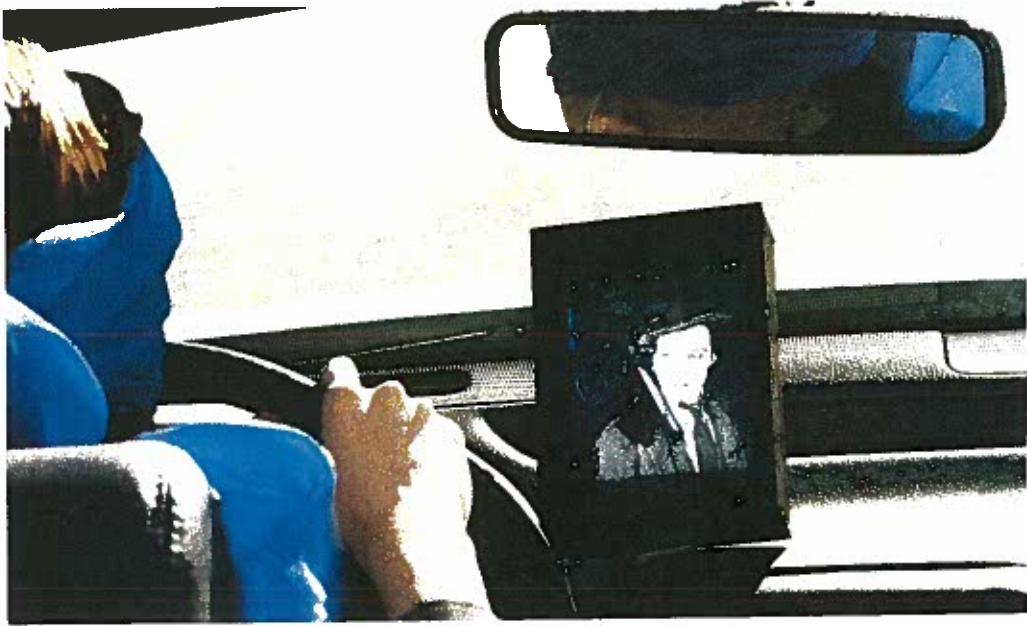
Existential Space in Art

We do not live in an objective world of matter and facts, as commonplace naïve realism assumes. The characteristically human mode of existence takes place in the worlds of possibilities, moulded by our capacities of fantasy and imagination. We live in worlds of the mind, in which the material and the mental, as well as the experienced, remembered and imagined, completely fuse into each other. As a consequence, the lived reality does not follow the rules of space and time as defined by the science of physics. In fact, we can say that the lived world is fundamentally 'unscientific' when measured by the criteria of Western empirical science. In its diffuse character, the lived

world is closer to the realm of dreams than to a scientific description. In order to distinguish the lived space from physical and geometrical space, it can be called lived, or existential space. Existential space is structured on the basis of meanings and values reflected upon it by an individual or group, either consciously or unconsciously; existential space is a unique experience interpreted through the memory and experience of the individual. On the other hand, groups, or even nations, share certain characteristics of existential space that constitute their collective identities and sense of togetherness. The experiential lived space, not physical or mathematical space, is the object and context of both the making and the experiencing of art and architecture. The task of architecture is 'to make visible how the world touches us', as Maurice Merleau-Ponty wrote of the paintings of Cézanne.¹¹ In accordance with the philosopher, we live in the 'flesh of the world'¹² and architecture structures and articulates this very existential flesh, giving it specific meanings. Architecture tames and domesticates space and time in the flesh of the world for human habitation. It frames human existence in specific ways and defines a basic horizon of understanding. In their widest and general meaning, architectural structures 'humanise' the world by giving it a human measure and cultural and human meanings. Architecture turns the soulless physical world into a home of man. We know and remember who we are and where we belong fundamentally through our cities and buildings, our constructed world, the human – architecturally humanised – microcosm.

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Art creates images and emotions that are experientially as true as the actual encounters of life.
Rebecca Horn, *Buster's Bedroom*, still from a film by the artist, 1990.

Emotion and Imagination

'This very power of synthesis points to a primordial unity of sensation and understanding brought about by the imagination prior to the functioning of either faculty. The synthetic role of imagination, presupposed by both faculties, is indeed so primordial that it operates behind our backs, as it were, unconsciously. A startling consideration this, which may explain why it took Western philosophy almost two thousand years to officially recognize its existence [...]'

Richard Kearney

Reality of Imagination

Imagination is usually associated specifically with the creative capacity, or the realm of art, but the faculty of imagination is the foundation of our mental existence and our way of dealing with stimuli and information. Research by brain physiologists and psychologists has shown that mental images are registered in the same zones of the brain as visual perceptions, and that these images possess all the experiential authenticity of those perceived by our own

eyes.² No doubt actual stimuli and imaginations in the other sensory realms are similarly close to each other, and thus, experientially, equally 'real'. This affinity or sameness of the external and internal experience is, of course, self-evident to any genuine artist without the proof of psychological research.

Experience, memory and imagination are qualitatively equal in our consciousness; we may be equally moved by something evoked by our memory or imagination as by an actual experience. Art creates images and emotions that are as true as the actual encounters of life. Fundamentally, in a work of art we encounter ourselves, our own emotions, and our own being-in-the-world in an intensified manner. A genuine artistic and architectural experience is primarily a strengthened awareness of self. An art work or building made thousands of years ago, or produced in a culture completely unknown to us, touches us because we encounter the timeless present of being a human being through the work, and consequently rediscover the actuality of our own being-in-the-world. One of the paradoxes of art and architecture is that although all moving works are unique, they reflect what is general and shared in the human existential experience. In this way, art is tautological; it keeps repeating the same basic expression over and over again: how it feels to be a human being in this world.

For me, one of the most touching and enigmatic works of architecture that I have ever encountered is surely the Ryoan-ji Zen garden in Kyoto. The unexplainable richness, subtlety and poetry of this piece of garden art is that it does not present any argument or theory; the fifteen stones set on a raked sand surface simply exist and project a combined sense of mystery and dignity, vagueness and completeness.

Art and architecture offer us alternative identities and life situations, and this is their great mental task. Great works give us the possibility of experiencing our own existence through the existential experience of some of the most talented individuals of humankind. This is the miraculous and merciful equality in all art. All artistic effect or impact is based on the identification of self with the experienced object, or the projection of the self on the object, as Melanie Klein argues. Jorge Luis Borges points out the true locus of artistic experience: 'The taste of the apple [...] lies in the contact of the fruit with the palate, not in the fruit itself; in a similar way [...] poetry lies in the meeting of the poem and reader, not in the lines of symbols printed on the pages of a book. What is essential is the aesthetic act, the thrill, the almost physical emotion that comes with each reading.'³

We experience a work of art or architecture through our embodied existence and capacity of projection and identification. An artistic experience activates a primordial mode of embodied, undifferentiated and animistic experiencing; the separation and polarisation of subject and object is temporarily lost and the material world is encountered as if it had its own life force. Both the glorious beauty and the pitiful ugliness of the object of artistic representation are momentarily identified with our own embodied experience. Many of us can never mourn our personal loss or tragedy with the intensity we suffer the fate of the fictive figures of literature, theatre and cinema, distilled through the existential experience of a great artist. Architectural ugliness or existential falseness can make us experience alienation and weakening of the sense of self, and finally make us fall mentally and somatically ill.

The Gift of Imagination

The uniqueness of the human condition is this: we live in the manifold worlds of possibilities created and sustained by our experiences, recollections and dreams. The ability to imagine and daydream is surely the most human and essential of our mental capabilities. Perhaps, after all, we are humans not because of our hands or intelligence, but thanks to our capacity for imagination. We would definitely not use our hands meaningfully without being able to imagine the result of our action. But the deluge of excessive, non-hierarchical and meaningless pictures in our current culture of images – 'an unending rainfall of images', in Italo Calvino's words⁴ – flattens our world of imagination. No space is left for the imagination as everything imaginable is already here. In his preface to his novel *Crash*, JG Ballard argues: 'the relationship between fiction and reality is turning upside down [...] we are increasingly living in a world of fictions, and that is why the writer's task is not to invent fiction. Fictions are already here, and thus the writer's task is rather to invent reality.'⁵ I feel likewise that the architectural imagination today, assisted and empowered by the computer, is producing too much architectural fiction, and we need rather to design an 'architecture of reality', to paraphrase the title of Michael Benedikt's book.⁶ We already long for an architecture that brings us back to the concrete realities of our physical and material world. This is not a sentimental longing for a lost world, but a world revitalised and re-eroticised by an architecture that makes us experience the world rather than itself.

The image flood of television externalises images and makes them passive when compared with the interior and active imagery evoked by reading a book. There is a dramatic difference between the passive looking at pictures, on the one hand, and images created by one's imagination, on the other. The effortless images of entertainment imagine on our behalf. The mesmerising image flow of the consciousness industry detaches images from their historical, cultural and human context and thus 'liberates' the viewer from investing his/her emotions and ethical attitudes in what is perceived. Benumbed by mass communication, we can already watch the most outrageous cruelty without the least emotional involvement. The deluge of images that grows overwhelming for the senses and emotions, suppresses and dulls imagination, empathy and compassion.

As our imagination weakens, we are left at the mercy of an incomprehensible future. Ideals are projections of an optimistic imagination, and consequently the loss of imagination is bound to wipe out idealism, too. In my view, the lack of horizon, ideals and alternatives even in today's political thought is a consequence of a withering of political imagination. The overpowering sense of pragmatism and lack of stimulating visions today are likely to be consequences of an impoverished imagination. A culture that has lost its imagination can only produce apocalyptic visions of threat as projections of its repressed collective unconscious. A world devoid of imaginable alternatives, due to the absence of imagination, is the world of Aldous Huxley's and George Orwell's manipulated subjects.

The duty of education is to cultivate and support the human abilities of imagination and empathy, but the prevailing values of culture tend to discourage fantasy, suppress the senses, and petrify the boundary between the world and the self. The idea of sensory training is nowadays connected solely with artistic education proper, but the refinement of sensory literacy and sensory thinking has an irreplaceable value in all areas of human activity.

Reality of Art

The manner through which art affects our mind is one of the great mysteries of human communication. The understanding of the essence and mental workings of art has become confused and blurred by the superficial use of the notions of 'symbolisation' and 'abstraction' as well as the obsession

'When it rains in my films, it simply rains.'
Andrey Tarkovsky, *Nostalghia*,
1983 Domenico's leaking house.
Production: Opera Film (Rome)
for RAI TV Rete 2 in association
with Sovinfilm (USSR).



with newness. A work of art or architecture is not a symbol that represents or indirectly portrays something outside of itself; it is an image object that places itself directly in our existential experience. The notion of symbolisation should be viewed critically and with suspicion in the context of art. Andrey Tarkovsky, for one, whose films appear to be saturated with symbolic signification, strongly denies any specific symbolisation in his work. In his films rooms are flooded with water, water soaks through ceilings, and it is constantly raining. Yet he exclaims firmly: 'When it rains in my films, it simply rains.'⁷

Also Sartre is critical of the notion of symbolisation in artistic representation. In his view, art creates things rather than symbols: 'Tintoretto did not choose that yellow rift in the sky above Golgotha to *signify* anguish or to *provoke* it,' he writes. 'Not sky of anguish or anguished sky; it is an anguish become thing, an anguish which has turned into yellow rift of sky [...]. It is no longer *readable*.'⁸ Similarly, Michelangelo's stair hall of the Laurentian Library (1524–59) and the Medici Chapel (1505–34) with its allegorical sculptures are not symbols of melancholy, they are buildings that have fallen into a state of melancholy – or, more precisely, we lend these buildings our own sensation of metaphysical sorrow.

The buildings of Louis Kahn are not metaphysical symbols either; they are a form of metaphysical meditation through the medium of architecture, that leads us to recognise boundaries of our own existence and to deliberate on the essence of life. They direct us to experience our own existence with a unique intensity. Similarly, the masterpieces of early modernity do not represent optimism and love of life through architectural symbolisation. Even decades after these buildings were conceived, they evoke and maintain these positive sensations; they awaken and bring forth the hope sprouting in our soul. The joyful buildings of Gunnar Asplund's Stockholm Exhibition (1930) cannot be regarded as symbols of optimism, and Alvar Aalto's Paimio Sanatorium (1929–33) is not a mere metaphor of healing; even today these masterpieces offer the comforting promise of a better future.

A work of art may, of course, have conscious symbolic contents and intentions, but they are insignificant for the artistic impact or the temporal endurance of the work. Even the simplest work of art in terms of its external appearance is not devoid of meanings, or of connections with our existential and experiential world. An impressive work is always an image condensation that is capable of mediating the entire experience of being-in-the-world through a singular image. But, as Anton Ehrenzweig writes, 'Scientific abstraction differs from an empty generalization in the way in which potent abstract art differs from empty ornament.'⁹ In the words of Andrey Tarkovsky: 'The image is not a certain meaning, expressed by the director; but an entire world reflected as in a drop of water.'¹⁰

In a similar way, the mental impact of architecture does not derive from a formal or aesthetic game; it arises from experiences of an authentic sense of life. Architecture does not invent meaning; it can move us only if it is capable of touching something already buried deep in our embodied memories.

Art and Emotion

The art form of architecture mediates and evokes existential feelings and sensations. Architecture of our time, however, has normalised emotions and usually eliminates completely such extreme emotions as sorrow and bliss, melancholy and ecstasy.

The places and streets conceived by literature, painting and cinema are as saturated with emotion and as real as houses and cities built of stone. The

'invisible cities' of Italo Calvino enrich the urban geography of the world in the same way as the material cities built through the labour of thousands of hands. The commonplace and desolate rooms of Edward Hopper, or the shabby room in Arles painted by Vincent van Gogh, are as full of life and affect as the 'real' rooms in which we live. The 'Zone' in Andrey Tarkovsky's *Stalker*, which exudes an air of inexplicable threat and disaster, is certainly more real in our experience than the actual anonymous industrial structures in Estonia where the film was shot, because the landscape pictured by the masterful film director contains more significant human meanings than its physically real original. The mysterious 'Room' searched for by 'the Writer' and 'the Scientist' under the guidance of Stalker is finally disclosed as a very ordinary room, but the imagination of the travellers, as well as of the viewer of the film, has turned it into a centre point of metaphysical significance. This ordinary room has turned into Teilhard de Chardin's 'Omega', 'the point from which the world can be seen as a whole and correctly'.¹¹

Artistic Experience as Exchange

In the experience of art and architecture, a peculiar exchange takes place; I project my emotions and associations onto the work, or space, and it lends me its aura that emancipates my perceptions and thoughts. In Joseph Brodsky's view, a poem tells its reader: 'Be like me.'¹² The imaginary space suggested by the work turns real, and it becomes part of my experiential life world. As I experience the touching melancholy of Michelangelo's architecture, for instance, I am in fact moved by my own sense of melancholy as it is evoked and reflected back by the architectural work. I lend my melancholy to the Laurentian staircase in the same way that I lend Raskolnikov my experience of frustrated waiting while reading Dostoyevski's *Crime and Punishment*. This identification with the work of art, and the scene depicted by it, is so powerful that I find it unbearable to look at Titian's painting *The Flaying of Marsyas* (c. 1575), in which the satyr is skinned alive in Apollo's revenge, because I feel that my own skin is being violently peeled off.

An architectural work is not experienced as a series of isolated retinal pictures; it is touched and lived in its full and integrated material, embodied and spiritual essence. A profound work is always a world and a complete microcosm. It offers pleasurable shapes and surfaces moulded for the touch of the eye, but it also incorporates and integrates physical and



mental structures, giving our existential experience of being a strengthened coherence and significance. A great building enhances and articulates our understanding of gravity and materiality, horizontality and verticality, the dimensions of above and below, as well as the eternal enigmas of existence, light and silence.

Architectural melancholy of an artistic giant
Michelangelo Buonarroti and
Bartolomeo Ammannati,
Laurentian Library (Biblioteca
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1525. View of the stairs from
above

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Theory and Life

'Where is the life we have lost in living?
Where is the wisdom we have lost in knowledge?
Where is the knowledge we have lost in information?'

TS Eliot¹

Theory and Making

In his essay 'The sculptor speaks', Henry Moore – undoubtedly one of the greatest artists of the 20th century – writes about the relationship between creative work and its analysis: 'It is a mistake for a sculptor or a painter to speak or write very often about his work. It releases tension needed for his work. By trying to express his aims with rounded-off logical exactness, he can easily become a theorist whose actual work is only a caged-in exposition of conceptions evolved in terms of logic and words.'²

This passage expresses an artist's characteristic suspicion for the role of verbalisation, intelligence and theorisation in artistic work, or for analysing art intellectually. The sculptor does, however, connect the intuitive, spontaneous acts of making and the unconscious, synthesising essence of the creative act to man's rational analytic capacity: 'But though the non-logical, instinctive, subconscious part of the mind must play its part in his work, he also has a



Auguste Rodin, *The Cathedral*,
bronze, 65 x 30 x 30 cm, 1910

conscious mind, which is not inactive. The artist works with a concentration of his whole personality, and the conscious part of it resolves conflicts, organises memories, and prevents him from trying to walk in two directions at the same time.³

Matisse shares Moore's suspicion for verbal explanation of artistic intention in his rather shocking advice to aspiring young painters: 'First of all you must cut off your tongue because your decision takes away from you the right to express yourself with anything but your brush.'⁴

Today there seems to be a widespread confusion concerning the role of theory in the field of art, as well as the relationship of architectural theory and the making of architecture. Adopting a theoretical position and making explicit verbalised philosophical statements are often considered prerequisites for meaningful architecture. As a result, architectural education tends to be dominated by conceptualisation and the articulation of conscious intentions, and today's avant-garde architecture is often a mere medium for projecting intellectual or speculative ideas.



Henry Moore, *Sheep Piece*, height 435 cm. Photo Emily Peters.
The sculpture is placed in the grazing pasture beyond the studio where Moore used to work on his maquettes. The sculptor made the *Sheep Piece* as a gift to the sheep; the size of the sculpture allows the animals to walk under and rub against its inner surfaces.

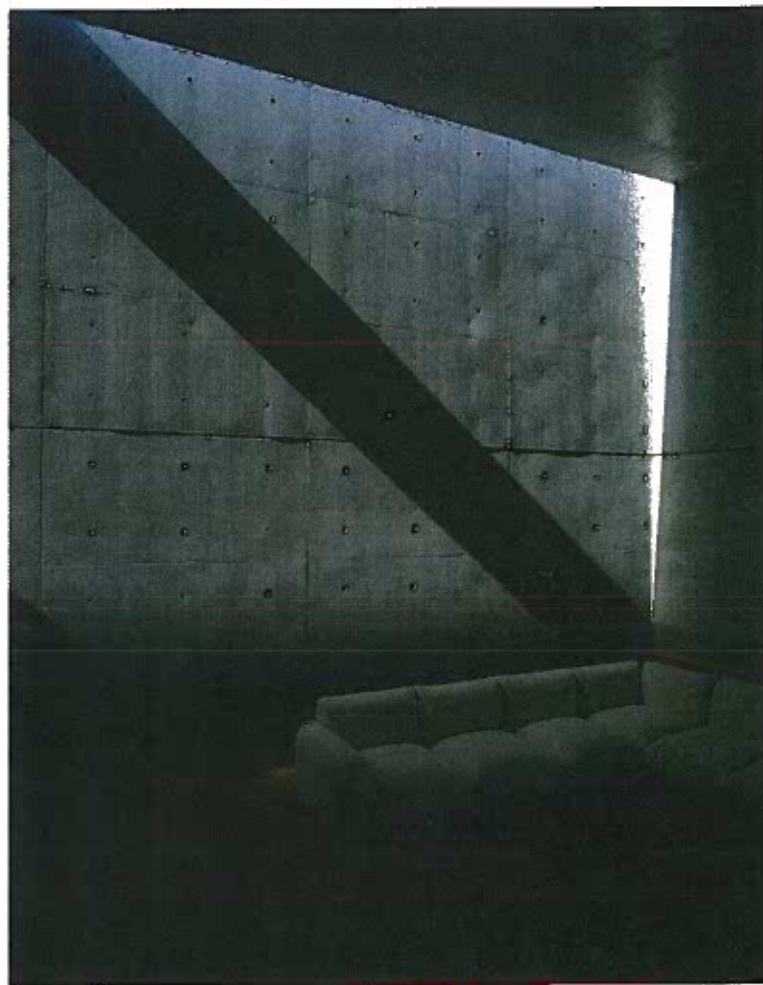
I, for one, do not believe in the need, or even the possibility, of a comprehensive, prescriptive theory of architecture by which design solutions and meanings could be generated. At the same time, however, I see distinct roles for the conceptual, philosophical and theoretical analyses of architecture. And, most importantly, designers need to be conscious enough of their aspirations and methods to 'prevent them from trying to walk in two directions at the same time', as the master sculptor advises.

Opposition of Theory and Making

Perhaps more interesting than the correlation of theory and architecture is the distance, tension and dialectic interaction between a theoretical point of departure and a creative exploration. Somewhat surprisingly, Tadao Ando has expressed a desire for a similar opposition between functionality and uselessness: 'I believe in removing architecture from function after ensuring the observation of functional basis. In other words, I like to see how far architecture can pursue function and then, after the pursuit has been made, to see how far architecture can be removed from function. The significance of architecture is found in the distance between it and function.'⁵

In any creative field the process of un-learning is just as important as learning, forgetting as important as remembering, uncertainty as important as certainty. Gaston Bachelard, for one, writes of the value of forgetting knowing: 'Knowing must [...] be accompanied by an equal capacity to forget knowing. Non-knowing is not a form of ignorance but a difficult transcendence of knowledge. This is the price that must be paid for an oeuvre to be, at all times, a sort of pure beginning, which makes its creation an exercise in freedom.'⁶ Knowledge becomes useful for a creative endeavour, when it is forgotten after it has turned into an ingredient of one's body and personality. The moment of looking at the world or at a specific task as if neither had been encountered before, *that* is the creative instant and frame of mind.

In the act of artistic making, a theoretical or intellectual consciousness has to be suppressed, if not entirely forgotten. More precisely, only embodied knowledge divorced from conscious attention seems to be useful in creative work. Jorge Luis Borges makes a provocative remark on his own habit of working: 'When I write something, I try not to understand it. I do not think



'The meaning of architecture lies in the distance between architecture and function.'
Tadao Ando, Koshino House, Ashiya, Hyogo, Japan, 1979–81, addition 1988–4. Photo Yoshio Takase.

intelligence has much to do with the work of a writer. I think one of the sins of modern literature is that it is too self-conscious.⁷ Even in the simple act of riding a bicycle, the theoretical knowledge of how the vehicle is kept upright is suppressed as the act is performed unconsciously through body memory; if you attempt to think what really takes place theoretically and factually in the dynamic and complex balancing act of bicycle riding, you are in immediate danger of tumbling over. Moshé Feldenkrais concludes:

The execution of an action by no means proves that we know, even superficially, what we are doing or how we are doing it. If we attempt to carry out an action with awareness – that is, to follow it in detail – we soon discover that even the simplest and most common of actions, such as getting up from a chair, is a mystery, and that we have no idea at all how it's done.⁸

The poet, sculptor or architect works through his/her entire physical and mental being, not primarily through the intellect, theory or acquired professional skills. In fact, what has been learned has to be forgotten in order to be useful. 'In my work I have never had any use for things I have known before I start my new artistic project,' the great Basque sculptor Eduardo Chillida said to me once in conversation.⁹

This paradoxical simultaneity of forgetting and knowing, learning and unlearning, is conceivable when we understand the fundamental unconscious and collaborative nature of any artistic or creative endeavour. The true artist and maker collaborates with the silent tradition of the craft and deploys the tacit knowledge that has accumulated in the tradition of the craft in question.

Artistic work is always bound to be a collaboration simultaneously on several levels. As John Dewey informs us in his seminal book *Art as Experience* (1934),¹⁰ the artistic dimension arises from the encounter of the work and its reader/viewer. The artistic experience is a collaborative effort of the writer and the reader, the painter and the viewer, the architect and the occupant. As Sartre argues: 'It is the joint effort of author and reader which brings upon the scene that concrete and imaginary object which is the work of the mind. There is no art except for and by others.'¹¹

'Great poetry is possible only if there are great readers,' argues Walt Whitman significantly.¹² It is equally evident that there are good buildings only as long as there are good dwellers and occupants; but aren't we, citizens of this obsessively materialist consumer world, losing our capacity to dwell, and as a consequence becoming unable to promote architecture as great readers/users of architectural spaces and narratives? Ludwig Wittgenstein suggests in one of his notes that this could, indeed, be the case: 'Architecture immortalises and glorifies something. Hence, there can be no architecture, where there is nothing to glorify.'¹³ Haven't we lost the dimensions in our culture and personal lives that could be worthy of glorification? Haven't we lost the dimension of 'ideals in our obsessively

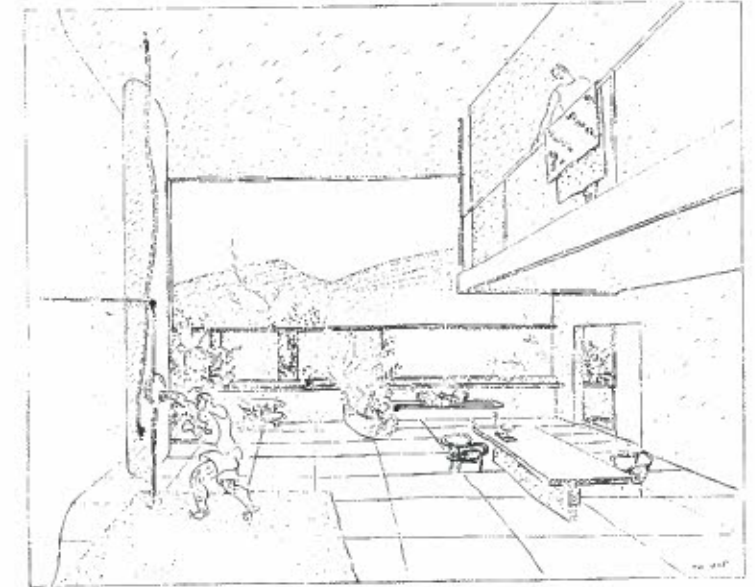
materialist world? Architectural thought arises from given conditions, but it always aspires to an ideal. Hence, the loss of the ideal dimension of life implies the disappearance of architecture.

Architectural works are rarely built by the architect alone; buildings arise from the collaborative effort of tens, often thousands of individuals, experts, builders, craftsmen, engineers and investors. But architecture is collaboration also in another, perhaps more fundamental sense. Meaningful buildings arise from tradition and they constitute and continue a tradition. In his book *The Art of the Novel*, Milan Kundera writes of 'the wisdom of the novel';¹⁴ he argues that all great writers listen to this wisdom, and as a consequence all great novels are wiser than their writers. No doubt there is also a 'wisdom of architecture', and all profound architects listen to this wisdom in their work. No architect worthy of his craft works alone; he works with the entire history of architecture 'in his bones';¹⁵ as TS Eliot writes about the tradition-conscious author. The great gift of tradition is that we can choose our collaborators; we can collaborate with Brunelleschi and Michelangelo if we are wise enough to do so.

In his first book, Vitruvius emphasises the importance of setting manual skill alongside a theoretical foundation: 'So architects who without culture aim at manual skill cannot gain a prestige corresponding to their labours, while those who trust to theory and literature obviously follow a shadow and not reality. But those who have mastered both, like men equipped in full armour, soon acquire influence and attain their purpose.'¹⁶

In my view, the discipline of architecture has to be grounded on a trinity of conceptual analysis, the making of architecture, and experiencing – or encountering – it in its full mental, sensory and emotional scope. The point that I wish to emphasise is that an emotional encounter with architecture is indispensable both for creating meaningful architecture and for its appreciation and understanding. Design practice that is not grounded in the complexity and subtlety of experience withers into dead professionalism devoid of poetic content and incapable of touching the human soul, whereas a theoretical survey that is not fertilised by a personal encounter with the poetics of building is doomed to remain alienated and speculative – and can, at best, only elaborate rational relationships between the apparent elements of architecture. But there are no 'elements' in artistic phenomena, as the parts derive their entire meaning from the whole.

The profound works of Modernism are images of a living reality and the newly emancipated lifestyle. Architecture is not a mere aestheticised object, but a staging and forming of life. Le Corbusier, *Jardin suspendu*, project, 1928–9.



Architecture as Image of Life

Architecture provides our most important existential icons by which we can understand both our culture and ourselves. The important works of Modernism are images of a living reality and a newly emancipated lifestyle; those of our own time are frequently merely self-referential pictures of architecture itself. Today's celebrated works frequently deal with philosophical issues of representation more than with mental content; they are discourses within the discipline itself without reflecting true life.

I would be the last one to question the validity of scholarly and theoretical investigation, but I want to promote investigations carried out with perceptive senses and a receptive and empathetic heart. Architecture is an art form of the eye, the hand, the head and the heart. The practice of architecture calls for the eye in the sense of requiring precise and perceptive observation. It requires the skills of the hand, which must be understood as an active instrument of processing ideas in the Heideggerian sense. As architecture is an art of constructing and physical making, its processes and origins are essential ingredients of its very expression. The architect needs his/her head for clear thinking – great architectural works never arise from

muddled thought. Yet architecture requires a special category of thinking, an embodied thinking through the very medium of architecture itself. Finally, the architect needs his/her heart in order to imagine situations of real life and to feel compassion for the human destiny. In my view, the gift of heart is most underestimated as a prerequisite for architecture in our times of self-centredness and false self-assurance.

The Task of Art

As today's consumer, media and information culture increasingly manipulate the human mind through thematised environments, commercial conditioning and benumbing entertainment, art has the mission to defend the autonomy of individual experience and provide an existential ground for the human condition. One of the primary tasks of art is to safeguard the authenticity and independence of human experience.

The settings of our lives at large are irresistibly turning into a mass-produced and universally marketed kitsch. In my view, it would be ungrounded idealism to believe that the course of our obsessively materialist culture could be altered within the foreseeable future. But exactly because of this pessimistic view of the future of technologically advanced cultures, the ethical task of architecture and art is so important. In a world where eventually everything becomes similar, insignificant and of no consequence, they have to maintain differences of meaning and, in particular, the criteria of sensuous experiential and existential quality. It remains the responsibility of the artist and the architect to defend the enigma of life and the eroticism of the life world.

'Only if poets and writers set themselves tasks that no one else dares imagine will literature continue to have a function,' Italo Calvino states. 'The grand challenge for literature is to be capable of weaving together the various branches of knowledge, the various "codes" into a manifold and multifaceted vision of the world.'¹⁷ Confidence in the future of architecture must, in my view, be based on the very same knowledge of its specific task; architects need to set themselves tasks that no one else knows how to imagine. Existential meanings of inhabiting space can be articulated by the art of architecture alone. Thus architecture continues to have a great human task in mediating between the world and ourselves and in providing a horizon of understanding the human existential condition.

'My confidence in the future of literature consists in the knowledge that there are things that only literature can give us, by means specific to it,'¹⁸ writes Italo Calvino in his *Six Memos for the Next Millennium*; and he continues (in another chapter):

In an age when other fantastically speedy, widespread media are triumphing, and running the risk of flattening all communication onto a single, homogenous surface, the function of literature is communicating between things that are different simply because they are different, not blunting but even sharpening differences between them, following the true bent of written language.¹⁹

In my view, the task of architecture is to maintain the differentiation and hierarchical and qualitative articulation of existential space. Instead of participating in the process of further speeding up the experience of the

Andy Goldsworthy, *Dandelion flowers pinned with thorns to wind-bent willowherb stalks laid in a ring*, Yorkshire Sculpture Park, West Bretton, England, 1 May 1987



world, architecture has to slow down experience, halt time, and defend the natural slowness and diversity of experience. Architecture must defend us against excessive exposure, noise and communication. Finally, the task of architecture is to maintain and defend silence.

Art is generally viewed as a means of reflecting reality through the artistic artefact. The art of our time thought-provokingly often reflects experiences of alienation and anguish, violence and inhumanity. In my view, mere reflection and representation of prevailing reality is not a sufficient mission for art. Art should not increase or reinforce human misery, but should alleviate it. The duty of architecture and art is to survey ideals and new modes of perception and experience, and thus open up and widen the boundaries of our lived world.

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¹⁴ Milan Kundera, *Romaanin taide [The Art of the Novel]*,

Werner Söderström (Helsinki), 1986, p. 165 (translated by Juhani Pallasmaa).

¹⁵ TS Eliot, 'Tradition and the Individual Talent', *Selected Essays*, Faber & Faber (London), 1948, pp. 14–15. 'The historical sense involves a perception not only of the pastness of the past, but of its presence. The historical sense compels a man to write, not merely with his own generation in his bones, but with a feeling that the whole of literature [...] has a simultaneous existence and composes a simultaneous order.'

¹⁶ Frank Granger, *Vitruvius on Architecture*, William Heinemann (London) and Harvard University Press (Cambridge, Massachusetts), 1955, p. 7.

¹⁷ Italo Calvino, *Six Memos for the Next Millennium*, Vintage Books (New York), 1993, p. 112.

¹⁸ Ibid.

¹⁹ Ibid, p. 45.

Shirin Neshat, *Bonding*, resin-coated print and ink, 86.1 x 130.8 cm, 1995. Edition of 10+1 AP. Photo taken by Kyong Park.

